

Privacy-Preserving Machine Learning with Federated Learning and Differential Privacy: A Comprehensive Survey of Mechanisms, Trade-offs, and Frontiers

1 Foundations, Architectural Paradigms, and Threat Models

1.1 Core Definitions and the Privacy-Utility-Efficiency Trilemma

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Federated Learning (FL) has emerged as a transformative paradigm in distributed machine learning, enabling multiple participating parties to collaboratively construct a global model without disclosing their private raw data. In this architecture, data remains localized on client devices or within institutional silos, while only model updates—such as gradients or parameters—are transmitted to a central aggregator. Although this fundamental shift aims to bolster data privacy by design, the mere retention of data locally does not inherently guarantee complete confidentiality. Recent investigations have demonstrated that adversaries can still infer sensitive information by analyzing shared model updates through techniques such as gradient inversion or membership inference attacks. Consequently, fulfilling the rigorous requirements of modern data protection necessitates the integration of robust protection mechanisms into the FL framework. These mechanisms must reconcile the often opposing demands of preserving strict privacy and maintaining high model utility. Furthermore, for any FL system to be viable in large-scale deployments, it must achieve high efficiency regarding communication bandwidth, computational resources, and training latency. The intricate interplay between these three critical pillars—privacy, utility, and efficiency—forms the cornerstone of theoretical analysis in Privacy-Preserving Machine Learning (PPML) and serves as the foundational context for the architectural taxonomies discussed in the subsequent section.

To formally address privacy concerns within this distributed setting, Differential Privacy (DP) has become the gold standard for providing rigorous, mathematical guarantees. DP ensures that the output of an algorithm remains statistically indistinguishable whether or not any single individual’s data is included in the training set. In the context of FL, this is typically achieved by injecting controlled randomness, or noise, into the model updates before they are shared. While DP offers a robust defense against various inference attacks, its implementation introduces a complex optimization landscape. The introduction of noise, while essential for privacy, inevitably perturbs the learning trajectory, potentially degrading the accuracy and convergence properties of the global model. This creates an inherent tension where strengthening privacy guarantees through increased noise levels or stricter privacy budgets directly correlates with a reduction in model utility.

The theoretical underpinnings of this tension are rigorously formalized in the concept of the “No-Free-Lunch” (NFL) theorem for federated learning systems. As proposed in recent foundational work, it is fundamentally unrealistic to expect an FL algorithm to simultaneously provide excellent privacy, high utility, and maximum efficiency across all scenarios [1]. This theorem posits that improvements in one dimension inevitably necessitate compromises in at least one of the others. For instance, enhancing privacy through mechanisms like randomization, homomorphic encryption, secret sharing, or compression invariably incurs costs in terms of utility loss or efficiency reduction [1]. The NFL theorem serves as a critical guide for system designers, indicating that the selection of protection parameters is not merely a tuning exercise but a strategic decision that defines the operational boundaries of the system.

Recent research has expanded this trilemma analysis beyond static bounds to dynamic, efficiency-constrained optimization problems. In Differentially Private Federated Learning (DPFL), previous studies often focused narrowly on the trade-off between utility and privacy, frequently overlooking the critical role of training efficiency, which is paramount for timely model completion [2]. However, efficiency is not a monolithic metric; it is influenced by multiple factors including the noise level (σ), the total number of communication rounds (T), and the client sample ratio (q). A comprehensive theoretical analysis reveals that these factors are deeply interconnected. For example, increasing the noise level to satisfy stringent privacy requirements may necessitate more communication rounds to achieve convergence, thereby reducing efficiency. Conversely, increasing the sample ratio can improve convergence speed but may alter the privacy accounting dynamics. By formulating this as an efficiency-constrained utility-privacy bi-objective optimization problem, researchers have derived analytical solutions for the Pareto front, demonstrating that optimal parameter design requires a holistic view of all three dimensions [2].

The challenge is further exacerbated in realistic scenarios characterized by data heterogeneity and resource constraints. In settings where user data is highly non-IID (non-independent and identically distributed), the privacy-utility trade-off becomes even more pronounced. Algorithms like DP-SCAFFOLD have been developed to incorporate DP constraints while addressing heterogeneity, yet they clearly highlight that privacy guarantees come at the cost of utility, particularly as the level of data heterogeneity and the number of local updates increase [3]. Moreover, in resource-constrained edge environments, the computational overhead of DP mechanisms can lead to severe performance degradation and transmission bottlenecks. Innovations such as integrating Lottery Ticket Hypothesis pruning with zero-concentrated DP aim to mitigate these issues by reducing model size and tightening privacy analysis, yet the fundamental trade-offs remain [4].

Furthermore, the pursuit of optimality in this trilemma has led to the development of unified frameworks that leverage learning theory to quantify these objectives. Approaches such as FedPAC utilize Probably Approximately Correct (PAC) learning to transform the multi-objective trade-off problem into a more tractable single-objective optimization, quantifying utility loss, privacy leakage, and efficiency in terms of sample complexity [5]. These theoretical advancements underscore that achieving a balanced system is not about eliminating trade-offs but about navigating the feasible region defined by the NFL theorem. Whether through adaptive noise scheduling, hierarchical privacy structures, or game-theoretic pricing models, the core objective remains the identification of an optimal operating point that satisfies specific application requirements without violating the fundamental limits imposed by the privacy-utility-efficiency trilemma [6]. Thus, understanding these core definitions and their intrinsic conflicts provides the necessary theoretical lens to evaluate the specific architectural paradigms—ranging from Horizontal and Vertical FL to Split and Decentralized systems—explored in the following section, where the manifestation of these trade-offs varies significantly based on data partitioning and network topology.

1.2 Taxonomy of Federated Architectures and Data Partitioning

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Building upon the theoretical constraints of the privacy-utility-efficiency trilemma established in Section 1.1, the practical realization of Privacy-Preserving Machine Learning (PPML) is fundamentally shaped by the architectural choices regarding data distribution and network topology. These structural decisions are not merely engineering preferences; they dictate the specific threat models, the nature of privacy leakage risks, and the applicability of Differential Privacy (DP) mechanisms.

As the following section will detail, the vulnerability of a system to honest-but-curious or malicious actors varies drastically depending on whether the architecture is Horizontal, Vertical, Split, or Decentralized. Consequently, a rigorous taxonomy of these paradigms is essential for understanding how the abstract trade-offs of Section 1.1 manifest in concrete system designs.

Horizontal Federated Learning (HFL) represents the most canonical form of federated training, characterized by sample partitioning where clients share a common feature space but possess disjoint sets of data samples. This architecture is prevalent in cross-device settings, such as mobile keyboard prediction or IoT sensor networks. However, the assumption that local model updates are inherently privacy-preserving is often flawed; research indicates that even in HFL, private information can be divulged by analyzing uploaded parameter weights, necessitating the integration of DP mechanisms to mitigate membership inference and model inversion attacks [7]. The efficacy of these defenses in HFL is highly sensitive to data heterogeneity; empirical benchmarks demonstrate that non-i.i.d. data distributions and small local datasets lead to the most significant performance degradation when DP noise is introduced [7]. Furthermore, the sheer volume of data generated by smart devices in these horizontal settings has spurred the development of Local Differential Privacy (LDP) variants, which perturb data on the client side before transmission to ensure robust privacy assurances regardless of server trustworthiness [8].

In contrast, Vertical Federated Learning (VFL) addresses scenarios where data is partitioned by features rather than samples. Here, multiple organizations hold different attributes for the same set of users, a common setup in finance and healthcare for tasks like fraud detection or disease prediction. The privacy dynamics in VFL differ significantly from HFL because the exchange of intermediate gradients or feature embeddings can reveal sensitive underlying data structures. For instance, shared feature embeddings in VFL may inadvertently leak information under privacy attacks, prompting the need for specialized defenses such as norm clipping and adaptive feature embeddings to maintain a delicate equilibrium between privacy and task utility [9]. The complexity of VFL is further compounded during model evaluation, where standard metrics like the Area Under Curve (AUC) can leak label information if not computed with specific label-DP algorithms designed for vertical settings [10]. To address the computational bottlenecks and privacy risks simultaneously, recent advancements have introduced secure aggregation protocols that offer massive speedups compared to Homomorphic Encryption while preserving the confidentiality of vertically partitioned data [11]. Additionally, frameworks utilizing secret sharing have been proposed to mitigate straggler effects in split VFL models while ensuring information-theoretical privacy against colluding clients and curious servers [12].

Split Learning (SL) and its hybrid variants introduce a third architectural paradigm, where a deep neural network is partitioned between clients and servers. In this setup, clients transmit “smashed data” (activations from a cut layer) rather than full model updates. While this reduces the computational load on edge devices, it introduces distinct vulnerability surfaces tailored to specific threat profiles. Security analyses reveal that the lower dimensionality of updates in Split Learning can sometimes offer greater robustness against model poisoning attacks compared to traditional FL, as the reduced parameter space limits the attack surface [13]. However, the transmission of intermediate activations remains a vector for reconstruction attacks, requiring specialized countermeasures. To counter this, novel frameworks like Λ -Split partition generative models into three segments, ensuring that raw inputs and outputs remain on the local device while only hidden layer representations traverse the network, thereby fortifying privacy against eavesdropping in cloud-powered generative AI services [14]. Hybrid architectures, such as Hybrid Split FL (HSFL), have also emerged to leverage the strengths of both FL and SL in Device-to-Device (D2D) heterogeneous

networks, aiming to reduce communication costs and training delays while navigating the complex privacy landscape of distributed feature learning [15].

Finally, Decentralized or Peer-to-Peer (P2P) Federated Learning eliminates the central aggregator, relying instead on a graph topology where nodes communicate only with their neighbors. This architecture enhances robustness and scalability but introduces a critical, often overlooked privacy variable: the network topology itself. Recent theoretical work establishes that the privacy bound in decentralized FL is strictly dependent on the size of “honest components”—maximally connected subgraphs remaining after excluding untrustworthy nodes [16]. Consequently, attack-resilient network topologies provide superior privacy guarantees, whereas less robust structures, such as power-law networks, are prone to revealing more private information. To optimize this, topology-aware DP mechanisms like Top-DP have been designed to leverage unique communication features to reduce noise scales and improve model usability without compromising privacy guarantees [17]. Furthermore, decentralized optimization algorithms employing relay communication mechanisms, such as DP-RECAL, have been developed to minimize local communication involvement, thereby reducing the overall privacy budget required to defend against deep leakage from gradients [18]. The interplay between decentralization and privacy is further complicated by the need for asynchronous updates; frameworks like A(DP)²SGD integrate Rényi DP into asynchronous decentralized parallel SGD to prevent inference attacks while maintaining optimal convergence rates in heterogeneous environments [19]. Ultimately, the choice of architecture dictates the fundamental trade-offs, requiring designers to carefully align data partitioning strategies and network topologies with appropriate DP mechanisms to ensure viable privacy-utility balances [20]. By defining these structural landscapes, we establish the necessary context to analyze how specific adversary capabilities—from passive observation to active manipulation—exploit the unique weaknesses of each paradigm, as explored in the subsequent section on threat modeling.

1.3 Threat Modeling: From Honest-But-Curious to Malicious Actors

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While Section 1.2 established how architectural choices and data partitioning define the structural landscape of Privacy-Preserving Machine Learning (PPML), the actual security and privacy guarantees of these systems are fundamentally contingent upon the specific threat models they aim to address. The topology of a federated network dictates the attack surface, but the adversary’s capability—ranging from passive observation to active manipulation—determines the severity of the risk. A rigorous analysis must therefore distinguish between passive adversaries who seek to infer sensitive information through observation and active adversaries who deliberately manipulate the learning process to compromise integrity or reconstruct private data. This spectrum spans from the “honest-but-curious” server, which adheres to protocol specifications while attempting to glean insights from client updates, to fully malicious clients capable of executing sophisticated poisoning, backdoor, and gradient inversion attacks. Understanding these threat vectors is essential for contextualizing the theoretical guarantees discussed in Section 1.4, as the efficacy of Differential Privacy (DP) mechanisms varies significantly depending on whether the adversary is bounded by computational limits or possesses adaptive, malicious intent.

The honest-but-curious threat model remains a primary concern in centralized FL architectures, particularly those described in Horizontal and Vertical settings where a central server aggregates updates. In this scenario, the server faithfully executes the aggregation algorithm but utilizes the received gradients or model parameters to infer the raw training data of individual clients. This

vulnerability is starkly illustrated by Deep Gradient Leakage (DGL) attacks, where adversaries can recover high-fidelity private training images solely from shared gradient vectors. Recent systematizations of these attacks reveal that while early iterations relied on ideal settings, modern techniques have evolved to function under more practical constraints, although their efficacy can be sensitive to model architecture and post-processing defenses [21]. The severity of this threat is underscored by the development of scalable model inversion attacks, such as Scale-MIA, which can efficiently reconstruct user data from aggregated updates even when secure aggregation protocols are in place, by exploiting the latent space representations within the model architecture [22]. Furthermore, the risk extends beyond simple reconstruction; leaked data can be utilized to train downstream models that outperform benign federated strategies, demonstrating that privacy breaches have direct utility implications for attackers [23]. These passive inference capabilities highlight why DP mechanisms, often introduced in the architectural phase, are critical for obscuring individual contributions, even when the server follows the protocol.

Beyond passive inference, the landscape includes active adversaries who act as malicious clients or compromised servers, introducing threats to model integrity that passive protections alone cannot address. These actors aim to degrade model utility, inject backdoors, or bias the global model through data poisoning and model poisoning attacks. Backdoor attacks are particularly insidious, as they allow an adversary to embed a hidden trigger that causes the model to misclassify specific inputs while maintaining normal performance on benign data. The threat landscape has expanded to include client-targeted backdoor attacks, such as the “Sniper Backdoor,” which leverages generative adversarial networks and Siamese neural networks to compromise a single victim client without affecting others, thereby evading many standard defense mechanisms [24]. The sophistication of these attacks continues to rise, with recent works demonstrating that backdoors can be embedded via invisible perturbations or handcrafted weight modifications that bypass traditional detection methods based on trigger inversion or weight analysis [25; 26]. In decentralized or split architectures, where trust is distributed, the potential for such active manipulation requires robust verification mechanisms that go beyond the confidentiality provided by standard DP.

A critical evolution in threat modeling involves the complex interplay between privacy mechanisms and robustness against these active attacks. While Differential Privacy is often deployed to mitigate gradient leakage from honest-but-curious observers, it introduces nuanced trade-offs regarding poisoning attacks. Some research suggests that the noise injection and gradient clipping inherent in DP-SGD can inadvertently facilitate stealthy model poisoning by masking malicious updates, creating a dual-edge scenario where privacy defenses may reduce robustness against integrity attacks. Conversely, other studies argue that gradient shaping, a core component of DP mechanisms, can enhance robustness against indiscriminate poisoning by bounding gradient magnitudes and altering orientations, though it may remain ineffective against strong, targeted attacks [27]. This tension highlights the necessity of viewing privacy and security not as isolated goals but as interconnected properties that must be optimized jointly, a challenge that becomes even more pronounced when theoretical privacy bounds are applied to real-world systems.

The capability of active adversaries also extends to adaptive attacks designed specifically to circumvent proposed defenses, revealing gaps between theoretical assumptions and practical resilience. For instance, while dropout was initially proposed as a lightweight defense against gradient leakage, adaptive attackers have developed “Dropout Inversion Attacks” that jointly optimize for client data and dropout masks, effectively neutralizing the protection offered by stochastic regularization [28]. Similarly, attackers have devised methods to bypass robustness-aware perturbation defenses in Natural Language Processing models by controlling the robustness gap between poisoned and

clean samples through adversarial training steps [29]. The emergence of large language models (LLMs) has further expanded the attack surface, with new poisoning strategies tailored to exploit the instruction tuning process, allowing adversaries to compromise model outputs with minimal data contamination [30]. These adaptive strategies demonstrate that static defenses are insufficient, necessitating dynamic threat models that account for an adversary’s ability to learn and evolve.

Finally, the threat model must account for the potential of Sybil attacks and collusion, where multiple malicious entities coordinate to overwhelm aggregation rules or reconstruct data through ensemble methods, a risk particularly relevant in peer-to-peer and decentralized topologies. Ensemble inversion techniques demonstrate that targeting multiple models jointly can significantly improve the quality of reconstructed training data compared to single-model attacks, posing a severe risk in distributed settings where model diversity might be assumed to offer protection [31]. Additionally, the concept of “client-side gradient inversion” challenges the assumption that server control is necessary for data reconstruction, showing that malicious clients can amplify the loss of specific classes to dominate aggregated updates and recover private inputs from other participants [32]. These diverse and evolving threat vectors necessitate a comprehensive defense-in-depth strategy that addresses both the confidentiality of data against honest-but-curious observers and the integrity of the learning process against active, adaptive adversaries. However, as the following section will explore, bridging the gap between these rigorous threat models and the actual privacy guarantees provided by theoretical frameworks remains a significant challenge, often leading to a disconnect between worst-case bounds and empirical reality.

1.4 Limitations of Theoretical Guarantees and the Gap to Empirical Reality

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While Section 1.3 established a comprehensive threat model ranging from honest-but-curious observers to adaptive, malicious actors, translating the rigorous mathematical guarantees of Differential Privacy (DP) into robust defenses against these real-world adversaries remains a formidable challenge. The theoretical framework of DP provides worst-case upper bounds on privacy leakage, ensuring that the presence or absence of any single individual in a dataset has a negligible impact on the output distribution. However, a significant disconnect often exists between these conservative theoretical guarantees and the actual privacy leakage observed in practical machine learning deployments. While DP offers a mathematically sound shield against adversaries with arbitrary auxiliary knowledge, the resulting privacy parameters, particularly the privacy budget ϵ , can be overly pessimistic or even vacuous in real-world scenarios. This section critically analyzes this divergence, examining how realistic threat models and empirical auditing techniques reveal that standard guarantees may not accurately reflect the true resilience of a system against the practical adversaries described previously, setting the stage for the trust assumptions discussed in Section 1.5.

A primary source of this gap lies in the worst-case nature of standard DP definitions, which assume an adversary possessing near-complete knowledge of the dataset except for the target record. In practice, such omnipotent adversaries are rare; as noted in Section 1.3, real-world attackers often operate under informational constraints. Research has shown that practical attackers frequently lack exact knowledge of the private data, rendering the worst-case guarantees of large ϵ values seemingly vacuous from a theoretical standpoint, yet empirically effective against state-of-the-art membership inference attacks (MIAs). To bridge this understanding, recent work introduces con-

cepts like Practical Membership Privacy (PMP), which models the attacker’s actual uncertainty about the data contents. This approach demonstrates that a large DP parameter can translate into a much smaller PMP parameter, effectively guaranteeing strong privacy against practical MIAs even when theoretical bounds suggest otherwise [33]. This highlights that the theoretical upper bound is often a loose estimate of the actual risk faced in typical deployment contexts, suggesting that systems deemed insecure by strict ϵ bounds may still offer substantial protection against the passive inference attacks detailed earlier.

Furthermore, the interpretation of privacy loss is complicated by the distinction between the privacy of the training mechanism and the privacy of a specific trained model instance. Traditional DP analysis focuses on bounding the leakage of the algorithm itself over all possible random seeds and datasets, whereas empirical auditing often seeks to measure the leakage of a concrete model instance deployed in the wild. New metrics, such as Epsilon*, have been proposed to quantify the privacy risk of a single model instance using black-box access, providing an empirical lower bound that contrasts with the theoretical upper bounds of the training mechanism. Studies utilizing such metrics reveal that models trained with DP can exhibit significantly reduced empirical privacy risks compared to non-private baselines, suggesting that the theoretical conservatism of DP mechanisms may obscure their practical efficacy [34]. Similarly, the instantiation of hypothetical adversaries to establish lower bounds on distinguishing games has shown that while theoretical bounds for algorithms like DP-SGD are tight in the worst case, practical restrictions on adversary capabilities create a substantial gap where differential privacy proves to be highly conservative [35].

The challenge of validating these guarantees is further exacerbated by flaws in empirical evaluation methodologies, which often fail to account for the sophistication of the adaptive attackers described in Section 1.3. Many existing studies rely on weak attack vectors or fail to characterize the privacy leakage of the most vulnerable samples, leading to misleading conclusions about the effectiveness of empirical defenses compared to formal DP baselines. Rigorous evaluations have demonstrated that when stronger attacks and proper baselines are employed, many ad-hoc empirical defenses fail to compete with properly tuned DP-SGD, even when the latter operates with vacuous provable guarantees [36]. This underscores the necessity of robust auditing frameworks that can accurately measure privacy leakage without relying on unrealistic assumptions about adversary limitations. Techniques such as one-shot empirical privacy estimation allow for the systematic auditing of privacy loss during a single training run, addressing the scalability issues of previous methods that required thousands of retraining steps [37], thereby providing a more reliable metric for assessing resilience against the evolving threat landscape.

Moreover, the assumption of data independence in standard DP mechanisms often fails to account for temporal correlations and dependencies inherent in real-world data streams, a factor that adaptive adversaries could potentially exploit. When data points are correlated, the privacy leakage can accumulate over time, a phenomenon known as temporal privacy leakage, which traditional DP bounds do not adequately capture. Analyses using Markov models have revealed that privacy loss may increase continuously in continuous data release scenarios, necessitating specialized mechanisms to bound leakage under temporal correlations [38]. Additionally, the impact of prior knowledge and data correlation on privacy leakage requires a unified analysis, leading to proposals like Prior Differential Privacy (PDP) to evaluate leakage considering the exact prior knowledge possessed by an adversary [39]. These nuances are critical when considering the active, adaptive strategies discussed in the previous section, as they reveal gaps where static theoretical bounds may underestimate the risk posed by coordinated or iterative attacks.

Finally, the abstract nature of the privacy budget ϵ makes it difficult for practitioners to contextu-

alize risk and make informed decisions about trust deployment. While theoretical bounds provide a number, they often fail to communicate the probabilistic nature of the risk to stakeholders, complicating the selection of appropriate trust architectures as will be discussed in Section 1.5. Efforts to visualize privacy-utility trade-offs and translate ϵ into bounds on Bayesian posterior beliefs or identifiability scores aim to make these guarantees more interpretable and actionable [40][41]. Ultimately, closing the gap between theory and practice requires a holistic approach that combines tighter theoretical bounds tailored to realistic threat models, such as those based on hypothesis testing interpretations of reconstruction attacks [42], with rigorous empirical auditing. Only by aligning theoretical guarantees with the empirical reality of adversarial capabilities can we establish the foundational trust necessary for the intermediaries and architectural choices explored in the following section.

1.5 Trust Assumptions and the Role of Intermediaries

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While the previous section highlighted the gap between theoretical privacy guarantees and empirical reality, the practical realization of these guarantees in Federated Learning (FL) is fundamentally contingent upon the underlying trust assumptions regarding system intermediaries. The deployment of FL architectures spans a spectrum of trust models that dictate both the security posture and the feasibility of privacy mechanisms. At one extreme lies the assumption of a trusted central aggregator, a paradigm where the server is presumed to faithfully execute aggregation protocols without attempting to infer private information from received model updates. However, this assumption is increasingly viewed as fragile in real-world scenarios involving heterogeneous stakeholders, such as competing healthcare institutions or financial entities, where the central coordinator may be semi-honest or even malicious. In such untrusted environments, reliance on a single point of trust creates significant power imbalances, necessitating a shift toward robust defensive mechanisms that do not depend on the benevolence of intermediaries. Literature indicates that basic FL systems often lack sufficient anonymization, leaving them vulnerable to a potentially malicious service provider capable of accessing individual user updates [43]. Consequently, the role of intermediaries has evolved from passive aggregators to either active guardians or primary adversaries, requiring a critical re-evaluation of how trust is established and maintained.

To address the vulnerabilities inherent in trusting a central server, researchers have developed architectures that minimize or eliminate trust requirements through cryptographic primitives and decentralized topologies. For instance, the integration of Trusted Execution Environments (TEEs) offers a hardware-based approach to confidentiality, allowing server-side computations to occur within isolated enclaves verifiable by external parties. This approach mitigates the risk of an honest-but-curious server by ensuring that aggregation logic and intermediate states remain shielded from the operating system and the service provider itself [43]. Furthermore, advancements in TEE design, such as microarchitectural isolation and secure shared memory, aim to protect against sophisticated side-channel attacks that could otherwise compromise enclave integrity [44]. By leveraging TEEs, the trust boundary shifts from the software and organizational level to the hardware level, though this introduces new challenges regarding the trusted computing base (TCB) and the necessity for remote attestation protocols to verify the execution environment’s integrity [45].

In contrast to hardware-based trust, cryptographic solutions like Secure Multi-Party Computation (SMPC) and Local Differential Privacy (LDP) operate under the assumption that no intermediary can be trusted. LDP, in particular, enforces privacy at the client level before any data leaves

the device, rendering the central aggregator oblivious to individual contributions. This is crucial in industrial settings, such as cross-silo healthcare applications, where distributed entities may be untrusted or subject to regulatory constraints prohibiting the sharing of raw gradients [46]. However, adopting LDP often incurs a cost to model utility due to the high noise levels required to satisfy strict privacy bounds. To bridge this gap, hybrid approaches have been proposed that combine the strengths of different trust models. For example, hierarchical FL introduces intermediary levels where calibrated noise can be added, offering a balance between the strong privacy of LDP and the higher utility of Central DP, effectively creating a tiered trust model that mitigates disparate utility-privacy trade-offs [47].

The power dynamics between clients and intermediaries are further complicated by the potential for active attacks, including model poisoning and backdoor insertion. In scenarios where the central aggregator is untrusted or where malicious clients aim to disrupt the learning process, the system must employ robust aggregation rules and verification mechanisms. Blockchain technology has emerged as a promising tool to decentralize trust and provide immutable audit trails, ensuring that the aggregation process is transparent and resistant to tampering. Frameworks utilizing blockchain can prevent malicious participants from influencing the global model by enforcing smart contracts that validate updates before aggregation, thereby reducing reliance on a single authoritative entity [48]. Moreover, decentralized FL architectures remove the central server entirely, distributing the aggregation task across the network. While this eliminates the single point of failure, it introduces new trust challenges related to network topology and the potential for adversarial node placement, where attackers strategically position themselves to maximize influence or privacy leakage [49].

Furthermore, the concept of “zero trust” has gained traction, advocating for the continuous verification of all participants regardless of their presumed status. This approach is particularly relevant in dynamic environments like 5G networks and IoT ecosystems, where devices frequently join and leave the network. Trust-as-a-Service frameworks propose reputation-based mechanisms to dynamically evaluate the reliability of network providers and clients, adapting to changing threat landscapes without relying on static trust assumptions [50]. Such systems often incorporate moving target defense strategies, where network topologies or communication paths are dynamically altered to confuse adversaries and limit their ability to exploit trusted relationships [51].

Ultimately, the choice of trust assumption dictates the feasible defense mechanisms and the resulting system performance. Whether relying on hardware enclaves, cryptographic protocols, or decentralized consensus, the goal is to create a resilient FL ecosystem capable of withstanding both passive inference attacks and active adversarial manipulation. The interplay between these trust models and the role of intermediaries remains a critical area of research as the industry moves toward more scalable and secure deployments in sensitive domains. As demonstrated by recent studies, no single solution offers a panacea; instead, a defense-in-depth strategy that layers multiple trust mechanisms—such as combining TEEs with differential privacy or blockchain with robust aggregation—is often necessary to address the complex threat models of modern distributed learning [52]. This multifaceted approach ensures that even if one layer of trust is compromised, the overall integrity and privacy of the federated learning process remain intact.

2 Advanced Differential Privacy Mechanisms and Utility Optimization

2.1 Advanced Noise Injection Strategies and Sensitivity Bounding

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While the theoretical frameworks for measuring privacy loss are critical, the practical realization of Differential Privacy (DP) in machine learning hinges on the mechanisms used to inject noise and bound sensitivity. The foundational approach has traditionally relied on adding independent and identically distributed (i.i.d.) Gaussian noise to gradients, calibrated to the global sensitivity of the query. However, this one-size-fits-all strategy often fails to account for the intricate geometric structures of high-dimensional data, the discrete nature of certain model updates, and the stringent requirements of real-world deployment scenarios where hard bounds on error are mandatory. Consequently, recent research has pivoted toward advanced noise injection strategies and sophisticated sensitivity bounding techniques that exploit coordinate-wise disparities, structural correlations, and distribution-specific properties to achieve a superior privacy-utility trade-off. These mechanism-level optimizations provide the essential input parameters—such as noise scale and sensitivity bounds—that the tighter accounting methods discussed in the subsequent section rely upon to minimize cumulative privacy loss.

A primary limitation of standard mechanisms is the assumption of isotropic sensitivity, which ignores the fact that different coordinates of a gradient vector may exhibit vastly different magnitudes and sensitivities. To address this, researchers have explored the addition of independent but not identically distributed (i.n.i.d.) noise. By exploiting coordinate-wise disparity, it is possible to tailor the noise scale to the specific sensitivity of each dimension. Theoretical analyses and numerical simulations have demonstrated that i.n.i.d. Gaussian and Laplace mechanisms can achieve tighter privacy-accuracy trade-offs compared to their i.i.d. counterparts [53]. Notably, this approach challenges the conventional wisdom that Gaussian noise is superior in high dimensions; under conditions of irregular coordinate-wise sensitivities, the Laplace mechanism can outperform Gaussian mechanisms, offering a robust alternative for high-dimensional empirical risk minimization [53].

Beyond scalar adjustments, the geometry of the sensitivity set itself offers opportunities for optimization. Traditional l_2 or l_1 norm clipping often suffers from the curse of dimensionality, necessitating excessive noise. Advanced approaches revisit the geometry of high-dimensional sensitivity sets to characterize non-asymptotically optimal Gaussian noise for Rényi Differential Privacy (RDP). When sensitivity sets exhibit asymmetry across orthogonal bases, optimal noise bounds can be derived that are independent of dimension or rank, significantly mitigating the utility loss associated with high-dimensional spaces [54]. Furthermore, novel sampling strategies such as “twice sampling,” which implements both sample-wise and coordinate-wise sampling, enable Gaussian noise to align more closely with the sensitivity geometry, yielding asymptotic improvements in privacy amplification, particularly for small sampling rates [54].

In scenarios involving matrix-valued queries, such as those found in collaborative filtering or specific neural network layers, adding i.i.d. noise to each matrix element is suboptimal as it forfeits the exploitation of structural characteristics. The Matrix-Variate Gaussian (MVG) mechanism addresses this by adding noise drawn from a matrix-variate Gaussian distribution. This approach not only preserves (ϵ, δ) -differential privacy but also introduces the concept of directional noise, allowing the impact of perturbation on utility to be moderated based on the matrix’s structural properties [55]. Similarly, for problems involving sum, count, and vote, optimal pure differentially private K -norm mechanisms and concentrated differentially private elliptic Gaussian noise have been developed.

These problem-specific sensitivity space analyses provide closed-form expressions and faster sampling algorithms, yielding meaningful accuracy improvements over spherical Gaussian noise [56].

The requirement for hard bounds on deviations in real-time systems has spurred the development of truncated additive mechanisms. Unlike infinitely wide distributions, truncated noise ensures that the deviation from the true query output never exceeds a specified limit. Recent advances introduce gradient-descent-based tools to learn truncated noise patterns that optimize for differential privacy under sequential composition. These learned discrete noise patterns can replicate the shape of truncated Gaussians while offering strong utility bounds, and the methodology has been extended to incorporate mechanism output events with varying input-dependent zero occurrence probability [57]. Theoretical work has further established the optimality of the truncated Laplacian mechanism, showing that it closes the multiplicative gap between lower and upper bounds in various high privacy regimes, outperforming the optimal Gaussian mechanism across all privacy settings [58].

For federated learning environments involving integer updates or secure aggregation, the Skellam mechanism presents a compelling discrete alternative. Based on the difference of two independent Poisson random variables, the multi-dimensional Skellam mechanism provides privacy guarantees comparable to the continuous Gaussian mechanism while being naturally suited for low-precision communication constraints. Its closure under summation and the efficiency of Poisson sampling make it an attractive practical choice for distributed settings [59]. Additionally, hybrid distributions that combine the sharp center of the Laplace distribution with the light, sub-Gaussian tails of the Gaussian distribution have been proposed. This “grafted” mechanism leverages the best characteristics of both distributions to achieve a better trade-off between privacy and accuracy, with rigorous conditions derived for both one-dimensional and high-dimensional settings [60].

Finally, sensitivity bounding has evolved from static global clipping to adaptive and batch-wise strategies, directly influencing the magnitude of noise required. Standard individual clipping in DP-SGD can introduce significant bias and prevent the use of crucial components like Batch Normalization. Batch clipping, which averages and clips batches of gradients rather than individual gradients, alongside Adaptive Layerwise Clipping (ALC), allows for rigorous DP proofs while enabling the convergence of deep neural networks that fail under traditional clipping schemes [61]. Adaptive methods such as AdaCliP utilize coordinate-wise adaptive clipping to provably add less noise than previous methods, empirically demonstrating improved model accuracy [62]. Moreover, automated frameworks like Auto DP-SGD estimate clipping thresholds and noise multipliers dynamically based on gradient norms, further refining the balance between privacy preservation and model utility [63]. These advancements collectively signify a shift from generic noise addition to context-aware, geometrically optimized, and adaptively bounded privacy mechanisms, laying the groundwork for the precise privacy accounting techniques detailed in the following section.

2.2 Tight Privacy Accounting and Adaptive Composition Theorems

2.2 Tight Privacy Accounting and Adaptive Composition Theorems

Building upon the context-aware noise injection and adaptive sensitivity bounding mechanisms detailed in Section 2.1, the efficacy of Privacy-Preserving Machine Learning (PPML) ultimately depends on the precision with which cumulative privacy loss is quantified. In Federated Learning (FL) environments, where models undergo thousands of communication rounds, traditional composition theorems for (ϵ, δ) -differential privacy (DP) often yield overly conservative bounds that accumulate linearly or with a square-root factor. This conservatism forces practitioners to inject excessive noise—counteracting the optimizations achieved in the previous section—to satisfy a fixed

privacy budget, thereby severely degrading model utility. To bridge this gap, recent research has pivoted toward tighter accounting frameworks and adaptive composition strategies that accurately reflect the actual privacy loss incurred during iterative learning, serving as the critical analytical backbone for the optimization-centric techniques discussed in Section 2.3.

A cornerstone of this advancement is the adoption of Rényi Differential Privacy (RDP), which provides a more granular measure of privacy loss via Rényi divergence. RDP allows for significantly tighter composition bounds compared to standard (ϵ, δ) -DP, particularly when applied to subsampled mechanisms like DP-Stochastic Gradient Descent (DP-SGD). The theoretical underpinnings of RDP have been strengthened by unified frameworks that jointly consider group privacy and amplification by subsampling. For instance, recent work has developed a unified framework for deriving amplification by subsampling guarantees specifically for RDP, demonstrating that joint consideration of these properties yields provably tight group privacy amplification guarantees that surpass existing principles [64]. Furthermore, the conversion between RDP and other privacy notions has been optimized; researchers have developed machinery to optimally relate approximate DP to RDP based on the joint range of f -divergences, enabling the derivation of optimal approximate DP parameters from a given RDP level. This optimization can effectively allow for approximately 100 more stochastic gradient descent iterations for the same privacy budget compared to state-of-the-art methods at the time [65; 66].

Parallel to RDP, the framework of f -Differential Privacy (f -DP) has emerged as a powerful tool for handling composition algebraically. Inspired by hypothesis testing, f -DP avoids the information loss inherent in divergence-based relaxations when converting back to (ϵ, δ) -guarantees. A central result in this area is the Central Limit Theorem for DP, which posits that the privacy guarantees of any hypothesis-testing-based definition converge to Gaussian Differential Privacy (GDP) under composition. This convergence provides a computationally inexpensive tool for analyzing exact composition, offering a coherent and analytically tractable alternative to traditional accountants [67]. Building on this, privacy profiles—which describe δ as a function of ϵ —have been utilized to bound the privacy of private selection mechanisms like Report Noisy Max. Numerical approaches leveraging these profiles have shown substantial improvements over RDP-based accounting in end-to-end private learning experiments, particularly in regimes where RDP conversions are known to be lossy [68].

Beyond analytical frameworks, numerical accountants leveraging the Fast Fourier Transform (FFT) have revolutionized the precision of privacy loss estimation. These methods operate on the privacy loss distribution (PLD) formalism to compute tight (ϵ, δ) bounds. Initially designed for homogeneous compositions, FFT-based accountants have been extended to handle heterogeneous compositions, where different mechanisms or privacy parameters are used across iterations—a scenario increasingly common given the adaptive noise strategies from Section 2.1. Rigorous error analysis in these extensions allows practitioners to choose algorithm parameters that guarantee desired accuracy, often providing tighter bounds than moment accountants while managing computational complexity through techniques like the Plancherel theorem [69]. Similarly, numerical accountants for discrete-valued mechanisms and subsampled Gaussian mechanisms have been proposed, utilizing FFT to provide strict lower and upper bounds for privacy parameters, thereby enabling significant reductions in noise variance—up to 75% in some cases—without compromising privacy guarantees [70; 71]. Another analytical approach, the Edgeworth Accountant, employs Edgeworth expansions to approximate the distribution of privacy-loss log-likelihood ratios, offering non-asymptotic bounds with essentially constant computational cost regardless of the number of compositions [72].

The frontier of privacy accounting also addresses the critical need for adaptivity, aligning closely

with the dynamic parameter adjustment trends seen in modern optimization. In realistic FL scenarios, privacy parameters (such as noise scale or batch size) may need to be tuned dynamically based on intermediate model performance. Standard composition theorems often require these parameters to be fixed a priori, limiting flexibility. To overcome this, the concepts of privacy filters and privacy odometers have been introduced. Privacy filters act as stopping rules to ensure the total privacy budget is not exceeded, while odometers provide time-uniform bounds on realized privacy loss. Recent advancements have constructed filters that match the rates of advanced composition even with adaptively chosen parameters, closing the gap between theoretical flexibility and practical efficiency [73; 74]. These concepts have been extended to concurrent compositions of interactive mechanisms, proving that full adaptivity can be maintained without compromising privacy guarantees [75].

Furthermore, the scope of accounting has expanded from global dataset guarantees to individual-level precision. Traditional DP accounts for the worst-case privacy loss over all individuals, which can be overly pessimistic for “typical” data points. Individual privacy accounting methodologies, such as those utilizing a Rényi filter, enable the tracking of personalized privacy loss estimates for each participant. This approach allows for tighter accounting where individuals with lower actual privacy loss can participate in more rounds or tolerate less noise, effectively improving the overall privacy-utility trade-off [76]. Complementing this, data-dependent frameworks like per-instance DP and Fisher Information Loss have been explored to provide fine-grained guarantees. Innovations such as bounded support Gaussian mechanisms have been shown to amplify privacy guarantees under these data-dependent accounting frameworks, reducing per-instance ϵ bounds by up to 30% without negative effects on utility [77]. Additionally, the concept of partial sensitivity has been introduced to determine the influence of specific input features on an individual’s privacy loss, adding another layer of granularity to the accounting process [78]. Finally, randomized verification paradigms like Estimate-Verify-Release (EVR) and saddle-point accountants offer new ways to convert privacy estimates into strict formal guarantees with high efficiency, further refining the landscape of tight privacy accounting [79; 80].

Collectively, these advancements in tight accounting and adaptive composition provide the rigorous theoretical infrastructure necessary to validate the complex, multi-stage privacy operations employed in modern PPML. By minimizing the overestimation of privacy loss, these frameworks enable the aggressive utility-enhancing strategies discussed in the subsequent section, ensuring that the trade-off between privacy preservation and model performance is managed with maximal precision.

2.3 Optimization-Centric Utility Enhancement and Convergence Stabilization

2.3 Optimization-Centric Utility Enhancement and Convergence Stabilization

While Section 2.2 established rigorous frameworks for quantifying and tightening privacy budgets, the practical deployment of these guarantees often incurs a severe penalty on model utility. The mechanical application of Differential Privacy (DP)—typically involving per-sample gradient clipping and Gaussian noise injection—frequently disrupts optimization trajectories, leading to sluggish convergence and degraded generalization. To bridge this gap, recent research has pivoted from purely accounting-centric solutions to optimization-centric innovations. These approaches embed privacy mechanisms directly into the training dynamics, aiming to stabilize convergence, mitigate the bias introduced by privacy operations, and leverage the geometry of the loss landscape. By doing so, they narrow the privacy-utility gap without compromising the tight theoretical guarantees discussed

previously.

A primary strategy involves exploiting the geometry of the loss landscape to enhance robustness against DP-induced perturbations. Sharpness-Aware Minimization (SAM) has emerged as a pivotal technique, operating on the premise that flat minima offer superior generalization and stability. In DP contexts, flat minima are particularly advantageous as they exhibit lower sensitivity to parameter shifts, rendering models more resilient to the noise required for privacy. Standard DP training often converges to sharp landscapes, exacerbating performance loss; algorithms like DP-FedSAM address this by integrating SAM into Federated Learning to foster local models with improved weight perturbation robustness [81]. By targeting flatter regions, these methods effectively reduce the norm of local updates, permitting either less aggressive noise injection or greater tolerance to existing noise levels. Theoretical analyses support this, suggesting SAM implicitly regularizes the loss function in alignment with private learning requirements, though practical implementations with constant perturbation sizes may converge only to neighborhoods of optima rather than exact stationary points [82].

Complementing geometric approaches, correcting the statistical bias inherent in gradient clipping and noise addition is critical for stabilizing convergence. Standard DP-SGD yields a biased gradient estimate that can stall optimization progress. Bias-Aware Minimisation (BAM) tackles this by explicitly modeling the relationship between per-sample gradient norms and estimation bias, offering a provable reduction in bias to enhance utility [83]. Furthermore, the intersection of DP and adaptive optimizers like Adam introduces unique challenges; independent noise injection in DP-Adam can skew the second-moment estimator, causing divergence from non-private behavior. The DP-AdamBC algorithm was specifically developed to correct this bias, restoring the expected dynamics of Adam and significantly boosting accuracy across diverse tasks [84].

To further refine optimization under privacy constraints, adaptive strategies have been re-engineered to function effectively amidst high noise levels. Since standard adaptive preconditioners often degrade when overwhelmed by DP noise, frameworks like AdaDPS utilize non-sensitive side information to precondition gradients, enabling effective adaptation while reducing the necessary noise magnitude [85]. Alternatively, methods such as DP² (Differentially Private adaptive training with Delayed Preconditioners) leverage the observation that adaptive methods can tolerate stale preconditioners. By constructing delayed but less noisy preconditioners, DP² achieves convergence speeds up to four times faster than non-adaptive baselines without relying on auxiliary public data [86]. These advances are supported by rigorous theoretical bounds for algorithms like DP-RMSProp and DP-Adam, which demonstrate sharper population gradient bounds compared to prior art [87].

The sophistication of these optimization techniques extends to the dynamic adjustment of privacy parameters during training. Static clipping thresholds and noise multipliers often fail to capture the evolving distribution of gradients throughout the learning process. Dynamic DP-SGD approaches propose adjusting these parameters on the fly while adhering to a cumulative privacy budget, thereby stabilizing updates and improving accuracy, particularly in stringent privacy regimes [88]. Automating this further, Auto DP-SGD introduces mechanisms for automatic clipping threshold estimation based on gradient norms alongside an automatic noise multiplier decay strategy. This dual improvement has demonstrated superior performance on complex datasets like CIFAR-100 while substantially reducing the required privacy budget [63]. Additionally, online sensitivity optimization treats the clipping threshold as a learnable parameter, optimizing it via gradient descent to dynamically balance the trade-off between clipping bias and sanitization noise [89].

Finally, the theoretical foundations of these optimization methods are being fortified through ad-

vanced accounting and convergence analyses that build upon the frameworks in Section 2.2. The integration of Rényi Differential Privacy (RDP) and f -DP allows for tighter accounting, which is essential when employing the adaptive composition strategies where privacy parameters shift per iteration [90]. Moreover, viewing private learning algorithms through the lens of noisy fixed-point iterations provides a unified framework for deriving both privacy and utility guarantees. This perspective recovers popular methods like DP-SGD while enabling the design of novel private optimizers such as private ADMM [91]. Collectively, these optimization-centric innovations demonstrate that by carefully tailoring the optimization process to the specific constraints of differential privacy, it is possible to achieve stable convergence and high utility. This sets the stage for the next frontier: leveraging communication-efficient perturbations, such as sparsification and quantization, to simultaneously address bandwidth constraints and privacy vulnerabilities, as explored in the following section.

2.4 Communication-Efficient Perturbation via Sparsification and Quantization

2.4 Communication-Efficient Perturbation via Sparsification and Quantization

Building upon the optimization-centric strategies detailed in Section 2.3, which stabilize convergence and mitigate bias through geometric awareness and adaptive preconditioning, this section addresses the concurrent challenges of communication overhead and privacy vulnerability in distributed settings. While Section 2.3 focused on refining the internal dynamics of the training process, practical Federated Learning (FL) deployments face external constraints: the transmission of high-dimensional gradient vectors incurs significant latency and bandwidth costs, particularly in edge computing environments, while simultaneously exposing clients to sophisticated reconstruction attacks where adversaries invert shared updates to recover sensitive data. Recent advancements demonstrate that integrating Differential Privacy (DP) with communication compression strategies—specifically random sparsification and quantization—offers a synergistic solution. These techniques are no longer viewed merely as efficiency tools but as integral components of privacy-preserving mechanisms that simultaneously mitigate bandwidth constraints and fortify defenses against inference threats.

Random sparsification, which involves transmitting only a subset of gradient coordinates, has emerged as a potent method for amplifying privacy guarantees while reducing data volume. Theoretical and empirical analyses reveal that sparsification interacts non-trivially with the noise injection required for DP. Contrary to the intuition that compression invariably degrades model utility, studies indicate that aggressive sparsification can actually improve test accuracy in differentially private training settings. This counter-intuitive phenomenon occurs because sparsification reduces the variance in estimating the average gradient, thereby making the model less susceptible to the detrimental effects of Gaussian noise added for privacy [92]. By randomly selecting gradient components before clipping and noisification, systems can adjust the trade-off within the convergence bound, leading to a smaller upper bound when noise is the dominant factor [93]. Furthermore, this approach inherently strengthens privacy against reconstruction attacks; the sparse nature of the transmitted updates obscures the complete gradient landscape, making it significantly harder for adversaries to perform deep gradient leakage attacks compared to dense gradient sharing [93].

Quantization serves as another critical pillar in this domain, transforming continuous gradient values into discrete representations to minimize bit-width requirements. Advanced quantization schemes have been designed to shape quantization errors to mimic specific probability distributions, such as the Gaussian distribution, thereby satisfying Client-Level Differential Privacy (CLDP)

without the need for explicit, separate noise addition layers. For instance, Gaussian Layered Randomized Quantization (Gau-LRQ) utilizes a layer multishift coupler to compress raw gradients while ensuring the resulting error distribution adheres to privacy constraints [94]. This approach allows for dynamic allocation of privacy budgets and quantization bits, optimizing the trade-off between communication cost, privacy levels, and convergence performance. Theoretical frameworks utilizing f -differential privacy have further demonstrated that discrete-valued mechanisms, including binomial noise and sign-based compression methods, can provide tight privacy guarantees that break the traditional three-way trade-off between privacy, communication, and accuracy [95].

However, the integration of compression and privacy is not without risks, and naive implementations can lead to a false sense of security. Empirical studies warn that standard gradient obfuscation techniques, such as simple quantization or sparsification, may not suffice against advanced semantic reconstruction attacks if not rigorously combined with formal DP mechanisms [96]. To address this, unified frameworks like SoteriaFL have been proposed, which accommodate general compression operators alongside local differential privacy to achieve superior communication complexity without sacrificing utility or privacy guarantees [97]. Additionally, research into adaptive differential quantization has shown that allowing nodes to optimize their quantization parameters dynamically can achieve optimum solutions with low communication costs while keeping private data unrevealed under both passive and eavesdropping adversary models [98].

The interplay between compression and privacy extends to specialized architectures and optimization strategies, bridging the gap between the adaptive methods of Section 2.3 and the structural innovations to follow. For large embedding models, naive application of DP-SGD can destroy gradient sparsity, leading to inefficiencies; novel algorithms like DP-FEST and DP-AdaFEST have been developed to preserve this sparsity during private training, achieving massive reductions in gradient size while maintaining accuracy [99]. Similarly, in decentralized nonconvex optimization, the timing of gradient clipping relative to communication compression plays a pivotal role in stabilizing training and enabling local differential privacy analysis [100]. Furthermore, the combination of variational modeling with partial gradient perturbation has been shown to prevent deep gradient leakage more effectively than compression alone, suggesting that hybrid approaches leveraging both structural stochasticity and sparse updates offer the most robust defense [101].

Ultimately, the convergence of sparsification, quantization, and differential privacy represents a frontier where efficiency and security are mutually reinforcing rather than competing objectives. By leveraging the inherent noise-like properties of compression errors and strategically aligning them with formal privacy accounting, modern FL systems can achieve high-fidelity learning outcomes with minimal communication overhead and robust protection against data reconstruction. These communication-efficient perturbations set the stage for the next evolution in the field: moving beyond gradient-based backpropagation entirely. As explored in the following section, alternative perturbation paradigms and forward-pass mechanisms aim to overcome the fundamental memory bottlenecks of traditional DP-SGD, offering scalable pathways for Large Language Models that complement the bandwidth savings achieved here.

2.5 Alternative Perturbation Paradigms and Forward-Pass Mechanisms

2.5 Alternative Perturbation Paradigms and Forward-Pass Mechanisms

Building upon the communication-efficient compression strategies discussed in Section 2.4, which leverage sparsification and quantization to harmonize bandwidth constraints with privacy, this section addresses a fundamental computational bottleneck inherent in standard differentially private training: the reliance on backpropagation. While Differentially Private Stochastic Gradient Descent (DP-SGD) remains the de facto standard, its requirement to compute and store per-example gradients for clipping purposes introduces prohibitive memory overheads, particularly when scaling to Large Language Models (LLMs) or operating under strict resource constraints. To overcome these limitations, recent research has pivoted toward alternative perturbation paradigms that decouple privacy mechanisms from the backward pass. These approaches—encompassing output perturbation, objective perturbation, and novel forward-pass injection strategies—aim to circumvent the scalability barriers of methods while maintaining rigorous privacy guarantees. Notably, these paradigms set the stage for the architectural innovations and transfer learning techniques explored in Section 2.6, offering distinct pathways to mitigate the privacy-utility trade-off without solely relying on model structure or public data priors.

Output perturbation, a foundational technique historically constrained by loose sensitivity bounds and the non-convex nature of deep learning loss landscapes, has been revitalized through integration with convexified loss functions and refined sensitivity analyses. Frameworks that inject noise into randomly sampled neurons at the output layer, trained using convexified loss functions, have demonstrated superior privacy-utility trade-offs compared to standard DP-SGD implementations [102]. By establishing tighter upper bounds on the global sensitivity of model parameters, these methods control overall noise injection more effectively, achieving better accuracy against black-box membership inference attacks. Furthermore, the theoretical understanding of output perturbation has been expanded to encompass broader classes of convex, Lipschitz loss functions, providing the first known DP excess risk and runtime bounds for optimizing such general families [103]. This evolution allows output perturbation to be applied to objectives beyond average performance, including fairness and adversarial robustness, thereby broadening its utility in complex learning scenarios.

Parallel to refinements in output perturbation, objective perturbation has seen significant extension beyond its traditional reliance on convexity. Originally limited to convex and smooth objective functions, new algorithms now operate effectively over discrete and continuous domains without assuming convexity, requiring only boundedness and Lipschitz continuity of the loss function [104]. These oracle-efficient algorithms leverage random linear terms augmented to the optimization problem, enabling their application to non-convex tasks such as learning linear classifiers with 0/1 loss, where they can outperform standard convex-surrogate approaches. Additionally, the interplay between stability and output perturbation has been elucidated, showing that algorithms with uniform stability can achieve differential privacy with reduced noise scales, effectively generalizing privacy guarantees to implicitly stabilized empirical risk minimization procedures [105]. This connection highlights that enhancing algorithmic stability is a viable pathway to reducing the privacy cost inherent in output perturbation mechanisms, a principle that complements the stability-focused architectural designs discussed later.

Perhaps the most transformative shift in alternative paradigms is the emergence of forward-pass perturbation mechanisms, specifically designed to address the scalability challenges of Large Lan-

guage Models (LLMs). Traditional backpropagation-based DP methods struggle with the memory demands of large Transformers, prompting the development of techniques that inject noise directly during the forward propagation stage. The DP-Forward Propagation (DP-FP) paradigm, for example, employs representation clipping and noise addition in the forward pass, coupled with micro-batch subsampling to amplify privacy signals [106]. This approach eliminates the need for per-example gradient instantiation, resulting in memory costs comparable to non-private training while significantly outperforming state-of-the-art DP-SGD variants on tasks involving models like RoBERTa-large. Similarly, the DP-Forward framework perturbs embedding matrices directly in the forward pass of LMs, satisfying strict local DP requirements for both training and inference data [107]. By utilizing an analytic matrix Gaussian mechanism, DP-Forward not only achieves utility levels nearly matching non-private baselines but also provides robust defense against embedding inversion and sensitive attribute inference attacks, vulnerabilities that gradient-based methods often fail to mitigate.

The exploration of non-backpropagation methods extends further into zeroth-order optimization and node perturbation strategies, offering biologically plausible and computationally distinct alternatives. DPZero introduces a private zeroth-order algorithm that fine-tunes LLMs without backpropagation, leveraging the memory efficiency of forward-only passes to achieve nearly dimension-independent privacy rates [108]. This is particularly critical as model sizes continue to grow, rendering gradient storage increasingly impractical. Moreover, effective learning via node perturbation has been revisited, where noise is injected into network activations rather than gradients. By aligning node perturbation with directional derivatives and incorporating input decorrelation mechanisms, researchers have achieved convergence rates and test performances approaching those of backpropagation, offering a biologically plausible and computationally distinct alternative for training noisy or discontinuous systems [109]. These methods share conceptual ground with the parameter-efficient fine-tuning strategies discussed in the subsequent section, as both seek to minimize the footprint of private updates.

Finally, the use of synthetic loss surrogates and randomized smoothing represents another frontier in alternative perturbation, bridging the gap between optimization dynamics and model robustness. Techniques such as DPLIS construct smooth loss functions that favor noise-resilient models residing in flat regions of the loss landscape, thereby boosting utility and training stability under fixed privacy budgets [110]. Additionally, the integration of synthetic data generation within knowledge distillation frameworks allows for the transfer of teacher model knowledge to student models using differentially private synthetic text, effectively compressing autoregressive LLMs while preserving data privacy [111]. Collectively, these alternative paradigms illustrate a maturing field that is moving beyond the constraints of gradient clipping and backpropagation. By offering diverse, efficient, and robust mechanisms, they provide the essential algorithmic foundation upon which the data-centric and architecture-centric strategies of Section 2.6 are built, collectively driving the field toward practical privacy-preserving machine learning in the era of massive models.

2.6 Leveraging Public Data, Transfer Learning, and Model Architecture Design

2.6 Leveraging Public Data, Transfer Learning, and Model Architecture Design

While Section 2.5 detailed algorithmic innovations that circumvent the computational bottlenecks of backpropagation through forward-pass mechanisms and alternative perturbation paradigms, a

complementary strategy to bridge the privacy-utility gap lies in restructuring the learning context itself. Even with efficient optimization algorithms, standard Differentially Private Stochastic Gradient Descent (DP-SGD) often suffers from severe performance degradation in deep neural networks due to the compounding effects of gradient clipping and noise injection. To address this, recent research has pivoted toward three synergistic structural strategies: leveraging public datasets for robust pre-training priors, employing transfer learning for parameter-efficient fine-tuning, and designing specialized model architectures that are intrinsically robust to privacy noise. These approaches collectively aim to reduce the sensitivity of the learning process and minimize the amount of private information required to achieve state-of-the-art performance, setting the stage for the fairness-centric discussions in Section 2.7 by establishing high-utility baselines that are less susceptible to the disparate impacts of privacy mechanisms.

The utilization of public data has emerged as a critical paradigm for enhancing private learning, operating on the hypothesis that non-convex loss landscapes require a two-phase optimization process: first, selecting a suitable “basin” in the loss landscape, and second, solving the optimization within that basin. The former is significantly more challenging with limited, noisy private data, whereas public pre-training effectively solves this initialization problem [112]. Consequently, fine-tuning models pretrained on large-scale public datasets allows private learning algorithms to operate in a more favorable region of the parameter space, where the impact of added noise is minimized. This approach has yielded remarkable improvements, with studies demonstrating that carefully selecting which layers to fine-tune can establish new state-of-the-art trade-offs between privacy and accuracy [113]. For instance, accelerated fine-tuning recipes have achieved near-perfect accuracy on benchmark datasets like CIFAR-10 under strict privacy budgets by driving model parameters rapidly to stable regions [114]. However, the reliance on public pre-training is not without controversy or nuance. While it boosts utility, researchers caution against viewing models pretrained on web-scraped data as inherently “private,” noting that such practices could erode trust if the distinction between public pre-training and private fine-tuning is not clearly communicated [115]. Furthermore, the effectiveness of this paradigm depends heavily on the similarity between the public source domain and the sensitive target domain; significant distribution shifts can limit generalization capabilities. Despite these caveats, the integration of public priors remains a dominant strategy, with methods like DP-RandP exploring the generation of priors from random processes to further decouple performance from the availability of massive labeled public datasets [116].

Beyond data strategies, the design of model architecture plays a pivotal role in mitigating the utility loss associated with DP, echoing the sparsification themes introduced in Section 2.4 but applied structurally rather than communicatively. Traditional architectures optimized for non-private settings often possess excessive capacity and unbounded activation functions, which increase gradient sensitivity and necessitate higher noise levels. Research indicates that sparser models are fundamentally better suited for DP-SGD, as they require proportionally less perturbation to achieve the same privacy guarantee. The introduction of equivariant convolutional networks exemplifies this shift; by embedding symmetry constraints directly into the architecture, these models achieve higher accuracy with substantially fewer parameters and reduced computational overhead compared to standard dense networks [117]. Similarly, the choice of activation functions is critical. It has been observed that unbounded activations like ReLU can lead to unstable gradient norms, whereas bounded activation functions, such as tempered sigmoids, strictly limit the sensitivity of individual training points, consistently outperforming their unbounded counterparts in private settings [118]. Automated architecture search has also been employed to identify structures inherently friendly to differential privacy. Frameworks like DPNAS (Neural Architecture Search for Deep Learning with Differential Privacy) automatically design networks that balance expressivity with low sensi-

tivity, achieving superior privacy-utility trade-offs across various benchmarks [119]. Additionally, techniques such as pre-pruning and gradient-dropping have been proposed to reduce the effective parameter space during training, thereby lowering the cumulative privacy cost while maintaining classification accuracy [120].

Transfer learning techniques further refine these architectural advantages by enabling parameter-efficient fine-tuning, creating a natural bridge to the zeroth-order and forward-pass methods discussed in Section 2.5. Instead of updating all model weights, which accumulates significant privacy budget, modern approaches focus on updating only small subsets of parameters or adapters. This is particularly relevant in the context of Large Language Models (LLMs), where simpler and sparser algorithms for differentially private fine-tuning have demonstrated that larger models can better maintain their accuracy when privacy constraints are introduced, provided that the fine-tuning is parameter-efficient [121]. Innovations such as Zeroth-order optimization (DP-ZO) have pushed this frontier further by privatizing only scalar step sizes rather than full gradient vectors, enabling the private fine-tuning of massive models like OPT-66B with minimal performance degradation [122]. Moreover, frameworks like Offsite-Tuning allow for the adaptation of billion-parameter foundation models without the data owner ever accessing the full model weights, preserving both data and model privacy while achieving computational efficiency [123].

In summary, bridging the privacy-utility gap requires a holistic approach that moves beyond merely tuning hyperparameters of standard algorithms. By strategically leveraging public data to initialize models in optimal loss basins, designing sparse and equivariant architectures to minimize sensitivity, and applying advanced transfer learning techniques to restrict the scope of private updates, the field is making significant strides toward practical and high-performing privacy-preserving machine learning. These strategies collectively demonstrate that with the right structural and data foundations, the cost of privacy can be dramatically reduced without compromising the integrity of the learned models. However, while these architectural and data-centric optimizations improve overall utility, they do not inherently guarantee equitable outcomes across all demographic groups. As the field advances, it becomes imperative to examine whether these gains are distributed fairly, a critical inquiry that leads directly to the exploration of fairness-aware clipping and holistic evaluation frameworks in the following section.

2.7 Fairness-Aware Clipping and Holistic Utility-Privacy Evaluation Frameworks

2.7 Fairness-Aware Clipping and Holistic Utility-Privacy Evaluation Frameworks

While the architectural innovations and transfer learning strategies discussed in the previous section significantly mitigate the general utility loss associated with Differential Privacy (DP), they do not fully address a more insidious consequence: the disparate impact of privacy mechanisms on underrepresented subgroups. The integration of DP into machine learning pipelines has revealed that standard implementations, particularly Differentially Private Stochastic Gradient Descent (DP-SGD), often inadvertently exacerbate existing biases. Although DP provides rigorous mathematical guarantees against membership inference and data reconstruction, its core operations—gradient clipping and noise injection—disproportionately degrade model utility for minority groups compared to majority groups. Research demonstrates that in neural networks trained with DP-SGD, the accuracy drop is significantly more severe for underrepresented classes and complex subgroups, effectively widening the fairness gap present in non-private models [124]. This inequity arises because samples from minority groups frequently possess larger gradient norms or reside in more complex regions of the loss landscape, making them uniquely susceptible to the truncation

effects of uniform clipping bounds and the obscuring effects of additive noise.

To counteract this challenge, recent advancements have pivoted toward fairness-aware clipping strategies and adaptive noise schemes designed to decouple the privacy-utility trade-off across different demographic groups. A primary approach involves modifying the gradient clipping mechanism to account for group-specific characteristics rather than applying a monolithic global threshold. For instance, the DPSGD-F algorithm introduces a modified update rule that adaptively adjusts the clipping norm for different groups, thereby removing the potential disparate impact of differential privacy on protected groups [125]. By ensuring that clipping bias does not systematically disadvantage specific subpopulations, such mechanisms aim to equalize utility loss across groups without compromising the overall privacy budget. Furthermore, alternative architectures such as FairDP propose independently training models for distinct individual groups using group-specific clipping terms to assess and bound disparate impacts before progressively integrating knowledge to formulate a comprehensive model [126]. This stratified approach allows for more nuanced control over the privacy-utility-fairness trilemma, ensuring that the defensive capabilities of DP do not come at the expense of equitable performance.

Beyond clipping mechanics, the allocation of the privacy budget itself has emerged as a critical lever for improving fairness. Traditional DP mechanisms often expend the privacy budget uniformly across features or samples, which can undervalue the conditional probabilities critical for predictive quality in minority groups. Innovative ensemble methods have been developed to distribute the privacy budget both “wisely” to maximize predictive accuracy and “fairly” to bound potential disparities in accuracy across groups [127]. These methods leverage feature importance and per-group performance objectives to dynamically allocate noise, ensuring that obfuscation does not render the data useless for underrepresented populations. Additionally, the concept of stratification has been revisited as a simple yet powerful baseline; by computing independent private estimates for each group and recombining them, practitioners can often achieve high-accuracy population-level statistics while mitigating the disparate impact seen in naive global aggregation [128].

However, optimizing algorithms for fairness is insufficient without robust, holistic evaluation frameworks capable of assessing both utility and defense capabilities in a domain-agnostic manner. Historically, the field has lacked comprehensive benchmarks that simultaneously measure privacy leakage, model utility, and fairness metrics. To fill this gap, new evaluation frameworks like DPMLBench have been introduced to perform holistic measurements of improved DPML algorithms. These frameworks taxonomize improvements across the machine learning lifecycle and jointly evaluate algorithms on utility and defensive capabilities against attacks such as Membership Inference Attacks (MIAs) [129]. Such tools are essential for verifying whether a proposed fairness-aware clipping mechanism actually reduces bias or merely shifts the vulnerability. Empirical studies using these frameworks have revealed complex interactions; for example, while sensitivity-bounding techniques are crucial for defense, some Label DP algorithms may achieve less utility loss but remain fragile to MIAs, highlighting the need for multi-dimensional assessment [129].

Moreover, the evaluation of fairness in DP systems must extend beyond static accuracy metrics to include confidence intervals and uncertainty quantification. Given that modern training relies on stochastic optimization, the fairness properties of the learned model are themselves random variables. Recent work proposes constructing confidence intervals for test unfairness in group-fairness-aware classifiers trained via online SGD, providing a statistical basis for certifying fairness rather than relying on point estimates [130]. This is particularly important when comparing different privacy mechanisms, such as DP-SGD versus PATE, where empirical evidence suggests that while both exhibit disparate impacts, the severity varies significantly, necessitating rigorous comparative

analysis [131]. Furthermore, evaluation must consider long-term and societal impacts, recognizing that post-processing of differentially private data can itself introduce new forms of unfairness in downstream decision-making tasks [132].

Ultimately, the path toward trustworthy privacy-preserving machine learning requires a paradigm shift from viewing privacy and fairness as competing objectives to treating them as co-design constraints. Holistic frameworks must incorporate metrics that capture the intersectional nature of bias, the robustness of defenses against adaptive adversaries, and the stability of fairness guarantees under composition. As the field evolves, the development of standardized auditing tools capable of automatically detecting disparate impacts introduced by DP mechanisms will be critical for deploying equitable AI systems in high-stakes domains. The synthesis of fairness-aware clipping, adaptive budget allocation, and rigorous, multi-faceted evaluation represents the frontier of research aimed at ensuring that the right to privacy does not come at the cost of the right to fair treatment.

3 Synergistic Integration with Cryptography and Hybrid Defense Frameworks

3.1 Trusted Execution Environments as the Core of Confidential Federated Learning

3.1 Trusted Execution Environments as the Core of Confidential Federated Learning

While the synergistic integration of Differential Privacy (DP) and Secure Multi-Party Computation (SMPC) discussed in the following section addresses utility gaps through cryptographic and statistical means, an alternative paradigm shifts the trust assumption from mathematical proofs to hardware guarantees. The integration of Trusted Execution Environments (TEEs) into Federated Learning (FL) architectures represents this paradigm shift, moving from purely cryptographic or statistical privacy to hardware-enforced confidentiality. Unlike DP, which incurs significant utility loss due to noise injection to bound inference risks, TEEs—such as Intel Software Guard Extensions (SGX) and ARM TrustZone—provide secure enclaves where code and data are isolated from the host operating system, hypervisors, and even privileged administrators. This capability is particularly critical in FL scenarios involving honest-but-curious servers, where the central aggregator adheres to the protocol but attempts to infer sensitive information from received model updates. By executing aggregation logic within a TEE, the system ensures that raw gradients or model parameters remain encrypted outside the enclave and are decrypted only within the secure boundary, effectively mitigating server-side inference attacks without degrading model accuracy through noise.

However, deploying TEEs in distributed learning systems introduces distinct challenges centered on performance bottlenecks, memory constraints, and software development complexity. The strict memory isolation models inherent in architectures like Intel SGX often result in substantial overheads. For instance, the spatial isolation model, which prevents enclaves from sharing memory regions with the operating system or other enclaves, necessitates expensive data copy operations that can slow execution by one to two orders of magnitude. To address these inefficiencies, recent research has proposed novel memory models such as Elasticlave, which allows for selective and temporary memory sharing between enclaves and the OS, demonstrating performance improvements of up to two orders of magnitude while maintaining security guarantees [133]. Such architectural innovations are essential for making TEE-based FL viable in latency-sensitive edge computing environments where resources are constrained.

In the context of ARM-based edge devices, ubiquitous in IoT and mobile FL deployments, TrustZone technology offers a viable alternative to SGX. TrustZone separates the processor into a “secure world” and a “normal world,” providing hardware-isolated areas for sensitive code. However, performance studies indicate that switching between these worlds and managing secure storage incurs non-negligible energy and latency costs, which must be carefully optimized for battery-powered devices [134]. Furthermore, the monolithic nature of traditional Trusted OS implementations in TrustZone can expand the Trusted Computing Base (TCB), increasing the attack surface. Solutions like ReZone propose restructuring the monolithic TEE into multiple sandboxed domains to restrict privileges, thereby mitigating a significant percentage of known vulnerabilities while sustaining acceptable performance for applications like DRM-protected video encoding [135]. Similarly, uTango introduces a multi-world architecture for IoT devices, leveraging TrustZone-M primitives to create multiple equally secure execution environments, addressing architectural deficiencies in existing dual-world implementations [136].

To further harden FL systems against sophisticated attacks, researchers are increasingly combining TEEs with other technologies, such as blockchain, to create hybrid defense frameworks. A notable example is the integration of SGX with blockchain networks to secure model aggregation in Industrial Internet-of-Things (IIoT). In this setup, blockchain nodes host SGX-enabled processors to perform secure aggregation, while the blockchain ledger provides an immutable audit trail and consensus mechanism to verify the integrity of the global model [137]. This synergy addresses not only confidentiality but also the integrity and verifiability of the learning process, protecting against model poisoning and tampering. Additionally, frameworks like TZ4Fabric extend Hyperledger Fabric to execute smart contracts within ARM TrustZone, minimizing the TCB by isolating only the contract logic rather than the entire node, thus balancing security and performance trade-offs in supply chain applications [138].

Despite these advancements, the development of TEE-enabled FL applications remains hampered by complex programming models and the difficulty of porting existing machine learning libraries. The restrictive interfaces of TEEs often require developers to manually partition applications and rewrite code using low-level, error-prone languages. To lower this barrier, domain-specific languages like HasTEE have been introduced, embedding TEE programming within Haskell to automatically partition applications and enforce information flow control, as demonstrated in case studies involving federated learning and differentially-private data clean rooms [139]. Moreover, the supply chain for TEE development is critical; maintaining a robust ecosystem of third-party libraries, such as the extensive porting of Rust libraries to Intel SGX, is vital for reducing engineering costs and accelerating the adoption of privacy-preserving data processing [140].

Performance monitoring and optimization are also crucial for large-scale deployments. Tools like TEEMon provide continuous, fine-grained performance metrics for TEE-based applications, helping developers identify bottlenecks such as excessive system calls or paging issues within the enclave page cache [141]. Similarly, stress-testing frameworks like Stress-SGX allow for the evaluation of SGX-enabled nodes under various workloads, revealing insights into the limitations of legacy code porting and the performance disparities across different hardware generations [142]. For data-intensive FL tasks, access pattern leakage remains a concern even within TEEs. Frameworks like SGX-MR utilize MapReduce to regulate dataflows, offering a more efficient alternative to Oblivious RAM (ORAM) for protecting access patterns in data-intensive applications [143].

In conclusion, while TEEs offer a powerful mechanism for establishing confidential federated learning by protecting model updates against honest-but-curious servers without the utility penalty of noise, their practical efficacy depends on overcoming significant hurdles related to memory manage-

ment, development complexity, and performance overhead. Ongoing research into flexible memory models, hybrid architectures combining TEEs with blockchain, and developer-friendly tooling is essential to realizing the full potential of hardware-assisted privacy in distributed machine learning ecosystems. As the field evolves, the focus must remain on balancing the rigorous security guarantees of TEEs with the scalability and efficiency requirements of modern FL deployments. Notably, while TEEs eliminate the need for noise to protect against the server, they do not inherently protect against information leakage from the final model or side-channels; thus, the subsequent exploration of synergizing DP with SMPC offers a complementary approach to resolving the utility-privacy trade-off through cryptographic amplification rather than hardware isolation.

3.2 Synergizing Differential Privacy with Secure Multi-Party Computation

3.2 Synergizing Differential Privacy with Secure Multi-Party Computation

While the preceding section established Trusted Execution Environments (TEEs) as a hardware-centric solution for confidential federated learning—eliminating the need for noise to protect against honest-but-curious servers—it inherently relies on trust in silicon manufacturers and remains vulnerable to side-channel attacks or information leakage from the final model. To address these limitations without sacrificing utility, an alternative paradigm shifts the trust assumption from hardware guarantees to mathematical proofs through the synergistic integration of Differential Privacy (DP) and Secure Multi-Party Computation (SMPC). This hybrid approach is specifically designed to resolve the persistent utility gap that plagues standalone Local Differential Privacy (LDP) and Central Differential Privacy (CDP) models. While LDP offers robust protection against untrusted aggregators by perturbing data at the source, it often incurs prohibitive noise levels that degrade model accuracy, particularly in high-dimensional spaces. Conversely, CDP relies on a trusted central curator to add noise only after aggregation, achieving superior utility but creating a single point of failure vulnerable to insider threats. By combining DP and SMPC, cryptographic primitives ensure the confidentiality of individual updates during transmission and aggregation, allowing the system to operate under weaker trust assumptions while injecting significantly less noise than pure LDP protocols. This section analyzes the theoretical foundations, advanced mechanisms, and practical implications of such hybrid frameworks, serving as a statistical and cryptographic counterpoint to the hardware isolation discussed previously.

At the core of this synergy lies the concept of privacy amplification by shuffling or secure aggregation. In these hybrid models, users apply local randomization mechanisms that satisfy a relaxed form of LDP. These privatized messages are then submitted to a secure aggregation protocol, typically implemented via SMPC or a trusted shuffler, which computes the sum of the updates without revealing any individual contribution. The cryptographic guarantee that individual inputs remain hidden from the aggregator allows the system to leverage the “amplification by shuffling” effect. This phenomenon dictates that the privacy guarantee of the overall system is significantly stronger than the sum of the local guarantees, effectively bridging the utility gap between the local and central models. Research has demonstrated that by combining an LDP mechanism with a shuffler, privacy can be improved at no cost to the accuracy of operations insensitive to permutations, thereby enhancing utility in many tasks [144]. Furthermore, recent theoretical advancements have established strict bounds for DP guarantees in shuffle models, showing that for histogram queries, the statistical utility of the shuffle model can be nearly comparable to that of the central model under the same level of DP, despite the absence of a fully trusted curator [145].

A prominent example of this synergistic approach is the development of protocols that unify the benefits of accuracy-prioritized SMPC and privacy-prioritized DP. Traditional SMPC approaches achieve perfect output accuracy but may leak metadata or require complex interaction rounds, while pure DP approaches sacrifice accuracy for worst-case privacy. To address this, researchers have proposed unified frameworks such as Adaptive Differentially Quantized Subspace Perturbation (ADQSP). This approach combines quantization schemes with subspace perturbation to create a flexible mechanism that does not rely on heavy cryptographic primitives yet enjoys the benefits of both paradigms. By varying a single quantization parameter, ADQSP can smoothly transition between SMPC-type performances, characterized by high accuracy, and DP-type performances, characterized by strong privacy guarantees against powerful adversaries [146]. This adaptability is crucial for dynamic environments where the threat model or utility requirements may shift over time, offering a versatile alternative to the rigid hardware constraints of TEEs.

Beyond simple summation, hybrid protocols are increasingly addressing complex vector aggregation and high-dimensional learning tasks. The application of the shuffle model to vector aggregation has enabled private analysis across sophisticated structures like matrices and tensors, which are essential for deep learning. By utilizing advanced composition results, single-message protocols for real vector summation have been devised that maintain moderate noise requirements while unbinding users from their data [147]. In the context of federated learning, where communication efficiency is paramount, these hybrid methods are often coupled with compression techniques. For instance, schemes that jointly compress and privatize data using shared randomness have been shown to offer optimal privacy-accuracy-communication tradeoffs, compressing state-of-the-art LDP algorithms to the order of ϵ -bits of communication while preserving their guarantees [148].

The robustness of these hybrid systems against active adversaries is another critical area of focus. While SMPC protects against passive eavesdropping, active malicious clients can attempt to poison the aggregated model. Hybrid frameworks are being designed to withstand such general colluding attackers. For example, multi-group Differential Aggregation Protocols (DAP) have been introduced to mitigate the impact of poison values on estimated means under LDP, utilizing probing mechanisms to estimate attacker features without prior knowledge of attack patterns [149]. Moreover, the intersection of DP and cryptography extends to scenarios involving partially private data, where some statistics are released with DP and others without. Formalisms like Pufferfish privacy have been adapted to handle these mixed releases, ensuring that the combination of cryptographic secrecy and differential privacy provides a coherent and rigorous privacy account [150].

Furthermore, the emergence of “DP-Cryptography” highlights a growing trend where cryptographic primitives are explicitly allowed to leak differentially private outputs. This approach facilitates the implementation of alternatives to LDP that avoid the need for a trusted server while closing the accuracy gap associated with pure local models. These DP-cryptographic primitives offer performance improvements orders of magnitude better than standard cryptographic methods, making them suitable for resource-constrained environments [151]. The synergy also extends to bandwidth-efficient scenarios, such as wireless federated learning, where the superposition nature of the wireless channel provides a dual benefit of efficient gradient aggregation and strong DP guarantees. By incorporating user sampling with wireless aggregation, privacy leakage can be scaled down more aggressively than in orthogonal transmission schemes, demonstrating the physical layer’s potential to enhance cryptographic and differential privacy hybrids [152].

In summary, the synergistic integration of Differential Privacy and Secure Multi-Party Computation creates a robust defense-in-depth strategy that complements the hardware-assisted approaches of the previous section. It leverages the mathematical rigor of DP to bound information leakage

and the computational security of SMPC to protect data in transit and during processing, effectively mitigating the trust assumptions required in central models while drastically reducing the noise burden inherent in local models. As evidenced by advancements in adaptive quantization, shuffle models, and compressed aggregation, this hybrid frontier is essential for deploying scalable, accurate, and trustworthy machine learning systems in real-world, adversarial environments. While this approach successfully bridges the utility-privacy trade-off through cryptographic amplification, it still faces computational challenges in large-scale settings. Consequently, the following section explores an alternative trajectory: accelerating Fully Homomorphic Encryption (FHE) via hardware-assisted hybrid architectures to achieve strong confidentiality with improved efficiency, setting the stage for the decentralized trust models explored thereafter.

3.3 Accelerating Homomorphic Encryption via Hardware-Assisted Hybrid Architectures

3.3 Accelerating Homomorphic Encryption via Hardware-Assisted Hybrid Architectures

While the synergy between Differential Privacy and Secure Multi-Party Computation (SMPC) effectively bridges the utility gap through cryptographic aggregation and noise amplification, an alternative paradigm for privacy-preserving aggregation relies on Fully Homomorphic Encryption (FHE). FHE offers a distinct advantage by enabling computation directly on encrypted data without the need for interactive communication rounds or a trusted shuffler, theoretically providing the strongest possible confidentiality guarantees during model aggregation. However, the practical deployment of FHE in Federated Learning (FL) is severely hampered by its immense computational overhead, often resulting in performance degradation of several orders of magnitude compared to plaintext computation. This inefficiency stems primarily from the complexity of polynomial arithmetic and the necessity of the “bootstrapping” operation, a noise-reduction mechanism required to enable arbitrary-depth computations. To bridge the gap between these rigorous theoretical privacy guarantees and practical efficiency—thereby complementing the SMPC and DP strategies discussed previously—recent research has pivoted toward hardware-assisted hybrid architectures. These strategies leverage specialized hardware accelerators, Trusted Execution Environments (TEEs), and algorithmic optimizations such as quantization to make FHE viable for large-scale federated networks, setting the stage for the decentralized trust models explored in the subsequent section.

The most critical bottleneck in FHE schemes is the bootstrapping process, which refreshes ciphertext noise levels but consumes the vast majority of execution time and memory bandwidth. Traditional software implementations struggle with the massive data movement requirements associated with bootstrapping keys. To address this, specialized hardware accelerators have been developed to offload these intensive operations. For instance, the FPT accelerator introduces a fixed-point FPGA design specifically for Torus-based FHE (TFHE), exploiting inherent noise to utilize approximate fixed-point arithmetic rather than expensive floating-point units. This approach achieves a streaming pipeline that is compute-bound rather than memory-bound, delivering throughput improvements of two to three orders of magnitude over CPU-based implementations [153]. Similarly, the F1 accelerator utilizes a wide-vector processor architecture with functional units deeply specialized for FHE primitives like Number-Theoretic Transforms (NTT) and modular arithmetic. By explicitly managing the memory hierarchy to minimize data movement, F1 demonstrates speedups of up to 17,000x compared to state-of-the-art software, enabling real-time private deep learning scenarios that were previously infeasible [154].

Beyond dedicated accelerators, hybrid architectures that combine FHE with Trusted Execution Environments (TEEs) represent a promising middle ground that mirrors the hybrid philosophy of Section 3.2. TEEs, such as Intel SGX, provide a hardware-isolated enclave where code and data are protected from the host operating system and hypervisor. While TEEs alone protect against honest-but-curious servers, they do not support computation on data from mutually untrusted parties without decryption. A synergistic approach involves offloading the most expensive FHE operations, particularly bootstrapping, to a secure TEE. The TEEFHE framework exemplifies this strategy by moving the bootstrapping step into an SGX enclave, allowing the use of smaller ciphertexts and secret keys for the remaining homomorphic operations. This hybrid model significantly reduces both time and space complexity while mitigating side-channel leaks through careful memory access pattern obfuscation [155]. This division of labor allows the system to maintain the strong security guarantees of FHE for data transmission and storage while utilizing the relative efficiency of TEEs for critical noise management.

Furthermore, the efficiency of FHE in FL can be dramatically enhanced by integrating quantization-aware optimizations, addressing the tension between precision and performance often seen in privacy-preserving mechanisms. Neural networks inherently possess robustness against quantization errors, yet standard FHE implementations often rely on high-precision integer arithmetic to avoid accuracy loss, leading to bloated ciphertexts and slow computation. Research indicates that reducing integer sizes through quantization-aware training can yield substantial performance gains without sacrificing model utility. For example, reducing integer sizes by 33% to 43% in specific network architectures has been shown to decrease execution times by 40% to 80% under the BFV homomorphic encryption scheme [156]. Building on this, the SHE framework proposes a shift-accumulation-based deep neural network that leverages binary-operation-friendly encryption schemes to implement activation functions and pooling layers efficiently. By replacing expensive multiplications with cheap shifts and utilizing mixed bitwidth accumulators, SHE achieves state-of-the-art inference accuracy while reducing latency by over 90% compared to prior leveled-FHE approaches [157].

The architectural design of these accelerators must also address the memory bandwidth bottleneck, which often limits the efficacy of pure compute acceleration. Studies have shown that secure bootstrapping implementations are heavily bound by main memory bandwidth due to low arithmetic intensity and large cache requirements. Consequently, effective accelerators like ARK and BASALISC employ algorithm-architecture co-designs that focus on runtime data generation, inter-operation key reuse, and conflict-free memory hierarchies to minimize off-chip data movement [158] [159]. Additionally, emerging chiplet-based designs such as REED and CiFHER offer scalable solutions that mitigate wafer yield issues and reduce manufacturing costs while maintaining high throughput through non-blocking inter-chiplet communication [160] [161].

In the context of Federated Learning, these hardware-assisted hybrid architectures enable a new class of privacy-preserving protocols where the communication overhead is managed through compression techniques compatible with HE, and the computational burden is distributed across specialized edge or cloud accelerators. For instance, the FLASHE scheme demonstrates how additively symmetric homomorphic encryption can be optimized for cross-silo FL, minimizing training time increases to less than 6% with no additional communication overhead when integrated with industrial platforms [162]. By combining the algorithmic efficiency of quantization, the security isolation of TEEs, and the raw throughput of dedicated FHE accelerators, the field is moving toward scalable, secure, and efficient federated learning ecosystems. These advancements resolve the computational barriers of cryptographic privacy, thereby facilitating the transition to fully decentralized, verifiable

systems where trust is coordinated not just by hardware efficiency, but by distributed ledgers, as discussed in the following section.

3.4 Blockchain-Enabled Verifiability and Decentralized Trust Coordination

3.4 Blockchain-Enabled Verifiability and Decentralized Trust Coordination

Building upon the hardware-assisted hybrid architectures discussed in Section 3.3, which resolve the computational bottlenecks of Fully Homomorphic Encryption (FHE) to enable efficient cryptographic privacy, the next critical frontier is the decentralization of trust itself. While specialized accelerators and Trusted Execution Environments (TEEs) secure the computation and data transmission, traditional Federated Learning (FL) systems often remain reliant on a central aggregator, creating a single point of failure and necessitating strong institutional trust assumptions. The integration of blockchain technology represents a paradigm shift from these centralized models to decentralized, verifiable coordination mechanisms. By providing an immutable ledger for audit trails and facilitating decentralized consensus, blockchain addresses the vulnerabilities inherent in central aggregation, creating FL ecosystems where trust is algorithmically enforced rather than institutionally assumed. This evolution significantly enhances the robustness of collaborative learning against malicious actors and systemic failures, setting the stage for the complex interoperability challenges explored in the subsequent section.

A primary function of blockchain in this domain is the establishment of immutable audit trails that ensure the integrity of the entire learning lifecycle. By recording model updates, aggregation events, and participant contributions on a distributed ledger, blockchain systems create a transparent history that is resistant to tampering. This capability is crucial for detecting and deterring malicious behaviors such as model poisoning or data falsification. For instance, frameworks have been proposed where smart contracts automate the verification of model contributions, ensuring that only valid updates are incorporated into the global model [163]. Furthermore, the immutability of the ledger supports comprehensive data provenance management, allowing stakeholders to trace the origin and evolution of model parameters with high confidence [164]. This transparency extends to the evaluation of participant reputation, where blockchain-based voting and rating systems can intrinsically incentivize honest behavior while penalizing deviations, thereby sustaining a trustworthy collaborative environment [165].

Beyond auditing, blockchain enables verifiable secure aggregation through the deployment of smart contracts and Zero-Knowledge Proofs (ZKPs), complementing the cryptographic guarantees provided by the hardware layers. In many decentralized FL setups, the aggregation process itself must be verified to prevent a malicious aggregator from manipulating the global model or excluding specific clients. Recent advancements introduce ZKP-based aggregators that allow a server to prove the correctness of the aggregation algorithm without revealing individual client updates. The zkDFL framework exemplifies this approach, utilizing ZKPs to demonstrate that aggregation adheres to accepted protocols and that all client inputs are accounted for, all while managing the process via smart contracts to ensure scalability and reduced computational overhead [166]. Similarly, the zkFL system leverages ZKPs to guarantee that the aggregator executes intended behaviors faithfully, with miners validating these proofs on-chain without accessing sensitive model data [167]. These mechanisms effectively bridge the gap between privacy and verifiability, ensuring that the aggregation process is both confidential and mathematically provable.

Decentralized trust coordination is further reinforced through innovative incentive mechanisms and consensus protocols tailored for FL. The lack of fair and transparent reward schemes has historically

hindered the widespread adoption of FL, particularly in open networks where participants are self-interested. Blockchain-based solutions address this by encoding incentive rules into smart contracts, which automatically distribute rewards based on verified contributions. Systems like the “Proof-of-Contribution-Based Design” utilize distributed storage and on-chain smart contracts to assess trainer contributions and detect outliers, ensuring that honest participants are compensated fairly while malicious actors are slashed [168]. Moreover, novel consensus protocols such as “Proof-of-Useful-Work” have been adapted to treat FL training tasks as the computational challenge for block validation, thereby aligning the energy consumption of the blockchain with the productive goal of model training [169]. This alignment not only reduces the environmental footprint of the network but also creates a self-sustaining economy where computational resources are directly incentivized to improve model utility.

The role of blockchain also extends to securing client selection and managing cross-domain operations in hierarchical networks, addressing scalability concerns that parallel the hardware optimization strategies of Section 3.3. In scenarios where a central server might manipulate client selection to infer private data, blockchain-enforced random selection protocols provide a verifiable method to ensure fairness and privacy [170]. For large-scale IoT deployments, federated ledger architectures like Fed-DDM partition participants into permissioned domains with private intra-ledgers, coordinated by a public inter-ledger via smart contracts. This hierarchical structure guarantees the security of cross-domain operations without exposing sensitive information, effectively scaling FL to massive networks [171]. Additionally, the integration of ZKPs with blockchain oracles allows for the non-disclosing verification of external data and device credentials, extending the chain of trust from the data source to the final model [172].

Despite these advancements, challenges remain regarding the efficiency and latency of on-chain computations. High gas costs and limited throughput can bottleneck the FL process, particularly when dealing with high-frequency model updates. To mitigate this, researchers have proposed off-chain computation strategies where heavy lifting is performed externally, with only succinct ZKPs submitted to the blockchain for verification [173]. Furthermore, architectural innovations such as sharding and Direct Acyclic Graph (DAG) based consensus mechanisms are being explored to enhance scalability and reduce the impact of model staleness inherent in asynchronous decentralized settings [174]. As the field evolves, the convergence of blockchain, cryptography, and FL promises to deliver fully decentralized, trustworthy, and efficient AI ecosystems capable of operating in zero-trust environments. However, realizing this vision requires navigating the complexities of integrating these distinct technological stacks—blockchain, hardware enclaves, and cryptographic protocols—into cohesive systems, a challenge that defines the cross-technology interoperability landscape discussed in the following section.

3.5 Cross-Technology Interoperability and Heterogeneous Trust Frameworks

3.5 Cross-Technology Interoperability and Heterogeneous Trust Frameworks

While Section 3.4 established how blockchain can decentralize trust and provide verifiable audit trails, real-world Privacy-Preserving Machine Learning (PPML) ecosystems rarely operate within a single, homogeneous technological domain. Instead, they exist within highly fragmented landscapes characterized by disparate hardware capabilities, incompatible communication protocols, and divergent trust assumptions. A critical challenge in advancing Federated Learning (FL) and Differential Privacy (DP) lies in bridging these technological silos to create cohesive, interoperable systems. The integration of diverse privacy-enhancing technologies—ranging from Trusted Execution Environ-

ments (TEEs) and Homomorphic Encryption (HE) to the blockchain-based consensus mechanisms discussed previously—requires robust frameworks that can orchestrate cooperation among devices with fundamentally different security postures and computational constraints. Without such interoperability, the promise of decentralized intelligence remains confined to isolated domains, unable to leverage the collective power of heterogeneous networks.

A primary obstacle to cross-technology integration is the heterogeneity of TEE implementations across critical infrastructure and edge devices. Different manufacturers employ distinct isolation technologies, such as Intel SGX, ARM TrustZone, or RISC-V based enclaves, which often lack native mutual attestation mechanisms. This fragmentation prevents devices equipped with incompatible TEEs from establishing a unified trusted environment, thereby hindering collaborative learning tasks that require verified secure computation. To address this, decentralized coordination mechanisms have been proposed to act as a “translator of trust,” leveraging the immutable ledgers introduced in the previous section. For instance, recent work introduces decentralized frameworks that utilize blockchain technology to support key TEE functions, specifically attestation services, across heterogeneous environments [175]. In such architectures, a device possessing one type of TEE can interact securely with a distributed ledger to verify whether a collaborating device claiming a different TEE meets the requisite security standards. This approach decouples the trust verification process from specific hardware vendors, enabling a flexible ecosystem where new TEE schemes can be integrated without disrupting existing deployments.

Beyond hardware-level incompatibilities, the coordination of privacy protocols across varying network layers and identity management systems presents another significant hurdle. Traditional centralized identity providers often fail to support cross-domain interoperability due to geopolitical constraints, data protection legislations, or the sheer complexity of managing keys across multiple stakeholders. The emergence of Self-Sovereign Identity (SSI) and Decentralized Identifiers (DIDs) offers a promising pathway to resolve these issues by providing tamper-proof, cross-domain identity management that does not rely on centralized authorities [176]. By leveraging SSI, heterogeneous IoT devices and edge nodes can authenticate and authorize interactions autonomously, facilitating secure data aggregation and model updates even in dynamic networks where participants frequently join and leave [177]. This decentralized identity layer serves as a foundational prerequisite for interoperable PPML, ensuring that trust is established based on verifiable credentials rather than institutional affiliation.

Furthermore, the synergy between distributed ledger technologies (DLT) and privacy-preserving computation must be engineered to accommodate the varying resource constraints of participating nodes. In large-scale IoT deployments, directly integrating heavy cryptographic primitives like full Homomorphic Encryption or proof-of-work blockchains into resource-constrained devices is often infeasible. Hierarchical and federated ledger architectures offer a viable solution by stratifying the network into domains, extending the architectural concepts of Fed-DDM mentioned earlier. In such frameworks, efficient Byzantine Fault Tolerance (BFT) consensus protocols handle transactions within private intra-ledgers, while a public inter-ledger federates these domains without exposing sensitive data [171]. This stratification allows for the seamless interoperability of different consensus mechanisms and privacy tools, optimizing performance while maintaining security guarantees. Additionally, the integration of usage control mechanisms directly into DLT networks empowers devices to enforce granular privacy policies and data access conditions dynamically, creating a synergistic relationship between the ledger’s immutability and the device’s autonomy [178].

The complexity of these heterogeneous environments is further compounded by the need to manage varying levels of trust and privacy preferences among participants. In scenarios where some nodes

are fully trusted (e.g., via TEEs) while others rely solely on cryptographic guarantees or differential privacy, the system must adapt its aggregation and verification strategies accordingly. Decentralized approaches towards responsible AI emphasize the need for computational infrastructures that can bridge technical systems with social institutions, allowing for flexible intervention mechanisms that respect informational self-determination [179]. This involves developing protocols that can dynamically adjust privacy budgets and verification rigor based on the attested capabilities of each participant. For example, systems may employ lightweight privacy-preserving data aggregation strategies for low-power sensors while utilizing more robust secure multi-party computation for high-value nodes, all coordinated through a unified decentralized framework [180].

Moreover, the scalability of such interoperable systems depends on effective mechanisms for detecting and isolating malicious actors within a mixed-trust environment. In autonomous networks comprising diverse devices, the absence of a central trusted entity necessitates distributed detection schemes capable of identifying compromised nodes regardless of their underlying technology stack. Innovative schemes utilizing random elections and distributed consensus have demonstrated the ability to detect malicious devices efficiently in large-scale autonomous networks, ensuring the integrity of the collaborative learning process even when facing heterogeneous threats [181]. By combining these detection capabilities with cross-technology attestation and decentralized identity management, future PPML ecosystems can achieve a resilient foundation. However, as these heterogeneous components converge, they introduce new systemic vulnerabilities unique to hybrid architectures. The following section addresses these critical challenges, exploring defense-in-depth strategies required to protect against active adversaries exploiting the interplay between stateful hardware enclaves and decentralized consensus mechanisms.

3.6 Defense-in-Depth Against Active Adversaries and Systemic Vulnerabilities in Hybrid Systems

3.6 Defense-in-Depth Against Active Adversaries and Systemic Vulnerabilities in Hybrid Systems

As established in Section 3.5, the convergence of heterogeneous technologies—spanning Trusted Execution Environments (TEEs), blockchain consensus, and Differential Privacy (DP)—is essential for creating interoperable PPML ecosystems. However, this integration of diverse trust anchors and computational layers inevitably expands the attack surface, introducing systemic vulnerabilities unique to hybrid architectures. While the previous section outlined frameworks for bridging technological silos, it also highlighted that the interplay between stateful hardware enclaves and stateless or probabilistic consensus mechanisms creates new vectors for exploitation. Consequently, a robust defense-in-depth strategy is paramount to mitigate active adversaries who seek to compromise the entire system by targeting the interfaces between these layers. The primary challenge lies in addressing threats that arise specifically from these hybrid configurations, such as rollback attacks, microarchitectural side-channels, and the fragility of trust assumptions in decentralized environments.

One of the most critical systemic vulnerabilities inherent to TEE-blockchain hybrids is susceptibility to rollback attacks. Although TEEs provide strong isolation for code and data, they are inherently stateless regarding the global history of the blockchain unless explicitly engineered otherwise. In systems where consensus lacks immediate finality or relies on probabilistic mechanisms, an adversary controlling the network or hosting infrastructure can force a TEE to revert to a previous state, effectively undoing processed transactions or model updates. This capability enables attackers to

double-spend resources or manipulate federated learning aggregation by replaying advantageous historical updates. Research indicates that naive combinations of TEEs and blockchains fail to prevent such exploits because TEEs alone cannot detect ledger reorganization [182]. To counter this, hybrid systems must implement rigorous rollback protection mechanisms. For instance, the concept of “instantiation control” enforced through TEE-supported leases has been proposed to mitigate rollback and fork attacks without incurring the prohibitive computational costs of crypto-puzzles [183]. Furthermore, specific protocols have been developed to checkpoint the state of proof-of-stake chains onto proof-of-work chains, thereby leveraging the immutability of the latter to secure the former against long-range and rollback attempts [184]. These approaches underscore the necessity of coupling hardware-based isolation with cryptographic finality gadgets to ensure that the state within an enclave remains consistent with the canonical chain [185].

Beyond logical inconsistencies arising from state management, hybrid systems are acutely vulnerable to side-channel attacks that bypass traditional cryptographic protections. Even when data is encrypted and processed within a TEE, physical and microarchitectural side channels can leak sensitive information. Ciphertext side-channel attacks, for example, exploit deterministic memory encryption patterns to infer secret data, a vulnerability that persists despite constant-time coding practices [186]. In the context of IoT and edge computing, where federated learning is frequently deployed, Trusted OSes running inside TEEs have been shown to contain numerous unknown vulnerabilities exploitable via state-aware fuzzing, potentially compromising the entire secure world [187]. Moreover, memory isolation mechanisms in widely deployed architectures like ARM TrustZone have been found susceptible to buffer overflow attacks that allow malicious entities to access encrypted memory belonging to other trusted applications, illustrating that hardware boundaries are not impervious to sophisticated software exploits [188]. Addressing these issues requires a multi-layered defense that includes formal verification of hardware designs and runtime monitoring. Frameworks like SYNFI enable pre-silicon fault analysis to verify the effectiveness of countermeasures against fault injection, while tools like PASCAL provide design flows to eliminate timing side-channel vulnerabilities at the register-transfer level [189][190].

The complexity of these hybrid architectures also demands advanced runtime verification and compartmentalization to limit the blast radius of any single compromise. The interaction between TEEs and untrusted host services, such as file systems or network stacks, presents a significant attack vector where adversaries can manipulate service behavior to trigger bugs within the enclave. GateKeeper, a framework leveraging lightweight formal models, demonstrates how runtime validators can be automatically generated to guard service invocations, thereby preventing integrity violations caused by malicious external services [191]. Similarly, the Sirius framework introduces enclave-aware compartmentalization to control data flows between kernel objects, effectively eliminating a wide range of attacks that exploit the semantic gap between secure and normal worlds [192]. In decentralized settings, the reliance on a centralized attestation authority creates a single point of failure; thus, decentralized attestation schemes utilizing blockchain and smart contracts, such as JANUS, are crucial for establishing resilient trust networks that can withstand advanced attacks on the verification infrastructure [193].

Finally, the security of hybrid systems relies heavily on the correctness of the consensus protocols and smart contracts that orchestrate them. Statistical model checking and formal verification are indispensable for identifying elusive faults in blockchain consensus that could lead to systemic failures [194][195]. The adoption of accountable safety mechanisms and incentivized watchtower networks further strengthens the system against rational adversaries who might otherwise neglect their verification duties [196][197]. By synthesizing hardware hardening, cryptographic finality,

runtime monitoring, and formal verification, hybrid PPML systems can achieve a comprehensive defense-in-depth posture. This layered approach not only addresses the specific vulnerabilities introduced by cross-technology interoperability but also lays the groundwork for future research into adaptive security models capable of evolving alongside emerging adversarial tactics.

4 The Adversarial Landscape: Attacks, Vulnerabilities, and Empirical Auditing

4.1 Gradient Inversion and Data Reconstruction Vulnerabilities

4.1 Gradient Inversion and Data Reconstruction Vulnerabilities

The foundational premise of Federated Learning (FL) rests on the assumption that sharing model gradients rather than raw data preserves user privacy. However, this assumption has been fundamentally challenged by the emergence of Deep Gradient Leakage (DGL) attacks, which demonstrate that sensitive training samples can be reconstructed with high fidelity solely from shared gradient vectors. The seminal work [198] shattered the long-held belief in the safety of gradient exchange, empirically validating that attackers can recover pixel-wise accurate images and token-wise matching text from public gradients in both computer vision and natural language processing tasks. This vulnerability arises because gradients contain sufficient information to invert the optimization process, effectively allowing an adversary to solve for the input data that generated the specific weight updates. While the following section will address Membership Inference Attacks which focus on determining the presence of data, the threat modeled here is far more severe: the complete reconstruction of the private data itself.

To understand the theoretical underpinnings of these vulnerabilities, recent research has moved beyond empirical demonstrations to develop analytical frameworks that explain when and how leakage occurs. A critical advancement in this domain is the introduction of Inversion Influence Functions (I^2F), as proposed in [199]. This theoretical tool establishes a closed-form connection between recovered images and private gradients by implicitly solving the DGL problem. Unlike direct optimization-based attacks that are computationally expensive and opaque, I^2F provides a scalable method to analyze deep networks using only oracle access to gradients and Jacobian-vector products. This framework reveals that privacy leakage is not uniform; it is heavily influenced by model architecture, initialization, and the specific direction of gradient perturbations. Furthermore, [200] formulates the reconstruction problem as solving a linear system iteratively for each layer, accompanied by gradient matching corrections. This analysis suggests that the solvability of the reconstruction problem is primarily determined by the linear systems at each layer, allowing researchers to partially attribute data leakage risks to specific architectural choices. Similarly, [201] reinforces this by modeling data reconstruction as a layer-wise optimization problem, where the objective function depends on weights, gradients, and constraints pulled back from subsequent layers.

The threat landscape has evolved to address more complex scenarios, including transformer-based models and batched inputs, pushing the boundaries of what was previously thought defensible. [202] provides a comprehensive analysis of DGL specifically for transformers, rigorously investigating the conditions under which these attacks succeed and reevaluating the efficacy of adding noise as a defense within the differential privacy framework. For batched training, where multiple samples are processed simultaneously, attacks have become increasingly sophisticated. [203] addresses the semantic limitations of single-label inversions by introducing multi-label representations and edge detection regularization to reduce semantic errors in reconstructed images. Moreover, [204]

presents an algorithm capable of reconstructing high-resolution images even with large batch sizes, utilizing additional convolution blocks for label restoration and total variance regularization for image recovery, significantly reducing time costs while improving accuracy.

Adversaries have also developed techniques to bypass heuristic defenses and operate under weaker assumptions, rendering many ad-hoc protections obsolete. While defenses like dropout were initially thought to provide protection, [28] demonstrates that this stochasticity can be modeled during attack optimization. The authors propose a Dropout Inversion Attack (DIA) that jointly optimizes for client data and dropout masks, effectively bypassing the protection and reconstructing data with high fidelity. Similarly, the reliance on large auxiliary datasets for prior knowledge has been reduced by [205], which leverages anomaly detection models to capture data distributions from minimal data, posing a more substantial threat to real-world FL systems where extensive auxiliary data may not be available. Furthermore, [206] exploits multiple temporal gradients to recover data without any prior knowledge, using robust statistics to eliminate the impact of invalid gradients, thereby succeeding even in complex models and large datasets where single-iteration attacks fail.

In response to these escalating threats, various defense mechanisms have been proposed, though they often involve difficult trade-offs between privacy, utility, and efficiency. One promising direction is the manipulation of the data itself rather than just the gradients. [207] introduces a novel paradigm that constructs robust data to obfuscate attackers. By jointly optimizing for utility consistency and a semantic gap between robust data and raw data, Refiner effectively defends against state-of-the-art attacks. Another approach involves variational modeling, as seen in [101], which proposes PRECODE with Partial Perturbation (PPP). This method combines variational modeling to induce stochasticity with strategic partial gradient perturbation, preventing attack convergence while minimizing performance loss. Extending this, [208] introduces a Convolutional Variational Bottleneck (CVB) that can be placed early in the network to prevent leakage without the excessive parameter overhead of previous methods.

Despite these engineering efforts, standard differential privacy implementations often fall short of providing absolute security against reconstruction, highlighting the gap between theoretical guarantees and practical resilience. [209] argues that differentially private algorithms with fixed privacy parameters remain vulnerable, advocating instead for dynamic privacy parameters that adapt noise variance to the trend of gradient updates. Additionally, [210] suggests that gradient pruning, when modified into a Dual Gradient Pruning (DGP) strategy, can offer strong privacy guarantees while improving communication efficiency. However, the arms race continues, as [211] shows that even under defense settings, private data can be leaked using Generative Gradient Leakage (GGL), which leverages the latent space of GANs and gradient-free optimization to compensate for informational loss caused by gradient degradation. Finally, [212] offers a sobering theoretical result, proving that under mild conditions, training samples can be fully reconstructed from a single gradient query at a randomly chosen parameter value. This inherent vulnerability underscores the necessity for the rigorous theoretical bounds and refined privacy metrics discussed in the subsequent section, which aim to quantify exactly how much information leakage is permissible before membership and reconstruction risks become unacceptable.

4.2 Membership Inference Attacks and Theoretical Bounds

4.2 Membership Inference Attacks and Theoretical Bounds

While Section 4.1 demonstrated that gradient sharing can lead to the catastrophic reconstruction of raw training data, a distinct yet equally critical threat vector exists in the form of Membership Inference Attacks (MIAs). Unlike reconstruction attacks that seek to recover specific data points, MIAs aim to determine merely whether a specific record was part of a model’s training set. This distinction is vital: while reconstruction represents a total privacy breach, membership inference exposes the mere participation of an individual, which can itself reveal sensitive attributes (e.g., a patient’s presence in a disease-specific dataset). Differential Privacy (DP) has emerged as the gold standard for providing provable guarantees against such inference; however, a significant disconnect persists between the conservative theoretical bounds derived from DP parameters and the empirical success rates observed in practical attacks. This section critically examines this gap, exploring how traditional ϵ -DP bounds often overestimate privacy risks in realistic settings, and surveys emerging theoretical frameworks and metrics designed to provide tighter, more interpretable guarantees on adversary advantage.

The fundamental challenge lies in the worst-case nature of standard Differential Privacy. Theoretical guarantees based on (ϵ, δ) -DP are designed to hold uniformly over all possible datasets and adversaries with arbitrary background knowledge, leading to bounds on membership inference advantage that are often vacuous for the privacy budgets used in practice. For instance, analyzing a setting of DP-SGD with $\epsilon = 4$ using standard conversion methods yields an upper bound on the adversary’s advantage of approximately 0.97, suggesting near-total privacy failure, while tighter analysis yields a bound of approximately 0.36 [213]. However, empirical evaluations frequently demonstrate that state-of-the-art MIAs achieve significantly lower success rates under these same conditions, indicating that the theoretical bounds are overly pessimistic. To bridge this disparity, recent research has focused on deriving tighter bounds specifically tailored to the Gaussian mechanism and adaptive composition, which reveal that the actual advantage of an adversary can be orders of magnitude lower than what standard DP analysis suggests [213].

A pivotal advancement in reconciling theory with practice is the development of privacy notions that explicitly model the capabilities of realistic adversaries rather than worst-case scenarios. The concept of f -Membership Inference Privacy (f -MIP) offers a novel framework that provides interpretable privacy guarantees directly linked to the success rate of likelihood ratio-based attacks [214]. Unlike traditional DP, which can be difficult to interpret in terms of actual attack success, f -MIP allows for the derivation of parametric guarantees, such as μ -Gaussian Membership Inference Privacy (μ -GMIP), which quantify the indistinguishability of members and non-members under specific distributional assumptions. This approach highlights that models trained with standard Stochastic Gradient Descent (SGD) already possess an elementary level of protection due to inherent stochasticity, and that adding noise can amplify this protection in a quantifiable manner without the severe utility penalties associated with strict DP requirements [214]. Furthermore, the introduction of Practical Membership Privacy (PMP) addresses the observation that industrial deployments often utilize large ϵ values (e.g., $\epsilon \geq 7$) that theoretically offer weak guarantees yet empirically defend effectively against MIAs. PMP models the attacker’s uncertainty regarding the contents of the private data, revealing that a large DP parameter often translates into a much smaller PMP parameter, thereby explaining the robustness of such models in practical threat models [33].

Beyond redefining privacy notions, researchers have also developed closed-form bounds and information-theoretic approaches to evaluate privacy leakage without relying solely on the indirect translation from ϵ . Closed-form bounds for DP-SGD against record-level inference model the training process as an information-theoretic channel, allowing for the direct calculation of bounds on membership and attribute inference that match state-of-the-art empirical techniques while being computationally efficient [215]. Similarly, leveraging variants of Fano’s inequality within a multiple hypothesis testing framework has provided theoretical evidence that data reconstruction and membership inference remain bounded even at relatively large ϵ values, specifically when the underlying data domain size is considered [216]. These theoretical tools suggest that the privacy budget required to prevent significant inferential power scales logarithmically with the data domain size, offering a more nuanced understanding of privacy thresholds than binary pass/fail DP checks.

However, the landscape of theoretical bounds is complicated by data dependencies and heterogeneity, which are often ignored in standard assumptions. When training data exhibits statistical dependencies, the privacy parameter ϵ can effectively scale with the training set size, rendering standard DP guarantees meaningless and significantly increasing the performance of MIAs [217]. This highlights a critical vulnerability where theoretical bounds assuming independent and identically distributed (IID) data vastly underestimate the risk in real-world correlated datasets. Additionally, the phenomenon of disparate vulnerability indicates that privacy guarantees are not uniform across population subgroups; while DP bounds disparate vulnerability on average, specific subgroups may remain highly susceptible to inference attacks, a nuance often lost in aggregate theoretical metrics [218].

To address the limitations of both loose theoretical bounds and computationally expensive empirical audits, new metrics and auditing frameworks have been proposed. The Epsilon* metric provides an empirical lower bound on the privacy loss of a specific model instance by analyzing the true positive and false positive rates of a hypothesis test, allowing auditors to visualize the privacy-utility landscape without requiring access to the training mechanism [34]. Similarly, Differential Training Privacy (DTP) serves as an empirical metric to estimate privacy risk for classifiers that cannot be trained with formal DP, acting as a predictive measure for MIA success [219]. Despite these advances, caution is warranted, as some studies suggest that model owners can construct “proofs of repudiation” to plausibly refute MIA results, challenging the reliability of MIAs as a definitive privacy test [220]. Nevertheless, the convergence of tighter theoretical bounds like those in [213] and novel metrics like f -MIP [214] represents a crucial step toward a more accurate and practical understanding of membership privacy in machine learning systems. Having established the theoretical limits and empirical realities of passive inference threats, the discussion must now shift to the active threat landscape. The following section investigates how the very noise mechanisms introduced to satisfy these privacy bounds can be weaponized by active adversaries to compromise model integrity through poisoning and backdoor attacks.

4.3 Model Poisoning, Backdoors, and the Dual-Edge of Privacy Noise

4.3 Model Poisoning, Backdoors, and the Dual-Edge of Privacy Noise

While Section 4.2 established the theoretical and empirical nuances of defending against passive inference attacks, the landscape shifts dramatically when considering active adversaries capable of manipulating the training process. The integration of Differential Privacy (DP) into Federated Learning (FL) and centralized pipelines, primarily motivated by the need to protect data from the inference threats discussed previously, introduces a complex dual-edged sword regarding model

integrity. The very mechanisms designed to obscure individual contributions—specifically gradient clipping and the addition of random noise—create a paradoxical environment: while DP can inadvertently suppress certain malicious updates by dampening their signal, it simultaneously provides a powerful cover for sophisticated adversaries executing model poisoning and backdoor attacks. This section critically examines how privacy-induced noise can be exploited to mask adversarial perturbations, rendering traditional anomaly detection systems ineffective, and analyzes the nuanced trade-offs between privacy guarantees and robustness against active adversaries.

Historically, DP has been posited as a defensive mechanism against data poisoning. Theoretical arguments suggest that because DP limits the influence of any single data point or update, it should inherently restrict an adversary’s ability to skew the global model. Indeed, empirical studies have shown that DP-SGD (Stochastic Gradient Descent) can mitigate the success of standard backdoor attacks by clipping large gradients and obscuring the specific trigger patterns associated with malicious data [221]. Furthermore, hybrid approaches combining DP with data augmentation techniques, such as mixup, have been proposed to provably defuse poisoning attacks while maintaining model utility, suggesting that the regularization effects of privacy mechanisms can enhance generalization and reduce overfitting to poisoned samples [222]. Similarly, research into PATE (Private Aggregation of Teacher Ensembles) indicates that the bagging structure inherent in some DP frameworks offers natural resilience against backdoors, provided that hyperparameters are tuned correctly and the number of backdoors in the dataset remains limited [221].

However, this defensive narrative is increasingly challenged by findings that reveal the “dual-edge” nature of privacy noise, a reality that complicates the assumption of inherent robustness. In distributed settings, the stochastic noise added to gradients to satisfy DP guarantees can act as a smokescreen, allowing malicious clients to inject carefully crafted perturbations that mimic natural variance. Adversaries can exploit this by designing adaptive poisoning strategies that align their malicious updates with the expected distribution of DP noise, thereby evading norm-based or accuracy-based anomaly detection systems. For instance, recent work in cyber-physical critical infrastructures demonstrates that attackers can utilize the additional DP-noise to evade state-of-the-art anomaly detection techniques, successfully preventing the optimal convergence of the FL model while remaining undetected [223]. This phenomenon is not limited to federated settings; in cooperative multi-agent reinforcement learning, the noise introduced for privacy has been shown to give rise to novel, evasion-resilient localized poisoning attacks that capitalize on the inherent uncertainty to hinder model convergence [224].

The vulnerability is further exacerbated by the fact that DP mechanisms can alter the geometric properties of the loss landscape, potentially making models more susceptible to specific types of adversarial manipulation. Contrary to the assumption that privacy always enhances robustness, empirical observations indicate a trade-off where networks trained with DP can, in certain settings, become more vulnerable to adversarial examples and common corruptions compared to their non-private counterparts [225]. The components of DP training, such as gradient clipping, may inadvertently flatten the loss surface in ways that facilitate the persistence of backdoors or reduce the model’s capacity to reject out-of-distribution inputs associated with poisoning. Moreover, the effectiveness of DP as a defense is highly sensitive to hyperparameters; improper tuning can render the privacy mechanism ineffective against attacks while still incurring significant utility costs. Research highlights that while Label-DP algorithms might offer faster training and better accuracy, they can be more fragile to membership inference and potentially other adversarial exploits if not meticulously configured [129].

The interplay between privacy and poisoning also extends to the detection phase, creating a funda-

mental tension for defenders. Traditional defense mechanisms often rely on identifying statistical outliers in gradient updates. However, when DP noise is present, the variance of legitimate updates increases, widening the acceptance window for anomalies. This allows attackers to inject larger malicious signals that would otherwise be flagged in a non-private setting. Studies analyzing the statistical thresholds for detecting adversaries in Laplace mechanisms reveal that as the privacy parameter (ϵ) decreases (increasing privacy), the threshold for detecting an adversary’s bias must also shift, often making it statistically indistinguishable from natural noise [226]. Consequently, the defender faces a dilemma where strengthening privacy guarantees may simultaneously lower the detection rate of sophisticated poisoning attacks. Furthermore, the strategic selection of poisoned samples can amplify this effect; attackers employing confidence-driven sampling strategies, which target samples near the decision boundary, can create backdoors that are particularly stealthy [227]. When combined with the obfuscation provided by DP, these attacks become exceptionally difficult to mitigate. The challenge is compounded in scenarios involving continuous data release or temporal correlations, where privacy leakage and vulnerability to manipulation can accumulate over time, creating windows of opportunity for persistent adversaries [38].

In conclusion, while Differential Privacy offers a rigorous framework for limiting data leakage, its role in securing model integrity against poisoning and backdoors is ambivalent. It is not a panacea; rather, it introduces a new attack surface where privacy noise can be weaponized. The literature suggests that relying solely on DP for robustness is insufficient. Instead, a holistic defense strategy is required, one that accounts for the specific ways in which privacy mechanisms interact with adversarial optimization. This includes developing adaptive defense mechanisms, such as reinforcement learning-based policies for dynamic privacy level selection, which can respond to evolving attack vectors without human intervention [223]. Ultimately, navigating the dual-edge of privacy noise requires a careful calibration of privacy budgets, robust aggregation rules, and continuous empirical auditing. As we move from theoretical bounds and active attack vectors to the practical verification of these systems, the need for rigorous empirical auditing methodologies becomes paramount, a topic explored in the following section through canary-based measurements and poisoning verification tools.

4.4 Empirical Privacy Auditing and Canary-Based Measurement

4.4 Empirical Privacy Auditing and Canary-Based Measurement

As established in the preceding discussion on the dual-edge of privacy noise, relying solely on theoretical bounds or the inherent robustness of Differential Privacy (DP) mechanisms is insufficient for guaranteeing model integrity against sophisticated poisoning and backdoor attacks. Theoretical guarantees, while mathematically rigorous, often rely on worst-case assumptions that yield overly conservative privacy budgets, potentially masking the specific vulnerabilities exploited by active adversaries. Consequently, there is a critical need for empirical auditing methodologies that validate these theoretical bounds against actual privacy leakage in deployed systems. Empirical auditing serves as a practical lower bound on privacy loss, complementing theoretical upper bounds by measuring the real-world efficacy of privacy mechanisms under specific adversarial capabilities. Recent advancements have therefore shifted focus from generic membership inference attacks to sophisticated, statistically powerful frameworks designed to detect subtle violations and quantify leakage with high precision, directly addressing the verification gaps highlighted by the limitations of DP in active threat scenarios.

A prominent paradigm in this domain is the use of canary-based techniques, which involve injecting

carefully crafted data points, or “canaries,” into the training set to monitor their influence on the final model. This approach allows auditors to measure per-round exposure and determine if specific data instances are memorized or leaked, providing a direct metric for the success of potential data extraction or poisoning attempts. A significant contribution to this area is the work presented in [228], which introduces a rigorous methodology based on “Lifted Differential Privacy” (LiDP). This framework expands the traditional definition of differential privacy to accommodate randomized datasets, thereby granting auditors the freedom to design randomized canaries. By training models with K canaries versus $K - 1$ canaries and leveraging the symmetry of independently and identically distributed (i.i.d.) canaries, this method reuses privately trained models to run multiple statistical tests. The authors further introduce novel confidence intervals that adapt to empirical higher-order correlations, demonstrating significant improvements in sample complexity compared to previous approaches. This allows for more efficient detection of privacy violations without the prohibitive computational cost of retraining models thousands of times.

Addressing the scalability challenges inherent in federated learning (FL) and large-scale model training, researchers have developed “one-shot” estimation frameworks. Traditional auditing techniques often require retraining the model numerous times to estimate privacy loss, a process that is computationally infeasible when training runs take days or weeks. The paper [37] presents a novel approach that systematically addresses these challenges by allowing for the efficient auditing of privacy loss during a single training run. This method does not require a priori knowledge of the model architecture, task, or specific DP algorithm, making it highly versatile for diverse FL settings. The authors demonstrate that their one-shot approach provides provably correct estimates for privacy loss under the Gaussian mechanism and performs robustly across various adversarial threat models on established benchmarks. This represents a crucial step toward making empirical auditing a standard practice in production environments where computational resources are constrained.

Beyond canary insertion, poisoning-based verification tools have emerged as powerful instruments for auditing, creating a direct bridge to the active attack vectors discussed in Section 4.3. These methods leverage influence-based poisoning attacks to test the resilience of a system against privacy leaks. [229] proposes a comprehensive methodology that combines improved privacy search and verification methods with a toolkit of influence-based poisoning attacks. Unlike earlier works tailored to specific models or suffering from low statistical power, this framework offers a general approach to empirically evaluate privacy across various algorithms, including logistic regression, Naive Bayes, and random forests. It is particularly effective in detecting privacy violations caused by implementation errors or misuse, and when no violations are found, it aids in understanding the extent of information leakage given a specific dataset and privacy specification. Similarly, [230] extends these concepts to the inference phase, proposing the first framework for auditing private prediction algorithms. By instantiating adversaries with varying poisoning and query capabilities, this work evaluates the privacy leakage of algorithms such as PATE and Private-kNN, revealing that algorithms susceptible to poisoning often exhibit higher privacy leakage.

The landscape of auditing also includes black-box estimation techniques that do not require access to the internal training process, further broadening the scope of verification. [231] introduces a scheme that quantifies privacy leakage for large-scale models using generated data as “non-members.” This approach is particularly valuable for auditing models where the training process is opaque or retraining is impossible, requiring only access to a subset of the training data. Furthermore, [232] tackles the problem of measuring leakage in systems with large or continuous output spaces by exploiting an analogy between machine learning and leakage estimation. By utilizing universally consistent learning rules, such as nearest neighbor rules, F-BLEAU significantly reduces the number

of queries required for precise estimation compared to frequentist paradigms.

Despite these advancements, challenges remain in aligning empirical findings with theoretical metrics like Rényi Differential Privacy and ensuring the reliability of the audits themselves. [233] addresses this by presenting methods to assess privacy guarantees in black-box scenarios where only algorithmic outputs are available. The authors devise a new estimator for Rényi divergence that transforms into a statistical lower bound, applicable to a broad class of algorithms. Additionally, the reliability of these audits depends on the capability of the finite adversary employed. [234] investigates the conditions under which an unsuccessful audit by a finite adversary implies robustness against stronger adversaries, quantifying the requirements regarding network capacity and side information. Together, these methodologies form a robust toolkit for empirically validating privacy claims, bridging the gap between abstract theoretical guarantees and the tangible risks present in modern machine learning deployments. However, while these algorithmic auditing techniques provide crucial validation, they primarily operate under the assumption of idealized implementations. As we transition to the following section, it becomes evident that even rigorously audited systems can be compromised by the divergence between mathematical abstractions and the realities of software execution, hardware constraints, and data correlations.

4.5 Implementation Flaws, Side Channels, and Correlated Data Risks

4.5 Implementation Flaws, Side Channels, and Correlated Data Risks

While the empirical auditing techniques discussed in Section 4.4 provide crucial validation of theoretical privacy bounds, they primarily operate at the algorithmic level, often assuming idealized implementations. In practice, however, the translation of Differential Privacy (DP) mathematical abstractions into tangible software and hardware systems introduces a myriad of vulnerabilities that can render even rigorously audited guarantees vacuous. These implementation flaws create a significant divergence between promised privacy and actual security, stemming from three primary sources: finite-precision arithmetic, side-channel leaks, and the violation of independence assumptions in correlated data.

A fundamental category of vulnerability arises from the limitations of finite-precision arithmetic in digital computers. Theoretical DP mechanisms typically assume operations over real numbers with infinite precision; yet, practical implementations rely on floating-point representations inherently susceptible to precision-based attacks. Early work by Mironov highlighted how the discretization inherent in floating-point arithmetic could allow adversaries to invert the noise addition process, effectively recovering sensitive inputs. This vulnerability persists in modern systems, where attacks exploiting floating-point representation errors have demonstrated success rates as high as 92.56% against Differentially Private Stochastic Gradient Descent (DP-SGD) implementations [235]. Furthermore, these precision issues are not limited to continuous distributions; even discrete counterparts designed to mitigate floating-point risks can introduce new attack surfaces if not implemented with extreme care [236]. To address these fundamental flaws, researchers have proposed techniques such as interval refining and base-2 arithmetic implementations for the exponential mechanism, which aim to provide exact sampling and provable privacy within the constraints of finite computing architectures [237][236].

Complementing arithmetic vulnerabilities, side-channel attacks represent a critical threat vector that bypasses mathematical noise protection entirely by observing physical or temporal characteristics of system execution. Timing attacks, in particular, have emerged as a potent method for compromising DP systems. When noise generation algorithms, such as rejection sampling or inverse

transform sampling, exhibit data-dependent execution times, an adversary can infer the magnitude of the added noise or the underlying sensitive value by measuring the runtime. Studies have shown that naive implementations of discrete Laplace and Gaussian mechanisms are vulnerable to such timing side channels, with end-to-end attacks achieving success rates up to 99.65% in recovering sensitive attributes from private sums [235]. The risk extends to the sampling procedures themselves; unless the acceptance probability in rejection sampling is constant across all possible databases, the runtime inevitably leaks information, invalidating the ϵ -DP guarantee [238]. Mitigating these risks requires rigorous engineering, such as enforcing constant-time branching operations and utilizing specialized hardware analysis tools to detect microarchitectural timing vulnerabilities [239][240]. Additionally, power side-channel attacks pose similar threats in embedded and edge environments, where variations in power consumption during neural network inference can reveal model parameters or input data, necessitating countermeasures like masking and randomized clocking [241][242].

A third dimension of risk involves the erosion of privacy guarantees due to correlated data, particularly in continuous or streaming data release scenarios. Standard DP definitions often assume that data points are independent or that the adversary lacks knowledge of correlations between them. However, in real-world applications such as location tracking, health monitoring, or traffic flow analysis, data exhibits strong temporal correlations. When an adversary possesses knowledge of these correlations, often modeled via Markov chains, the privacy leakage can accumulate over time, leading to a phenomenon known as Temporal Privacy Leakage (TPL) [243][38]. In such settings, the privacy loss may increase unboundedly as more data points are released, rendering standard event-level DP protections insufficient. To counteract this, specialized mechanisms have been developed that explicitly account for temporal dependencies, converting existing DP mechanisms into ones robust against TPL by bounding the supremum of privacy loss over time [243]. Similarly, in longitudinal data collection under Local Differential Privacy (LDP), repeated queries can lead to a linear decay in privacy guarantees unless protocols exploit data sparsity or employ novel randomizers that correlate noise across time steps to break the linear dependence on the number of changes [244]. Furthermore, the assumption of independent user contributions in time-series data can be relaxed through subsampling and filtering strategies, which reduce the sensitivity of the output and allow for more efficient privacy mechanisms in aggregate time-series analysis [245].

The interplay between these implementation flaws and data correlations creates a complex adversarial landscape where theoretical bounds may become vacuous, posing unique challenges for the decentralized and local architectures discussed in the following section. For instance, floating-point errors can exacerbate the effects of temporal correlations by introducing predictable patterns in the noise that an adversary with prior knowledge of the data distribution can exploit. Moreover, the presence of correlated data can amplify the impact of side-channel leaks, as an attacker might combine timing information with temporal models to reconstruct entire trajectories of sensitive behavior. This is particularly pertinent in distributed settings where resource-constrained edge devices are more prone to side-channel leakage and where data correlations across a network graph can accelerate privacy degradation. Comprehensive evaluation frameworks are therefore essential to audit DP systems not just for their mathematical compliance, but for their resilience against these practical implementation failures and correlation-based attacks [129]. Addressing these challenges requires a holistic approach that integrates secure coding practices, constant-time algorithm design, and advanced privacy accounting methods capable of modeling complex data dependencies, ensuring that the promise of differential privacy is realized in the messy reality of deployed systems before transitioning to the specific architectural threats of decentralized networks.

4.6 Architecture-Specific Threats in Decentralized and Local Settings

4.6 Architecture-Specific Threats in Decentralized and Local Settings

While Section 4.5 exposed how implementation flaws and data correlations can erode privacy even in idealized theoretical models, the challenges intensify when Differential Privacy (DP) is deployed within specific architectural paradigms that lack a trusted central coordinator. The transition from centralized Federated Learning (FL) to decentralized and local settings fundamentally alters the threat model, shifting vulnerabilities from algorithmic precision and side channels to structural properties of network topologies and the granularity of data manipulation. In these architectures, privacy guarantees are no longer solely dependent on the magnitude of added noise but are critically contingent upon graph connectivity, peer-to-peer communication channels, and the absence of a trusted aggregator.

In Decentralized Federated Learning (DFL), the reliance on peer-to-peer communication over a graph topology—that privacy is not a global property enforced by a central mechanism but a local one dependent on the distribution of honest versus malicious nodes. Recent theoretical advancements highlight that privacy bounds in DFL are intrinsically linked to the underlying graph structure. Specifically, the minimum privacy loss at each aggregation step depends on the size of “honest components”—maximally connected subgraphs remaining after excluding untrustworthy participants [16]. This dependency implies that networks with weaker robustness, such as those following power-law distributions, inherently reveal more private information compared to more resilient topologies like Poisson networks. Furthermore, the topology itself can be exploited for reconstruction attacks; passive honest-but-curious adversaries can infer private data by observing sequences of privacy-preserving summations if the network contains short cycles. Research indicates that in subgraphs with as few as 18 users, a small coalition of adversaries can successfully reconstruct private data with non-negligible probability, provided the network topology allows for sufficient observational overlap [246]. Consequently, while acyclic networks offer theoretical immunity to such reconstruction, practical deployments often necessitate cyclic graphs for efficiency, thereby introducing a critical structural vulnerability that complements the implementation risks discussed previously.

The assumption that decentralization inherently provides superior privacy through obscurity has also been challenged, revealing that these architectures may paradoxically expand the attack surface. Contrary to early claims, rigorous analysis suggests that without a trusted server to mask individual contributions, every peer becomes a potential vantage point for an adversary, enabling gradient inversion attacks or even full control over a neighbor’s local model [247]. While concepts like “network differential privacy” attempt to formalize the privacy amplification gained from decentralization—suggesting that privacy leakage decays with the distance between nodes [248]—these gains are easily nullified by colluding attackers who occupy strategic positions in the graph. For instance, pairwise network differential privacy reveals that while random walk algorithms might offer better guarantees for nearby nodes compared to gossip protocols, the overall security remains fragile against coordinated adversaries who can isolate honest participants [249]. These structural weaknesses create a fertile ground for the domain-specific attacks explored in the subsequent section, particularly when decentralized nodes host sensitive applications.

Parallel to these topological threats, Local Differential Privacy (LDP) protocols face severe challenges regarding data integrity and robustness against fine-grained poisoning. In LDP, users perturb their data locally before transmission under the assumption that the server is untrusted. However, this architecture renders the system highly susceptible to data poisoning attacks where adversaries

inject carefully crafted, malformed data to skew global statistics. The very mechanism designed to protect privacy—local noise injection—can inadvertently amplify the impact of poisoning. Studies have shown that LDP protocols are more vulnerable to poisoning than their non-private counterparts because adversaries can directly manipulate the noisy responses rather than just the raw input [250]. This vulnerability extends to various analytical tasks, including frequency estimation, mean and variance calculation, and key-value data analytics. Attackers can formulate these manipulations as optimization problems to simultaneously maximize the frequency and mean values of target keys, effectively hijacking the output of the analytics pipeline [251].

The threat landscape in LDP is further complicated by the emergence of fine-grained attacks capable of manipulating specific statistical moments. The “Output Poisoning Attack” (OPA) demonstrates how adversaries can fine-tune bogus data in the output domain to distort both mean and variance estimates, exploiting the inherent characteristics of LDP mechanisms [252]. Moreover, the security of these protocols is not uniform; it depends heavily on design choices such as hash domain sizes in local-hashing-based schemes, which influence robustness beyond their known effects on utility [253]. Perhaps most insidiously, the presence of differential privacy noise can mask the signatures of malicious updates, making detection significantly harder. This phenomenon enables stealthy model poisoning attacks, such as Sybil attacks, where multiple fake clients manipulate noise levels to degrade model convergence or introduce backdoors without triggering standard anomaly detection systems [254]. In such scenarios, the privacy noise acts as a shield for the attacker, creating a paradox where stronger privacy guarantees can lead to weaker security postures against integrity attacks. This tension between privacy noise and model integrity sets the stage for the complex adversarial dynamics observed in domain-specific deployments, such as those involving Large Language Models, where the stakes of such manipulations are exceptionally high.

Ultimately, both decentralized and local architectures reveal that privacy and security are not monolithic guarantees but are deeply intertwined with system design. In decentralized settings, the graph topology dictates resilience against reconstruction and inference, requiring careful network design to prevent collusion from compromising honest components. In local settings, the lack of server-side validation opens the door to sophisticated poisoning strategies that exploit the probabilistic nature of local noise. Addressing these architecture-specific threats requires moving beyond generic privacy mechanisms to develop topology-aware defenses and robust aggregation protocols capable of distinguishing between legitimate privacy noise and malicious manipulation, a necessity that becomes even more critical when these architectures support the high-dimensional, semantically rich models discussed in the following section.

4.7 Domain-Specific Adversarial Landscapes and Large Language Models

4.7 Domain-Specific Adversarial Landscapes and Large Language Models

Building upon the architecture-specific vulnerabilities identified in decentralized and local settings, the integration of Federated Learning (FL) with Large Language Models (LLMs) introduces a distinct layer of complexity driven by domain-specific constraints and the unique semantic properties of language data. While previous sections highlighted how graph topology and local noise injection create attack surfaces, the convergence of FL and LLMs in high-stakes domains such as healthcare and cybersecurity expands the adversarial landscape further. Here, threats are not merely structural but are deeply embedded in the semantically rich parameter spaces and the critical nature of the applications, necessitating a rigorous re-examination of active membership inference threats and the practical efficacy of Differential Privacy (DP) defenses.

In the healthcare domain, the deployment of LLMs for clinical decision support and biomedical question answering has revealed profound brittleness when subjected to adversarial manipulation. Research indicates that named entities can serve as potent adversarial examples, where attackers utilize sophisticated sampling schemes to discover diverse adversarial entities that act as distractors [255]. These attacks transcend simple performance degradation; they can manipulate token-wise explanation methods, such as Shapley values, rendering them deceptive and undermining the trustworthiness of model predictions in life-critical scenarios. Furthermore, the challenge of ensuring safe natural language inference in clinical trials has led to the creation of specialized datasets designed to challenge LLMs with interventional and causal reasoning tasks, highlighting the vulnerability of these models to shortcut learning and factual inconsistency [256]. When such models are trained via federated protocols, the risk escalates as active membership inference attacks can theoretically guarantee success in determining whether specific patient records were part of the training set. Recent theoretical and practical analyses have demonstrated substantial privacy vulnerabilities in popular LLMs, including BERT and GPT variants, across real-world language datasets, even when state-of-the-art DP mechanisms are employed [257]. This suggests that standard DP configurations, often tuned for image classification, may provide vacuous guarantees when applied to the high-dimensional and semantically complex parameter spaces of medical LLMs.

Parallel challenges exist within the cybersecurity landscape, where LLMs are increasingly utilized for threat detection, code generation, and autonomous penetration testing. The dual-use nature of these models presents a paradox: while they enhance defensive capabilities, they also lower the barrier for adversaries to execute sophisticated attacks. Studies have shown that LLMs can be leveraged to automate post-breach exploitation, transforming rare, expert-led events into frequent, automated operations [258]. In a federated setting, where multiple organizations collaborate to train a robust cyber-threat detector, the system becomes susceptible to poisoning attacks that exploit the very noise introduced by DP mechanisms—a vulnerability echoing the integrity issues discussed in local settings. Specifically, adversaries can utilize adaptive model poisoning techniques to evade anomaly detection by manipulating the additional DP-noise, significantly reducing detection accuracy in critical infrastructure monitoring [223]. Moreover, the integration of LLMs into cybersecurity tools has exposed vulnerabilities where visual adversarial examples can induce models to misuse tools, leading to unauthorized actions such as leaking private conversations or deleting calendar events [259]. The complexity of these attacks is further compounded by the ability of LLMs to assist in generating attack code, as demonstrated by cases where models successfully broke established defenses like AI-Guardian with minimal human intervention [260].

The efficacy of Differential Privacy as a standalone defense in these domain-specific federated scenarios is increasingly being questioned. While DP provides theoretical bounds against membership inference, empirical audits reveal a significant gap between these guarantees and actual leakage in LLM contexts. The unique architecture of LLMs, characterized by their vast parameter counts and reliance on attention mechanisms, often renders standard gradient clipping and noise injection insufficient against determined adversaries equipped with auxiliary information. For instance, in federated large language models, active membership inference attacks have been designed with guaranteed theoretical success rates, exposing flaws in adapted FL protocols that fail to account for the specific statistical properties of language data [257]. Additionally, the trade-off between privacy and utility becomes more pronounced in specialized domains; excessive noise to ensure privacy can degrade the model’s ability to perform nuanced tasks such as pharmacovigilance reporting or psychological assessment, where precision is paramount [261][262].

Furthermore, the adversarial landscape is evolving to include attacks on the alignment and safety

guardrails of federated LLMs, extending the threat model beyond data privacy to behavioral integrity. Adversaries can exploit the collaborative nature of FL to inject backdoors during the instruction tuning phase, using gradient-guided triggers that maintain content integrity while compromising model outputs [30]. These backdoors can remain dormant until activated by specific prompts, posing a severe risk in domains like finance and critical infrastructure. The emergence of “jailbreak” attacks that bypass safety fine-tuning through disguise and reconstruction techniques further complicates the defense posture, achieving high success rates even on closed-source models [263]. Consequently, the community is moving towards holistic evaluation frameworks that integrate metrics for privacy leakage, fairness, and robustness, recognizing that no single mechanism like DP can address the multifaceted threats present in domain-specific federated LLM deployments [264]. Future research must therefore focus on developing adaptive privacy budgets, domain-aware clipping strategies, and hybrid defenses that combine cryptographic primitives with differential privacy to secure the next generation of collaborative AI systems in these critical sectors.

5 Addressing Data Heterogeneity, Personalization, Fairness, and Ethics

5.1 Architectural Strategies for Statistical Heterogeneity and Personalization

5.1 Architectural Strategies for Statistical Heterogeneity and Personalization

The pervasive challenge of statistical heterogeneity, characterized by non-independent and identically distributed (non-IID) data across decentralized clients, remains a primary obstacle to the widespread adoption of Federated Learning (FL). In real-world deployments, data distributions vary significantly due to user behavior differences, geographic locations, and device capabilities, often causing the global model trained via standard aggregation methods like FedAvg to converge sub-optimally or fail to generalize for specific users. To mitigate these issues, the research community has shifted from a “one-size-fits-all” paradigm toward sophisticated architectural strategies that emphasize personalization, clustering, and model decoupling. These approaches aim to balance the extraction of shared global knowledge with the retention of client-specific nuances, ensuring robust performance even in highly heterogeneous environments. Crucially, while these architectures enhance utility and adaptability, they also introduce new dimensions to the privacy landscape, as the mechanisms used to handle heterogeneity can interact complexly with privacy-preserving protocols—a tension that sets the stage for the privacy-fairness-utility trilemma discussed in the following section.

A dominant architectural strategy involves Clustered Federated Learning (CFL), which posits that clients can be grouped into clusters based on the similarity of their data distributions, allowing for the training of distinct global models for each cluster. This approach effectively decomposes a complex non-IID problem into several simpler IID sub-problems. Early implementations often relied on clustering clients based on model gradients or inference outputs; however, recent advances have refined these metrics to improve accuracy and efficiency. For instance, [265] introduces a two-stage decoupling algorithm that performs client clustering sequentially based on both inference outputs and model weights, utilizing Hopkins amended sampling to determine the optimal timing for clustering. Similarly, [266] leverages inference similarity to group clients with jointly trainable data distributions, capturing scenarios where different user groups possess distinct learning tasks. To address the computational cost of clustering during training, [267] proposes a method that identifies distribution similarities by analyzing principal angles between client data subspaces in a single-shot manner, significantly reducing the time required to stabilize cluster formation. Furthermore, [268] presents a flexible framework capable of handling arbitrary client participation rates and newly

joined clients, demonstrating promising clustering results even when the number of clusters is unknown.

Beyond simple clustering, Multi-Center Aggregation represents a significant architectural evolution where the system learns multiple global models simultaneously to serve as cluster centers. This method avoids the rigid assignment of clients to fixed groups by dynamically optimizing the matching between users and centers. [269] formulates this as a joint optimization problem solvable via a stochastic Expectation-Maximization (EM) algorithm, effectively capturing heterogeneity by assigning users’ gradients to different global models. Expanding on this, [270] integrates Graph Convolutional Networks (GCNs) with clustering to enable knowledge sharing across different domains, identifying domains via FL-compliant clustering and connecting domain-specific modules through a graph structure. In wireless network contexts, [271] explores the integration of clustered FL with non-orthogonal multiple access (NOMA), using spectral clustering to group users and optimizing sub-channel assignments to improve convergence rates under non-IID conditions.

Another critical architectural direction is the decoupling of model parameters to separate shared representations from client-specific layers. This strategy acknowledges that while lower layers of deep neural networks often learn general features applicable across clients, upper layers (such as classifiers) are highly sensitive to local data distributions. [272] argues for separating representation learning from classification learning, proposing a two-stage framework where a robust global representation is learned first, followed by the training of personalized classifier heads for each client. This concept is further refined in [273], which introduces a dynamic layer selection mechanism using Wasserstein distance to identify the optimal split point between cloud-based shared bodies and edge-based personalized heads, rather than rigidly assigning the last layer. Additionally, [274] employs hypernetworks to discern the importance of each layer from different clients, optimizing aggregation weights at a granular level to better exploit inter-user similarity.

Innovative approaches also explore parameter selection and structural personalization beyond simple layer splitting. [275] draws inspiration from the Lottery Ticket Hypothesis to incrementally expand subnetworks for personalization, allowing clients to personalize both parameters and sub-network structures concurrently. [276] integrates an attention mechanism to prioritize collaboration among clients with similar data distributions, effectively mitigating the negative impact of collaborating with dissimilar clients. Moreover, [277] proposes a paradigm where clients collaboratively train a generalized model while independently maintaining personalized and customized models, treating non-IIDness as a feature rather than a bug. For scenarios involving incomplete classes, [278] introduces “restricted softmax” and “inherited private models” to handle aggregation and personalization goals simultaneously. Finally, [279] emphasizes that architectural choices must be evaluated not just on average accuracy but also on fairness metrics, noting that the model with the highest average accuracy may not necessarily be the fairest across all users. These diverse architectural strategies collectively demonstrate that addressing statistical heterogeneity requires a move away from monolithic global models toward adaptive, modular, and cluster-aware frameworks. However, as these personalized and clustered models diverge from a unified global state, they create varying sensitivities to noise and disparate impacts across groups, directly influencing the delicate balance between privacy guarantees, fairness outcomes, and model utility explored next.

5.2 The Privacy-Fairness-Utility Trilemma: Theoretical Limits and Empirical Trade-offs

5.2 The Privacy-Fairness-Utility Trilemma: Theoretical Limits and Empirical Trade-offs

The architectural strategies for handling statistical heterogeneity discussed in the previous section, while enhancing model personalization and utility, inevitably introduce new vulnerabilities that complicate the preservation of privacy and fairness. As models diverge into cluster-specific or client-specific variants, they create varying sensitivities to noise and disparate impacts across user groups, giving rise to the complex triad known as the privacy-fairness-utility trilemma. While the trade-offs between privacy and utility, as well as fairness and utility, have been extensively studied in isolation, the mutual interactions among all three objectives remain less understood and frequently result in pernicious, undetectable biases when not explicitly managed [280]. A fundamental challenge arises because the mechanisms designed to ensure privacy, such as gradient clipping and the addition of random noise, do not affect all subgroups within a dataset uniformly. Specifically, underrepresented groups and those with complex data distributions—often the very groups targeted by personalized architectures—suffer disproportionate utility loss, leading to an exacerbation of existing disparities [125]. This phenomenon suggests that enforcing differential privacy can inadvertently amplify bias against unprivileged groups, creating a scenario where the pursuit of individual privacy compromises collective fairness [281].

Theoretical analyses have begun to map the boundaries of this trilemma, revealing inherent limitations in achieving simultaneous optimization. Research indicates that there is a fundamental tension where optimizing for one vector often penalizes another. For instance, while some claims suggest that predictive performance and fairness can be jointly optimized, empirical evidence demonstrates that this is frequently achievable only at the expense of data privacy [282]. Studies exploring this three-way tension have found that regardless of which objective is prioritized, one of the other vectors will inevitably be compromised, highlighting the necessity of impartial optimization frameworks that do not implicitly favor one goal over another [282]. To address this, recent work proposes the concept of “impartiality,” where machine learning pipelines are designed to avoid biasing any single objective, thereby allowing practitioners to recover accurate Pareto frontiers that visualize the true trade-offs between fairness, privacy, and utility [280].

A critical mechanism driving the degradation of fairness under differential privacy is the disparate impact of noise injection and gradient clipping. In standard differentially private stochastic gradient descent (DP-SGD), the clipping of gradients to bound sensitivity introduces statistical bias, while the subsequent addition of noise increases variance. This combination creates a “bias-variance-privacy” trilemma for statistical estimation, where no algorithm can simultaneously achieve low bias, low variance, and low privacy loss for arbitrary distributions [283]. The consequences of this are particularly severe for minority groups; the noise added to protect privacy can disproportionately impact the accuracy of decisions made for these groups, leading to significant outcome disparities in high-stakes applications such as resource allocation and voting rights benefits [284]. Furthermore, the convergence of gradients across different subgroups often varies, with DP mechanisms amplifying gender, racial, and religious biases during the fine-tuning of large language models (LLMs) due to these disparities in convergence rates [285].

However, the relationship between privacy and fairness is not universally antagonistic; it is highly context-dependent and influenced by the specific optimization strategies employed. Contrary to the prevailing understanding that differential privacy necessarily exacerbates unfairness, some em-

empirical analyses suggest that selecting an optimal model architecture specifically suited for the DP setting can mitigate these negative effects [286]. In certain scenarios, when the model architecture is optimized for DP-SGD, the differences in fairness metrics across groups may decrease or remain negligibly impacted compared to non-private baselines [286]. Additionally, the impact of DP on fairness can be reinterpreted through the lens of regularization. While DP tends to increase between-group performance differences in baseline settings, it has been observed to reduce these differences when combined with group distributionally robust training objectives, suggesting that the interaction is modifiable through careful algorithmic design [287].

The complexity of this trilemma is further compounded by the fact that fairness interventions themselves can introduce privacy risks. Efforts to minimize discrimination by imposing constraints on model behavior can alter the influence of training data points in a disproportionate manner, potentially increasing the information leakage about unprivileged subgroups [281]. This creates a paradoxical situation where the very groups intended to be protected by fairness constraints become more vulnerable to membership inference attacks. Consequently, the cost of achieving fairness is not distributed equally, and the privacy cost increases significantly for unprivileged subgroups, especially when the underlying training data is biased [281]. This bidirectional risk underscores the need for holistic evaluation frameworks that assess both privacy leakage and fairness metrics concurrently, rather than in isolation [288].

To navigate these challenges, researchers have developed various mitigation strategies and impartial optimization frameworks. One approach involves modifying the DP-SGD algorithm to adjust the contribution of samples based on group-specific clipping biases, effectively removing the disparate impact on protected groups [125]. Another strategy employs stratification, where independent private estimates are computed for each group and then recombined, offering a simple yet effective baseline for reducing utility disparities [128]. Furthermore, advanced frameworks like FairDP-SGD and FairPATE have been proposed to train impartially-specified models, enabling the discovery of Pareto-optimal solutions that balance the competing demands of the trilemma [280]. These methods often leverage Bayesian optimization to automatically discover trade-off points, providing practitioners with the tools necessary to make informed decisions about privacy budget allocation and fairness constraints [289; 290]. Despite these advances, the field continues to grapple with the theoretical limits of the trilemma, with ongoing research focused on developing algorithms that can converge stochastically while maintaining rigorous guarantees for both privacy and fairness across diverse and heterogeneous data distributions [291]. These theoretical insights and empirical trade-offs set the stage for the practical solutions discussed in the following section, where cryptographic protocols and differential privacy mechanisms are integrated to enforce fair aggregation without exposing sensitive demographic data.

5.3 Cryptographic and Differential Privacy Approaches to Fair Aggregation without Sensitive Data

5.3 Cryptographic and Differential Privacy Approaches to Fair Aggregation without Sensitive Data

Building on the theoretical limits and empirical trade-offs of the privacy-fairness-utility trilemma discussed in the previous section, a central practical challenge emerges: how to enforce group fairness constraints when the sensitive demographic attributes required for such auditing cannot be centrally accessed. In Federated Learning (FL), this creates a profound paradox: achieving group fairness typically necessitates knowledge of sensitive attributes (such as race, gender, or age)

to measure and mitigate disparate impacts, yet the core tenet of FL is to prevent exactly such data from leaving local client devices. Traditional approaches often force a binary choice between exposing sensitive attributes to a central aggregator to enforce fairness constraints or maintaining strict privacy at the cost of potential bias amplification. However, emerging hybrid frameworks demonstrate that this conflict is not insurmountable. By synergistically integrating Secure Multi-Party Computation (SMPC) and Differential Privacy (DP), it is possible to enforce group fairness constraints and compute critical importance weights without ever requiring clients to disclose their raw sensitive attributes in plaintext.

The foundational challenge lies in the necessity of knowing group membership to calculate fairness metrics like demographic parity or equalized odds. In a standard FL setting, if the server cannot see the sensitive attributes, it cannot verify if the global model is discriminating against specific subgroups. Recent advancements propose resolving this by leveraging SMPC to perform secure aggregation of statistics related to sensitive attributes while simultaneously employing DP to mask individual contributions within those aggregates. For instance, the [292] framework explicitly addresses this dichotomy by combining FL with SMPC and DP. This approach allows for the training of group-fair machine learning models in cross-device settings under complete and formal privacy guarantees. Crucially, [292] demonstrates that clients do not need to disclose their sensitive attribute values; instead, the system utilizes secure protocols to compute the necessary fairness adjustments collaboratively, ensuring that the server learns nothing about individual group memberships while still optimizing for global fairness objectives.

Beyond simple aggregation, the computation of importance weights for fairness-aware loss functions often requires complex interactions between parties holding vertically partitioned data or distinct feature sets. In scenarios where data is vertically partitioned, parties hold different features for the same set of individuals, making the identification of sensitive groups even more delicate. The [293] study introduces HDP-VFL, a framework that jointly learns generalized linear models from vertically partitioned data with negligible cost compared to non-private settings. While primarily focused on vertical partitioning, the underlying mechanism of combining cryptographic security with differential privacy offers a blueprint for fair aggregation. By analyzing how intermediate results can leak private information, such frameworks design DP-based algorithms that ensure data confidentiality while allowing for the secure computation of joint statistics necessary for fairness auditing. Similarly, [294] proposes a framework based on the functional mechanism, where the objective function is dissected into single-party and cross-party sub-functions. This allows local parties to compute and perturb polynomial coefficients securely, enabling the server to release a model that satisfies fairness criteria without ever reconstructing the sensitive input data.

The integration of DP into these cryptographic protocols serves a dual purpose: it provides a rigorous mathematical bound on privacy leakage that complements the computational security of SMPC, and it mitigates the risk of inference attacks that might arise from the output of the secure computation itself. This is particularly vital given the vulnerability disparities highlighted in the context of the trilemma, where aggregate statistics could otherwise allow an adversary with auxiliary knowledge to re-identify individuals within small or unique demographic groups. The [295] paper proposes a Secure Multiparty Differentially Private (SM-DDP) protocol that simultaneously achieves SMC and DP. This protocol utilizes homomorphic encryption for secure computation and the functional mechanism for DP, ensuring that parties cannot infer information about individuals even from the final constructed statistical model.

Furthermore, as established in the analysis of the trilemma, the tension between fairness and privacy is not merely technical but also statistical; enforcing privacy via noise injection can dis-

proportionately degrade utility for underrepresented groups, thereby exacerbating existing biases. To counteract this, hybrid approaches must incorporate adaptive mechanisms that account for group-specific sensitivities. The [125] work highlights that gradient clipping and noise addition in standard DPSGD disproportionately affect underrepresented classes. While this paper focuses on algorithmic modifications like DPSGD-F, its insights are critical for hybrid systems: any cryptographic protocol enforcing fairness must internally utilize adaptive clipping or weighting schemes that are themselves differentially private. Extending this, [296] utilizes Lagrangian duality to design neural networks that accommodate fairness constraints while guaranteeing the privacy of sensitive attributes. This method effectively embeds the fairness constraint into the optimization landscape in a way that respects DP bounds, offering a theoretical foundation for hybrid systems that must balance these competing objectives.

The complexity increases when considering that different stakeholders may have heterogeneous privacy requirements or trust levels, a factor that influences the fairness of the resulting model. The [297] framework addresses this by allowing each party to have a personalized privacy budget within a multi-party computation setting. This is essential for fair aggregation in real-world deployments where different demographic groups or data holders may demand varying levels of protection. By casting the utility loss minimization as a linear programming problem, TPMDP optimizes the noise injection process to maintain high utility for fairness metrics while respecting individual privacy thresholds. Additionally, the concept of “spending privacy budget fairly” is explored in [127], which argues that uniform budget allocation across features can undervalue critical conditional probabilities for minority groups. Hybrid frameworks must therefore adopt intelligent budget allocation strategies, potentially informed by secure multiparty computations of feature importance, to ensure that the privacy mechanism itself does not become a source of inequity.

Finally, the verification of fairness without compromising privacy remains a significant hurdle, bridging the gap between technical implementation and the ethical risks of data exposure. Querying for fairness metrics can inadvertently leak protected attributes if not carefully sanitized, a risk that foreshadows the ethical dilemmas of demographic data collection discussed in the following section. The [298] paper demonstrates that simply querying for metrics like statistical parity can allow model developers to reconstruct protected attributes. To prevent this, hybrid systems must employ techniques like Attribute-Conceal, which calibrates noise to the smooth sensitivity of bias queries. This ensures that the feedback loop used to enforce fairness—where the server adjusts model weights based on fairness metrics—does not become a side-channel for privacy violations. By integrating such smooth sensitivity mechanisms with SMPC protocols, systems can provide verifiable assurances of fairness to regulators and auditors without exposing the underlying sensitive data, thus closing the loop on trustworthy, private, and equitable machine learning while mitigating the ethical hazards of explicit demographic tracking.

5.4 Ethical Risks in Demographic Data Collection and Vulnerability Disparities

5.4 Ethical Risks in Demographic Data Collection and Vulnerability Disparities

While Section 5.3 demonstrated how cryptographic and differential privacy techniques can technically reconcile the conflict between fairness auditing and data confidentiality, a deeper ethical paradox remains: the very pursuit of algorithmic fairness often necessitates the collection and processing of sensitive demographic attributes, thereby introducing significant privacy risks and potential for harm. Traditional regulatory frameworks have frequently advocated for “fairness through unawareness,” arguing that removing sensitive attributes prevents discrimination; however, this approach is

increasingly recognized as obsolete in the era of big data, where highly correlated attributes allow for the accurate inference of protected categories [299]. Consequently, practitioners face intense pressure to collect explicit demographic data to audit and mitigate bias, yet this expansion of data collection infrastructure raises critical ethical concerns regarding surveillance, miscategorization, and the disparate vulnerability of marginalized groups to inference attacks—risks that technical safeguards alone cannot fully resolve.

A primary ethical risk associated with the aggregation of sensitive attributes is the expansion of surveillance infrastructures under the guise of social good. The imperative to measure fairness can inadvertently legitimize the collection of intimate personal data by private entities, leading to a normalization of demographic tracking that extends far beyond the original scope of fairness auditing. [300] argues that technical solutions often ignore broader questions of data governance and systemic oppression, warning that the drive to collect more data on demographics may reinforce systems of oppression rather than dismantle them. This concern is exacerbated by the inconsistent legal and corporate landscape governing such data; while some sectors are mandated to collect sensitive data for antidiscrimination purposes, others are prohibited from doing so, creating a fragmented environment where the definition of appropriate data use remains contested [301]. Furthermore, practitioners often face a dilemma where they lack the necessary data to detect bias, yet the collection of such data raises normative questions about consent, the definition of social categories, and the appropriateness of private companies inferring demographics [302].

Beyond the macro-level risks of surveillance, the process of collecting and categorizing demographic data introduces significant harms related to miscategorization and misrepresentation. Individuals are often forced into rigid, binary, or oversimplified demographic buckets that fail to capture the complexity of their identities, particularly for those with intersectional backgrounds. [303] demonstrates that focusing on single sensitive attributes can mask severe unfairness at the intersection of multiple categories, such as gender, age, and parental status, leading to imbalanced harm that standard fairness metrics overlook. When organizations attempt to fill gaps in demographic data through imputation or proxy variables, they risk cementing stereotypes and mischaracterizing what it means to belong to a specific group [300]. The reliance on proxies, such as geolocation or surname, to infer race or ethnicity has been shown to produce unidentifiable disparity measures and can introduce new biases if the proxy models themselves are flawed [304]. Moreover, the act of categorization itself can be harmful if it cedes the power to define identity away from the communities being analyzed, potentially leading to offensive labeling or the reinforcement of stigmatizing narratives [305].

Perhaps the most insidious ethical challenge lies in the unequal vulnerability of specific demographic groups to privacy attacks when sensitive data is utilized for fairness interventions. There is a growing body of evidence suggesting that the mechanisms designed to protect or ensure fairness for unprivileged groups can inadvertently increase their exposure to privacy breaches. [281] reveals that imposing group fairness constraints, such as equalized odds, can disproportionately increase the information leakage about training data for unprivileged subgroups via membership inference attacks. This creates a tragic trade-off where the populations most in need of fair treatment become the most vulnerable to privacy invasions. Similarly, [306] formalizes the concept of “vulnerability disparity,” demonstrating that membership inference attacks do not treat all demographic groups equally and that standard differential privacy defenses, while helpful, cannot completely eliminate these disparities without sacrificing model utility.

The risk is further compounded by the fact that fairness auditing tools themselves can be weaponized or manipulated to expose sensitive attributes. [298] demonstrates that even querying a

compliance team for aggregate fairness metrics can allow malicious model developers to reconstruct the protected attributes of individuals in the test dataset, effectively bypassing privacy safeguards. Additionally, [307] highlights that in-processing fairness algorithms often fail to ensure indistinguishable predictions across different values of sensitive attributes, leaving models vulnerable to attribute inference attacks that can exploit class imbalances to identify minority groups with high accuracy. The situation is dire for groups that are already marginalized; [308] notes that while some fairness methods degrade standard attack performance, novel attack strategies leveraging fairness discrepancies can specifically target these enhanced models, exposing the very groups the fairness constraints were meant to protect.

Ultimately, the ethical landscape of demographic data collection for fairness is fraught with tensions between the necessity of visibility for antidiscrimination and the imperative of privacy preservation. The assumption that more data inevitably leads to fairer outcomes is challenged by findings that data collection can expand surveillance, enforce harmful categorizations, and create asymmetric privacy risks. As noted in [309], addressing these issues requires not only technical fixes but a diversification of the computing community itself to better understand the socio-technical implications of these design choices. Without a rigorous framework that accounts for these ethical risks, the pursuit of algorithmic fairness may inadvertently perpetuate the very harms it seeks to eliminate, leaving vulnerable populations exposed to both discriminatory outcomes and privacy violations. These structural and ethical vulnerabilities set the stage for the subsequent discussion in Section 5.5, which explores how to ensure equitable participation and generalization for “silent” clients who, due to resource constraints or privacy concerns, may be unable or unwilling to contribute data, thereby preventing the systemic exclusion hinted at by these vulnerability disparities.

5.5 Equitable Participation, Incentive Mechanisms, and Generalization for Silent Clients

5.5 Equitable Participation, Incentive Mechanisms, and Generalization for Silent Clients

While Section 5.4 highlighted the ethical paradoxes inherent in collecting sensitive demographic data—where the pursuit of fairness can inadvertently exacerbate privacy risks and surveillance—this section addresses a complementary challenge: ensuring that the structural mechanics of Federated Learning (FL) systems do not themselves become sources of inequity. The realization of robust and fair FL systems hinges critically on overcoming the dual challenges of ensuring equitable client participation and maintaining model generalization for non-participating or “silent” clients. In realistic deployment scenarios, the assumption that all edge devices possess an intrinsic, undying desire to participate in training rounds is fundamentally flawed. Many devices face severe resource constraints, including limited battery life, intermittent connectivity, or insufficient computational power, rendering them unable to contribute consistently. Furthermore, rational clients often act as self-interested agents who weigh the costs of local computation and communication against the potential benefits derived from the global model. Without carefully designed incentive mechanisms, FL systems are prone to the “free-rider” dilemma, where clients consume the benefits of the aggregated model without contributing their data, ultimately leading to biased global models and degraded performance for the entire coalition. This dynamic risks creating a new form of disparity where only resource-rich participants shape the model, while marginalized or constrained devices are left with suboptimal outcomes, echoing the vulnerability disparities discussed previously.

To address these issues, recent research has shifted towards game-theoretic and economic frame-

works that align individual client incentives with global learning objectives. A pivotal advancement in this domain is the development of mechanisms that realistically model device utility to provoke genuine participation. For instance, the RealFM mechanism represents a significant departure from traditional approaches by modeling a non-linear relationship between model accuracy and device utility. This approach not only incentivizes data contribution and active device participation but also provably eliminates the free-rider dilemma while relaxing stringent assumptions regarding data homogeneity and monetary rewards [310]. By acknowledging that devices have varying valuations for the global model based on their local data quality and resource availability, such mechanisms ensure that participation is driven by mutual benefit rather than coercion, thereby fostering a more inclusive ecosystem that mitigates the risk of systemic exclusion.

Beyond binary participation decisions, the concept of randomized client participation has emerged as a sophisticated strategy to mitigate bias arising from system heterogeneity. Traditional incentive schemes often stimulate a fixed subset of high-resource clients, which can lead to a globally biased model if the excluded clients possess unique data distributions. To counteract this, game-theoretic incentive mechanisms have been proposed that employ customized pricing strategies to motivate different clients to join with specific participation probabilities. This approach allows the server to obtain an unbiased and high-performance model by analytically predicting how arbitrary participation levels and heterogeneous data affect convergence, revealing the possibility of bidirectional payments where clients may even pay for access to a superior global model [311]. Similarly, the integration of network pricing with participation incentives offers a holistic solution for managing communication overhead. By jointly designing participation incentives and dynamic network pricing, operators can align the interests of the server, users, and network providers, significantly reducing server costs while ensuring that users with varying private information are adequately compensated for their contributions [312].

The challenge of equity extends beyond active participants to include “silent” clients—those that are resource-constrained or choose not to train due to high local costs. Ensuring that these silent clients still benefit from a generalized model is essential for the ethical deployment of FL, particularly to prevent the amplification of inequities against those unable to participate actively. One promising direction involves dynamic personalization frameworks that allow clients to adaptively choose between training a personalized local model or relying on the global model. The DyPFL framework, for example, introduces a personalization rate that dynamically adjusts based on client willingness and performance, incentivizing participation in personalization while safeguarding the utility for those who opt for the global model [313]. This flexibility ensures that resource-constrained devices are not penalized for their inability to perform extensive local training, thereby promoting a more inclusive learning ecosystem that respects the limitations of vulnerable users.

Furthermore, achieving generalization for silent clients often requires robust aggregation strategies that can handle biased updates from a non-representative subset of participants. Methods that leverage client clustering and multi-center aggregation have shown efficacy in assigning silent or straggling clients to appropriate model centers based on their statistical similarity to active clusters. The Multi-Center Federated Learning approach, which clusters clients using model parameters to learn multiple global models, demonstrates that such strategies can effectively capture heterogeneity and improve personalization even when full participation is unattainable [269]. Additionally, contribution-based device selection schemes that balance exploration and exploitation can optimize the selection of active clients to maximize generalization for the entire population, including those currently silent [314].

The interplay between fairness and incentives is further complicated by the need to ensure equitable

outcomes across diverse demographic groups, linking back to the concerns of representation raised in the previous section. Mechanisms that incorporate fairness constraints into the incentive design, such as those utilizing artificial currencies to steer resource allocation towards system-optimal and equitable outcomes, are crucial for preventing the marginalization of minority groups [315]. Moreover, the emergence of tokenized incentive schemes backed by blockchain technology, such as FedToken, offers a transparent and trusted method for allocating rewards based on accurate data valuation, ensuring that even clients with smaller but high-quality datasets receive fair compensation [316]. These economic tools provide a structural safeguard against the kinds of disparate impacts identified in demographic data collection.

Ultimately, the path toward equitable participation and robust generalization requires a synthesis of economic incentives, adaptive personalization, and fairness-aware aggregation. By moving beyond the assumption of uniform cooperation and embracing the heterogeneous realities of edge devices, future FL systems can achieve a balance where both active contributors and silent beneficiaries thrive. As demonstrated by frameworks that utilize reinforcement learning to decipher client contribution dynamics, the integration of advanced algorithmic strategies with sound economic principles is paramount for building sustainable and trustworthy federated ecosystems [317]. These advancements collectively ensure that the promises of privacy-preserving machine learning are realized equitably across the entire spectrum of participating and non-participating entities, bridging the gap between technical efficiency and social justice.

6 System Efficiency, Scalability, and Resource-Constrained Deployments

6.1 Over-the-Air Computation for Low-Latency Aggregation

6.1 Over-the-Air Computation for Low-Latency Aggregation

While Federated Learning (FL) effectively addresses data privacy by decentralizing model training, its practical deployment in large-scale Internet of Things (IoT) ecosystems faces severe bottlenecks related to communication latency and massive connectivity. Traditional communication paradigms, which strictly adhere to a “compute-after-communicate” principle, require devices to wait for orthogonal time or frequency slots to transmit updates individually. Consequently, aggregation latency scales linearly with the number of participating nodes, rendering conventional orthogonal multiple access schemes inefficient for scenarios involving thousands of edge devices. To surmount these limitations, Over-the-Air Computation (AirComp) has emerged as a transformative technology that fundamentally shifts the operational paradigm from “compute-after-communicate” to “compute-when-communicate” [318]. By seamlessly integrating communication and computation procedures, AirComp exploits the inherent waveform superposition property of multiple-access channels (MAC) to perform function computation directly over the air, thereby enabling simultaneous data aggregation from massive devices with significantly reduced latency [319].

The core mathematical foundation of AirComp lies in the physical layer’s ability to naturally sum analog signals transmitted concurrently by multiple devices. Unlike traditional digital communication systems that treat signal interference as noise to be eliminated, AirComp leverages this interference as a computational resource. When multiple devices transmit their data signals simultaneously using analog amplitude modulation, the superposition of electromagnetic waves at the receiver naturally yields the sum (or a weighted average) of the transmitted values [319]. This capability allows the wireless channel itself to act as a distributed computer, drastically reducing the time required for wireless data aggregation (WDA), a critical operation for applications ranging from distributed sensing and control to federated learning [319]. Theoretical analyses have demon-

strated that under optimal transmission policies, the computation mean-squared error (MSE) of AirComp systems decays at a rate of $O(1/\sqrt{K})$ as the number of sensors K increases, highlighting the scalability and energy efficiency of this approach for large-scale networks [320].

Despite its theoretical promise, the practical deployment of AirComp faces significant challenges, primarily stemming from unfavorable wireless propagation conditions. The performance of AirComp is often bottlenecked by the device with the worst channel condition, as deep fading can severely distort the aggregated signal. To mitigate this, recent research has focused on advanced transceiver designs and the integration of reconfigurable intelligent surfaces (RIS). For instance, RIS-assisted AirComp systems have been proposed to reconfigure the propagation environment by smartly tuning phase shifts, thereby boosting received signal power and mitigating the performance bottlenecks caused by poor channel links [321]. Furthermore, novel alternating optimization frameworks have been developed to jointly design AirComp transceivers and RIS phase shifts, achieving near-optimal distortion performance with significantly reduced computational complexity [322].

While early AirComp implementations relied heavily on analog modulation, the limitations of analog transmission regarding noise resilience and compatibility with existing digital infrastructure have spurred the development of digital AirComp techniques. A prevailing belief that digital modulations are unfeasible for over-the-air computation due to meaningless signal overlapping has been challenged by innovative approaches such as ChannelComp. This method enables the use of digital modulations for computing arbitrary functions over-the-air, offering superior performance compared to analog counterparts, particularly for product functions, with improvements exceeding 10 dB in computation error [323]. Additionally, digital AirComp frameworks incorporating bit-slicing techniques have been proposed to allocate bits of data values to multiple symbols, thereby enhancing reliability and meeting the stringent requirements of 6G networks [324]. Prototype implementations using orthogonal frequency-division multiplexing (OFDM) have further validated that digital AirComp can overcome phase asynchrony issues that plague analog systems, ensuring accurate model aggregation even in practical multi-user environments [325].

The application of AirComp extends beyond simple averaging to complex distributed optimization and federated learning tasks. In the context of Federated Edge Learning (FEEL), AirComp facilitates the low-latency aggregation of model updates, effectively accelerating the training process. Studies on “Over-the-Air FedAvg” have shown that by jointly optimizing transmission power control and denoising factors, AirComp-based federated averaging can achieve an order-of-magnitude reduction in training latency compared to conventional digital orthogonal multiple access schemes [326]. Moreover, to address the straggler problem inherent in synchronous aggregation, semi-asynchronous mechanisms leveraging AirComp have been introduced, which adjust uplink transmit power based on model staleness to minimize convergence bounds while maintaining high efficiency [327].

Looking toward future network architectures, AirComp is being integrated with mobile platforms such as Unmanned Aerial Vehicles (UAVs) to further enhance performance. UAV-aided AirComp systems utilize the mobility of flying base stations to establish line-of-sight links and adaptively adjust distances, thereby minimizing time-averaged MSE through joint trajectory and power optimization [328]. Additionally, the convergence of AirComp with Integrated Sensing and Communication (ISAC) has given rise to Over-the-Air Integrated Sensing, Communication, and Computation (Air-ISCC), a holistic framework designed to improve spectrum efficiency and reduce latency by enabling simultaneous sensing, communication, and function computation [329]. As research progresses, the focus is shifting towards task-oriented designs where beamforming and precoding are optimized not just for signal fidelity, but directly for the accuracy of the downstream AI inference task, marking a definitive move toward task-centric network architectures [330]. While AirComp

effectively mitigates latency and connectivity bottlenecks through physical layer innovation, another class of challenges arises from computational heterogeneity and node failure; addressing these requires complementary strategies such as Coded Distributed Computing, which is discussed in the following section. Through these advancements, AirComp stands as a pivotal technology for realizing the low-latency, massive-connectivity vision of next-generation intelligent networks.

6.2 Coded Distributed Computing and Straggler Mitigation

6.2 Coded Distributed Computing and Straggler Mitigation

While Over-the-Air Computation (AirComp), discussed in the previous section, effectively addresses physical-layer latency and massive connectivity bottlenecks by leveraging waveform superposition, it does not fully resolve the challenges posed by computational heterogeneity and node failures at the application layer. In large-scale distributed machine learning and edge computing, system performance is frequently bottlenecked not by the average speed of computational nodes, but by the slowest participants, known as stragglers. These straggling nodes, which may suffer from hardware heterogeneity, network congestion, or intermittent connectivity, force synchronous systems to wait, thereby drastically increasing overall latency and reducing resource utilization. To address this critical challenge, Coded Distributed Computing (CDC) has emerged as a transformative paradigm that leverages coding theoretic techniques to introduce algorithmic redundancy, allowing systems to recover results from a subset of faster workers without waiting for the slowest ones. The fundamental premise of CDC is to treat slow or failed nodes as erasures in a communication channel, enabling the reconstruction of the final computation through error-correcting codes once a sufficient number of non-straggling nodes have completed their tasks [331].

The application of CDC extends beyond simple redundancy; it offers a sophisticated trade-off between computation load and communication overhead. By increasing the computation load during the mapping phase, CDC creates coded multicasting opportunities that can significantly reduce the communication load during the shuffle phase, a benefit particularly vital in bandwidth-constrained edge environments [332]. Early approaches often relied on Maximum Distance Separable (MDS) codes, such as Reed-Solomon codes, to achieve optimal straggler resilience. However, these dense linear combinations can be computationally prohibitive when dealing with sparse input matrices common in machine learning, as they destroy the inherent sparsity of the data and increase worker computation times [333]. Furthermore, standard MDS codes typically treat stragglers as complete erasures, ignoring any partial work completed by slow nodes before they lag behind, which represents a significant loss of computational potential in heterogeneous clusters [334].

To overcome the limitations of ignoring partial work, recent advancements have focused on hierarchical and approximate coding schemes. Hierarchical coded computation decomposes tasks into layers of subtasks with varying complexities, allowing the system to exploit the work completed by stragglers rather than discarding it entirely. This approach recognizes that the distinction between fast and slow nodes is often soft, and by designing codes that can integrate partial results, systems can achieve substantial reductions in expected finishing time [335]. Similarly, approximate gradient coding sacrifices a negligible amount of accuracy to gain significant robustness, utilizing sparse random graphs to ensure fast and approximately accurate distributed computation even in the presence of severe straggling [336]. These methods are particularly effective for iterative algorithms like gradient descent, where slight inaccuracies in intermediate steps can be tolerated without compromising final model convergence [337].

Complementing these reactive coding strategies are proactive predictive mechanisms designed to anticipate and mitigate straggler behavior before it impacts system performance. Traditional reactive models often incur delays by detecting stragglers only after they have slowed down the system. In contrast, modern approaches utilize deep learning techniques, such as Encoder Long-Short-Term-Memory (LSTM) networks, to analyze historical resource consumption patterns and predict which tasks are likely to become stragglers. By dynamically adapting scheduling based on these predictions, systems can proactively reassign tasks or adjust resource allocation, thereby reducing Service Level Agreement (SLA) violations and execution time [338]. Such predictive capabilities are essential for managing the volatility of task characteristics and host heterogeneity in cloud and edge environments.

The integration of CDC with Federated Learning (FL) introduces additional complexities due to the decentralized nature of data and the unreliable communication links between edge devices and central servers. In hierarchical FL setups, where intermediate helper nodes aggregate updates from edge devices, coding techniques must be applied to both the computation and communication phases to handle link failures and straggling devices. Schemes utilizing layered MDS codes and vector codes have been proposed to balance communication costs between edge nodes and helper nodes, ensuring graceful degradation in performance as link reliability fluctuates [339]. Moreover, in wireless edge networks where channel dynamics are unpredictable, joint optimization of coding structures and node scheduling via deep reinforcement learning has shown promise in minimizing learning delays by selecting the optimal set of edge nodes and coding schemes simultaneously [340].

Despite these advancements, challenges remain in adapting CDC to highly heterogeneous and dynamic environments. The assumption of homogeneous worker capabilities, prevalent in many theoretical CDC models, often fails to reflect real-world deployments where devices vary widely in computational power and energy constraints. Optimal load allocation strategies that account for group heterogeneity are necessary to prevent significant latency losses caused by uniform load distribution [341]. Additionally, the computational overhead of encoding and decoding must be carefully managed, especially for resource-constrained IoT devices. Lightweight designs, such as those based on combinatorial structures or low-complexity peeling decoders, are critical for making CDC viable in practice [342]. Future research directions point towards adaptive frameworks that can dynamically switch between different coding strategies based on real-time network conditions and straggler patterns, ensuring robust and efficient distributed learning across diverse and evolving edge infrastructures [343]. While CDC addresses straggler mitigation through algorithmic redundancy and error correction, it often operates within frameworks that still rely on defined synchronization points or assume specific completion thresholds. To further dismantle these synchronization barriers and enhance fluidity in distributed training, the following section explores Asynchronous Protocols and Communication-Computation Overlap, which offer a fundamentally different architectural approach to overcoming latency bottlenecks.

6.3 Asynchronous Protocols and Communication-Computation Overlap

6.3 Asynchronous Protocols and Communication-Computation Overlap

While Coded Distributed Computing (CDC), as discussed in the previous section, addresses straggler mitigation through algorithmic redundancy and error correction, it often operates within frameworks that still rely on defined synchronization points or assume specific completion thresholds. In contrast, Asynchronous Protocols offer a fundamentally different architectural approach to overcoming the latency bottlenecks inherent in large-scale distributed machine learning, particularly

within Federated Learning (FL) ecosystems deployed across heterogeneous edge devices. The synchronization barrier inherent in Bulk Synchronous Parallel (BSP) protocols frequently becomes the primary performance limiter, as traditional synchronous approaches require all participating clients to complete their local training iterations before the central server can aggregate updates. This forces faster nodes to remain idle while waiting for stragglers, drastically reducing resource utilization. To mitigate this inefficiency, asynchronous protocols have emerged as a critical paradigm, allowing devices to contribute model updates immediately upon completion of local computation without waiting for global synchronization. This shift not only alleviates the straggler problem more dynamically than reactive coding schemes but also enables a more continuous utilization of network and computational resources. However, the transition from synchronous to asynchronous updates introduces significant challenges regarding gradient staleness, convergence stability, and the complex interplay between communication latency and computation time.

The fundamental advantage of asynchronous mechanisms lies in their ability to decouple the update cycles of individual clients, creating a fluid data flow that contrasts with the rigid phases of coded or synchronous systems. In a fully asynchronous setting, the server aggregates incoming gradients or model parameters as soon as they arrive, maintaining a constantly evolving global model. Research has demonstrated that such approaches can significantly accelerate the learning process, especially in environments characterized by diverse computational capabilities and unreliable network conditions. For instance, the work on [344] highlights that while asynchronous settings effectively solve the straggler issue, they introduce the “stale model” problem, where slow clients utilize outdated global models for local training, potentially impeding convergence. To address this, effective aggregation strategies and client scheduling mechanisms are proposed to ensure that the global model converges with accuracy comparable to synchronous methods while achieving substantial speedups in the early stages of training. Similarly, [345] proposes a paradigm that eliminates waiting time completely by utilizing buffered results on the server, theoretically proving convergence and demonstrating speedups of up to 6x compared to synchronous strategies in both IID and Non-IID data scenarios.

Beyond merely decoupling update cycles, achieving full overlap of communication and computation is another pivotal strategy for enhancing system efficiency in asynchronous frameworks. In synchronous systems, communication and computation are often serialized; devices compute local updates, stop, communicate with the server, and then wait for the next round. Asynchronous protocols, conversely, allow these phases to proceed concurrently, effectively hiding communication latency behind ongoing local computations. By employing techniques such as one-sided Remote Memory Access (RMA) and non-blocking communication primitives, systems can maximize hardware utility. The study [346] illustrates how dedicated asynchronous progression engines can drive non-blocking communication operations, allowing computation to overlap efficiently with data transmission, thereby substantially reducing the total communication cost in realistic use cases. Furthermore, [347] emphasizes that integrating GPU-aware communication into asynchronous tasks enables automatic computation-communication overlap, which is essential for scaling performance on modern accelerated platforms. This approach minimizes the fine-grained overheads associated with host-device synchronization and leverages kernel fusion to maximize GPU utilization, offering a complementary efficiency gain to the load-balancing benefits of CDC.

Despite these benefits, the lack of strict synchronization introduces the challenge of staleness, where the gradient information received by the server may be based on an obsolete version of the global model. If not managed correctly, this can lead to divergence or oscillation in the optimization trajectory, a risk distinct from the erasure problems addressed by coding theory. Various the-

oretical and algorithmic solutions have been developed to bound the impact of staleness. [348] revisits asynchronous algorithms like FedBuff, extending theoretical analysis to remove boundedness assumptions on gradient norms, thereby providing robust convergence guarantees even under significant heterogeneity in data and delay. Additionally, [349] identifies that naively applying momentum in asynchronous settings introduces implicit bias due to staleness. The authors propose a momentum approximation technique that finds an optimal weighted average of historical updates to minimize this bias, compatible with secure aggregation and differential privacy, achieving notable speedups in convergence.

To further balance the trade-off between the robustness of synchronous methods and the speed of asynchronous ones, semi-asynchronous or bounded staleness approaches have gained traction. These hybrid models allow a degree of asynchrony but impose limits on how “old” an update can be before it is discarded or re-computed, effectively creating a soft synchronization barrier. [350] presents an adaptive strategy where the number of workers the parameter server waits for is dynamically selected to strike an optimal balance between straggling and staleness. This method requires workers with high staleness to restart computation, ensuring that the aggregated gradients remain relevant. Similarly, [351] introduces a token-control process to assemble gradients and decay those with severe staleness, enabling a tuning-free switch between training modes based on cluster status. This flexibility allows systems to adapt to dynamic resource constraints, leveraging asynchronous speeds when stragglers are present while maintaining the convergence properties of synchronous training when resources are abundant.

Moreover, the theoretical underpinnings of asynchronous convergence have been strengthened through novel analytical frameworks that support diverse network topologies. [352] proposes an algorithm that operates without a global clock, proving convergence to the exact solution under bounded delay assumptions by utilizing both primal and dual variables. In decentralized topologies, [353] introduces an algorithm that adaptively determines the number of neighbors to communicate with, achieving linear speedup in convergence while mitigating the effects of stale parameters from slow nodes. The integration of these asynchronous protocols with communication-efficient techniques, such as the partial-sharing principles discussed in [354], further reduces the communication load, making participation accessible for battery-constrained devices. By combining adaptive synchronization, staleness control, and computation-communication overlap, modern asynchronous protocols provide a scalable and resilient foundation for privacy-preserving machine learning. While these protocols optimize the timing and flow of computations, the following section explores an even more radical optimization: eliminating redundant computations entirely through compute-less networking and result reuse architectures.

6.4 Compute-Less Networking and Computation Reuse Architectures

6.4 Compute-Less Networking and Computation Reuse Architectures

While the asynchronous protocols and communication-computation overlap strategies discussed in the previous section effectively mitigate latency bottlenecks and straggler effects by optimizing the timing of data flow, they primarily focus on accelerating the execution of necessary computations. However, in the expanding Internet of Things (IoT) ecosystem encompassing billions of interconnected devices, a significant portion of computational tasks in domains such as augmented reality (AR), virtual reality (VR), and environmental monitoring involves repetitive inputs or highly similar data patterns that yield identical or near-identical outputs. In conventional architectures, including standard edge computing frameworks, servers often process these redundant requests from

scratch, leading to unnecessary expenditure of computational resources and increased response times even when transmission and scheduling are optimized. To address this fundamental inefficiency, the concept of “compute-less networking” has emerged as a transformative architectural paradigm that complements asynchronous execution. This approach shifts the operational model from “compute-after-communicate” to “compute-when-communicate,” or ideally, “compute-once-reuse-many,” thereby minimizing both communication overhead in the core network and redundant calculations at the edge by eliminating the need for re-execution entirely.

The cornerstone of compute-less networking is the mechanism of computation reuse, which relies on intelligent caching strategies to store the results of previously executed tasks. When a new request arrives, the system first checks if a similar or identical computation has already been performed; if so, the cached result is returned immediately, bypassing the need for re-execution. Research indicates that this strategy can drastically reduce task completion times and resource utilization beyond what asynchronous scheduling alone can achieve. For instance, implementations of network-based computation reuse architectures have demonstrated up to an 80% reduction in task completion time and a 60% reduction in resource utilization by effectively eliminating redundant executions at the edge server [355]. Furthermore, by ensuring that similar inputs trigger cache hits rather than full recomputation, these systems maintain high levels of correctness and accuracy, with studies reporting up to 90% computation correctness and accuracy in application-level evaluations [355]. Another investigation into the correctness of compute-less networking confirmed that task completion times could be reduced by up to 80% while preserving the integrity of the final output, validating the feasibility of this approach for latency-sensitive applications [356].

To operationalize computation reuse effectively within these optimized networks, sophisticated offloading mechanisms are required to match incoming service requests with the most appropriate edge servers capable of serving cached results or executing new tasks efficiently. Matching theory has been identified as a robust mathematical framework for designing such offloading schemes, working in tandem with the asynchronous aggregation principles described earlier. Specifically, the WHISTLE framework employs a vertical many-to-many matching scheme to offload the most frequently invoked and highly reusable services to optimal edge servers. This approach not only accelerates task completion by 20% but also reduces overall computation by up to 77% and communication overhead by up to 71% [357]. By extending this logic to horizontal one-to-many computation reuse sharing among edge servers, the architecture prevents redundant computations from bouncing back to the cloud, thereby optimizing the entire hierarchy of the network [357].

The benefits of compute-less architectures extend beyond mere speed and latency reduction; they are critical for energy-constrained environments, serving as a vital bridge to the microarchitectural optimizations discussed in the following section. In scenarios where IoT devices are battery-powered or rely on energy harvesting, minimizing the energy cost of data transmission and processing is paramount. Traditional offloading decisions often ignore the potential for result sharing, leading to suboptimal energy consumption even with efficient protocols. However, by leveraging the collaborative properties of applications with shared data, such as VR and AR, systems can ensure that shared data is transmitted only once, significantly lowering the energy burden on mobile users [358]. Moreover, the integration of service caching with task offloading allows edge servers to store application services and related databases, enabling the execution of corresponding tasks without retrieving data from distant clouds. Algorithms like OREO, which jointly optimize dynamic service caching and task offloading, have been shown to effectively reduce computation latency while maintaining low energy consumption in dense cellular networks [359].

Despite these advantages, the deployment of compute-less networking introduces complex chal-

lenges related to cache management, data similarity assessment, and system stability that must be addressed to ensure robust operation. The decision of what to cache and when to evict entries requires intelligent policies that account for user preferences, location awareness, and temporal dynamics. Deep learning-based approaches have been proposed to address these challenges, such as offline cache placement schemes that utilize stochastic quantization to optimize cache decisions based on location-aware user preference profiles [360]. Additionally, in environments with transient data generation, deep reinforcement learning (DRL) has been utilized to develop caching schemes that improve cache hit rates and reduce energy consumption while respecting data freshness constraints [361].

Furthermore, the stability of compute-less systems, particularly those employing random access for sporadic tasks, must be rigorously analyzed to prevent latency outages that could undermine the gains from redundancy elimination. Two-stage offloading approaches that employ multichannel random access have been proposed to stabilize systems with sporadic task arrivals, ensuring that the benefits of computation reuse do not come at the cost of system reliability [362]. As IoT networks evolve towards more transient and opportunistic execution models, particularly in batteryless systems powered by energy harvesting, the role of compute-less architectures becomes even more critical. Techniques that partition applications to minimize non-volatile data transfers and system activations align well with the goals of compute-less networking, potentially reducing required energy storage by over 94% [363].

In conclusion, compute-less networking and computation reuse architectures represent a pivotal shift in optimizing resource-constrained IoT deployments, moving the optimization frontier from *how fast* we compute to *whether* we need to compute at all. By transitioning from redundant processing to intelligent result caching and matching-based offloading, these architectures offer substantial improvements in latency, energy efficiency, and network bandwidth utilization, complementing the asynchronous timing optimizations of the prior section. The synergy between matching theory, deep reinforcement learning, and service caching provides a robust foundation for future IoT systems, enabling them to support computation-intensive and delay-sensitive applications sustainably. While these network-level strategies eliminate redundant workloads, the following section delves deeper into the device level, exploring specialized communication protocols and microarchitectural innovations designed to maximize the energy efficiency of the remaining essential computations on edge hardware.

6.5 Energy-Efficient Protocols and Microarchitectural Optimizations for Edge Devices

6.5 Energy-Efficient Protocols and Microarchitectural Optimizations for Edge Devices

While compute-less networking and computation reuse architectures effectively minimize redundant processing at the edge by leveraging cached results, the realization of Privacy-Preserving Machine Learning (PPML) and Federated Learning (FL) still places immense strain on the underlying edge devices themselves. These paradigms require frequent local model training and inference, which are fundamentally constrained by the severe energy limitations inherent to battery-powered sensors and Internet of Things (IoT) nodes. As the demand for real-time intelligence grows, the traditional von Neumann architecture often proves insufficient due to the high energy costs associated with data movement between memory and processing units. Consequently, bridging the gap between the high-level efficiency of compute-less networks and the low-level constraints of individual nodes necessitates a pivot toward specialized communication protocols and microarchitectural enhancements. These innovations prioritize energy efficiency without compromising the throughput required for

complex learning tasks, ensuring that the gains achieved through computation reuse are not negated by inefficient device-level operations.

A critical strategy in this domain involves the co-design of data compression algorithms with transmission policies to maximize node lifetime. Blindly applying compression can sometimes incur higher energy costs than transmitting raw data; therefore, optimal policies must jointly optimize compression and transmission rates based on channel conditions and delay constraints. Studies have demonstrated that such jointly optimized policies can extend sensor node lifetime by 90% to 2000% compared to transmission-only optimization, a gain that is particularly profound under stringent latency requirements [364]. Furthermore, in massive machine-type communication scenarios, protocols like NarrowBand IoT (NB-IoT) have been optimized to reduce signaling overhead for small data transmissions, achieving battery lifetime extensions of up to eight years through control plane optimizations [365]. These protocol-level optimizations serve as the foundational layer for efficient data exchange, complementing the broader network-level offloading strategies discussed previously.

Beyond protocol-level optimizations, microarchitectural innovations are essential for reducing the computational energy footprint of learning algorithms on the edge. Neuromorphic computing architectures, which emulate the sparse, event-driven nature of biological neural systems, offer a paradigm shift from continuous clock-driven processing to activity-dependent computation. This approach significantly reduces power consumption by activating only relevant neural subsets during inference. For instance, the integration of wake-up radios in neuromorphic split computing systems allows remote processing units to remain in low-power states until specific sparse impulse radio waveforms trigger activation, effectively translating input signal sparsity into direct energy savings for both computation and communication [366]. Similarly, the development of asynchronous neuromorphic hardware for tasks such as face detection has shown that power consumption is directly proportional to the number of synaptic operations, enabling ultra-low-power operation (approximately 20 mW) while maintaining competitive accuracy [367].

To further enhance efficiency, researchers are exploring heterogeneous many-core architectures tailored for spiking neural networks (SNNs). The “big.LITTLE” design philosophy, adapted for neuromorphic accelerators, utilizes cores with varying neuron and synapse capacities interconnected via parallel segmented buses. This configuration, managed by specialized runtime systems like SentryOS, allows for dynamic mapping of sub-networks to appropriate cores, resulting in energy reductions between 37% and 98% and latency improvements of up to 25% compared to traditional mesh-based Networks-on-Chip [368]. Additionally, the exploration of non-volatile memory (NVM) technologies, such as memristors and phase-change memory, enables the creation of dense, low-leakage synaptic storage. However, the reliability of these devices is sensitive to process variations and parasitic voltage drops. Advanced mapping techniques that account for endurance variations and thermal gradients within crossbar arrays can mitigate leakage power by over 50%, ensuring sustained performance and longevity for embedded ML systems [369] [370].

Parallel stream compression represents another vital microarchitectural optimization for handling high-volume data streams on multicore edge devices. Frameworks like CStream leverage hardware-conscious parallelization strategies and asymmetric multicore processors to achieve high throughput and low latency while drastically reducing energy consumption by up to 89% [371]. In the context of deep neural networks, systematic exploration of pruning, quantization, and deployment strategies allows models to be compressed to less than 10% of their original parameter count with negligible accuracy loss, making them viable for deployment on resource-constrained microcontrollers such as ARM Cortex-M series [372]. Automated frameworks like AdaDeep further refine this process by

selecting layer-wise compression techniques that holistically balance latency, energy, and storage constraints, achieving up to 9.8 times improvement in energy efficiency [373].

Moreover, the integration of compressed sensing techniques offers a dual benefit of data compression and encryption, which is particularly relevant for privacy-preserving applications. By leveraging the intrinsic sparsity of neural signals, compressed sensing can achieve sub-Nyquist sampling rates while simultaneously encrypting data through the sampling matrix, resulting in power savings of up to 35 times compared to conventional implementations [374]. For applications requiring extreme compression ratios, deep learning-based approaches using variational autoencoders have demonstrated the ability to compress physiological signal spectrograms by a factor of 293, vastly outperforming traditional codecs like JPEG2000 and H.264 while maintaining diagnostic integrity [375]. Finally, emerging technologies such as cryogenic neuromorphic hardware and photonic computing promise to push energy efficiency boundaries even further by minimizing resistance and leveraging ultra-fast optical processing, although significant engineering challenges regarding scalability and integration remain [376] [377]. Collectively, these specialized protocols and microarchitectural optimizations form the backbone of sustainable, scalable edge intelligence. By addressing the fundamental energy constraints at the device level, they enable the practical realization of privacy-preserving machine learning, thereby providing the robust, energy-aware foundation necessary for the intelligent resource orchestration across multi-tier fog and non-terrestrial networks discussed in the following section.

6.6 Intelligent Resource Orchestration in Multi-Tier Fog and Non-Terrestrial Networks

6.6 Intelligent Resource Orchestration in Multi-Tier Fog and Non-Terrestrial Networks

While the microarchitectural optimizations and energy-efficient protocols discussed in the previous section address the fundamental power and computational constraints of individual edge devices, realizing scalable Privacy-Preserving Machine Learning (PPML) and Federated Learning (FL) requires a higher-level strategy to manage these distributed resources dynamically. The proliferation of latency-sensitive Internet of Things (IoT) applications and the advent of sixth-generation (6G) networks have necessitated a paradigm shift from static resource allocation to intelligent, dynamic orchestration across multi-tier computing hierarchies. This evolution is particularly critical in environments integrating terrestrial fog computing with Non-Terrestrial Networks (NTN), where the heterogeneity of nodes—ranging from ground-based sensors to Unmanned Aerial Vehicles (UAVs), High Altitude Platforms (HAPs), and Low Earth Orbit (LEO) satellites—introduces profound complexity in managing communication and computation resources. Intelligent resource orchestration in these ecosystems relies heavily on predictive algorithms and advanced optimization frameworks to ensure seamless service delivery despite the spatiotemporal dynamics of network topologies and fluctuating service demands, effectively bridging the gap between device-level efficiency and system-wide performance.

In multi-tiered fog computing systems, the challenge lies in balancing the trade-off between processing latency and energy consumption while maintaining queue stability amidst temporal variations in system dynamics. Predictive offloading has emerged as a vital strategy to address these challenges. By formulating resource allocation as a stochastic network optimization problem, researchers have developed schemes that minimize time-average power consumption while guaranteeing stability for all system queues. For instance, predictive offloading and resource allocation schemes have demonstrated the ability to achieve near-optimal power efficiency with only mild requirements for predictive information, effectively reducing latency even in the presence of prediction errors [378].

Furthermore, the dynamic nature of fog networks, characterized by the unpredictable arrival and departure of neighboring nodes, necessitates online optimization frameworks. These frameworks utilize threshold-based algorithms to dynamically form local fog networks and optimize task distribution, significantly reducing maximum computational latency compared to baseline approaches [379].

The integration of NTN into this hierarchy expands the coverage horizon but intensifies the orchestration complexity due to long propagation delays, high Doppler shifts, and frequent handovers. In rural and remote areas where terrestrial infrastructure is sparse, HAPs and UAVs act as crucial edge servers, enabling vehicular edge computing (VEC) by processing tasks offloaded by energy-constrained ground vehicles. Analytical models characterizing these systems as queuing networks have been employed to assess optimal offloading factors that maximize the probability of real-time service under strict latency constraints [380]. Similarly, in Space-Air-Ground Integrated Networks (SAGINs), UAVs serve as mobile collectors that forward tasks from IoT devices to base stations or cloud servers. To handle the uncertainty in task arrival rates inherent in such dynamic environments, distributionally robust optimization techniques have been proposed. These data-driven approaches minimize expected system latency under worst-case probability distributions, ensuring robustness against environmental variability [381].

Deep Reinforcement Learning (DRL) has become a cornerstone for solving the NP-hard joint optimization problems associated with task offloading and resource allocation in these heterogeneous networks. In satellite-terrestrial edge computing scenarios, DRL-based algorithms have been designed to accelerate the learning process by adjusting candidate locations based on current channel states, achieving near-optimal offloading costs with reduced runtime consumption [382]. For more complex scenarios involving dependent tasks and collaborative UAV networks, meta-heuristic approaches combined with convex optimization have been utilized to minimize average user latency by jointly controlling offloading decisions and communication resource allocation [383]. Moreover, in integrated satellite-terrestrial networks, security-sensitive task offloading strategies modeled as Markov Decision Processes (MDP) and solved via Proximal Policy Optimization (PPO) have shown significant performance improvements by minimizing delay, energy consumption, and the risk of malicious attacks simultaneously [384].

The orchestration of resources in NTN also extends to maritime and aerial domains, where seamless connectivity is paramount. In maritime wireless networks, joint task computation and time allocation problems have been formulated to maximize weighted sum-rates for marine users, utilizing Bender and primal decomposition methods to achieve near-optimal solutions in polynomial time [385]. Similarly, for aerial-aided multi-access edge computing, re-configurable optimization frameworks continuously identify optimal MEC nodes and paths for applications such as in-flight entertainment, demonstrating substantial reductions in flow latency compared to static approaches [386]. The use of digital twins further enhances this orchestration by providing a virtual representation of the physical environment, allowing for the joint optimization of UAV trajectories and resource allocation to minimize total energy consumption while respecting processing delay constraints [387].

Despite these advancements, significant challenges remain in coordinating resources across disparate layers with varying trust levels and capabilities. The coupling of communication and MEC systems in MEC-empowered NTN requires process-oriented frameworks that utilize large-scale channel state information to characterize complex propagation environments, often decomposing non-convex latency minimization problems into solvable subproblems [388]. Additionally, the management of fog networks using reinforcement learning-based load balancing algorithms has proven effective

in minimizing processing time and overload probability by selecting optimal neighboring nodes without imposing specific assumptions on the network model [389]. As the industry moves toward 6G, the development of intelligent decision-making controllers leveraging AI and real-world open datasets will be pivotal in addressing the complexities of multi-constellation systems and ensuring fair, efficient resource distribution [390]. Ultimately, intelligent resource orchestration in multi-tier fog and NTN represents a convergence of predictive analytics, robust optimization, and adaptive learning. While this section establishes *where* and *when* computation should occur across the hierarchy, the sheer volume of data generated necessitates a complementary evolution in *how* data is processed within the network fabric itself, a topic explored in the following section through programmable data planes and in-network computing.

6.7 Programmable Data Planes and In-Network Computing for Real-Time Processing

6.7 Programmable Data Planes and In-Network Computing for Real-Time Processing

While intelligent resource orchestration across multi-tier fog and Non-Terrestrial Networks (NTN) effectively manages *where* and *when* computation occurs, the sheer volume of data generated by massive IoT ecosystems necessitates a fundamental rethinking of *how* data is transported and processed within the network fabric itself. The exponential growth of IoT deployments has created unprecedented demands for low-latency processing and efficient bandwidth utilization that traditional networking architectures struggle to meet. Conventional models, which strictly separate the control plane from a rigid data plane, often face bottlenecks where data generation rates outpace backbone transmission capacities. To address these limitations, the integration of Software-Defined Networking (SDN) with programmable data planes has emerged as a transformative paradigm. This approach shifts the computational burden from centralized servers or even edge nodes directly into the switching fabric, enabling in-network filtering, aggregation, and adaptation. By leveraging programmable data planes, networks can perform real-time processing tasks on forwarding devices, drastically reducing the volume of traffic traversing the core network and minimizing end-to-end latency for time-critical applications.

The evolution of the data plane from a static packet forwarder to a dynamic computing platform is central to this architectural shift, complementing the dynamic offloading strategies discussed in multi-tier systems. Early SDN implementations relied on protocols like OpenFlow, which offered flexibility in the control plane but maintained a relatively rigid data plane. However, recent advancements have introduced high-level programming languages and abstractions that allow for complex stateful processing within the switches themselves. A comprehensive analysis of data plane flexibility highlights the deviation from original architectures toward highly programmable environments capable of supporting diverse application logic [391]. This programmability is essential for implementing in-network computing strategies where switches do not merely route packets but actively participate in data manipulation. For instance, the development of abstract models for programmable data plane devices facilitates the creation of complex applications across heterogeneous hardware, enabling performance optimization and benchmarking that were previously unattainable [392].

One of the primary applications of programmable data planes in IoT contexts is the reduction of redundant traffic through in-network aggregation and filtering, a capability that significantly alleviates the bandwidth pressure on the fog and NTN layers described previously. In scenarios where multiple sensors generate correlated data streams, transmitting every raw packet to a central cloud processor or even an edge server is inefficient. Programmable switches can be configured to

aggregate these streams, perform initial filtering, or even execute lightweight inference models before forwarding only relevant summaries. The concept of “Dynamic Network Probes” exemplifies this capability by utilizing runtime interactive programming to optimize data plane resource allocation and support real-time network visibility, effectively turning the data plane into an active component of the analytics platform [393]. Furthermore, research into functional splits for deep learning suggests that decomposing neural networks and distributing sub-elements across the data plane can significantly enhance the efficiency of machine learning inference jobs in mobile networks, paving the way for 6G architectures [394].

Beyond simple aggregation, programmable data planes enable sophisticated traffic orchestration and content-aware communication, reinforcing the resource optimization goals of edge computing. In environments where computation resources are placed within a one-hop wireless topology, the ability to replicate and direct traffic based on content and computation needs is critical. Frameworks utilizing extended Berkeley Packet Filter (eBPF) within an SDN paradigm have demonstrated the ability to control traffic flows and packet replication for specific computation processes, resulting in significant savings in network bandwidth and system resources [395]. Similarly, the integration of publish/subscribe mechanisms with SDN allows for dynamic reaction to bandwidth sharing, mediating resource allocation between IoT aggregators and consumers to improve link utilization [396]. These capabilities are particularly vital for latency-sensitive applications, such as industrial monitoring or autonomous vehicle coordination, where delays introduced by round-trip communication to a central controller are unacceptable.

The synergy between SDN and programmable data planes also enhances the resilience and adaptability of IoT networks, addressing the spatiotemporal dynamics highlighted in multi-tier orchestration. Traditional networks often lack the agility to respond to sudden changes in traffic patterns or network topology without manual intervention or slow control plane updates. Programmable data planes allow for localized decision-making based on replicated global states, enabling switches to react instantaneously to network events such as congestion or link failures without waiting for controller instructions [397]. This local autonomy is complemented by the global view provided by the SDN controller, creating a hybrid architecture that balances scalability with responsiveness. For example, strategies involving in-network computing and programmable switches have been shown to achieve ultra-reliable low-latency communication (URLLC) by monitoring and modifying traffic at line speed, reducing latency by up to 50% compared to conventional architectures [398].

Moreover, the adoption of programmable data planes supports energy-efficient operations, a critical concern that aligns with the power optimization objectives of sustainable IoT and NTN deployments. By consolidating traffic and shifting workloads based on real-time conditions, programmable switches can reduce the activation of unnecessary network components and optimize power consumption. Approaches like P4Green leverage in-data plane registers to consolidate traffic onto the fewest number of switches and direct workloads to servers utilizing renewable energy, demonstrating that programmable data planes can contribute significantly to greener data centers [399]. This aligns with broader efforts to optimize network topologies and resource allocation dynamically, ensuring that the infrastructure adapts to both performance and sustainability goals [400].

Despite these advancements, challenges remain in fully realizing the potential of in-network computing for massive IoT. The complexity of programming heterogeneous data plane devices and ensuring security in a highly distributed processing environment requires robust abstraction layers and verification tools. Future research must focus on standardizing programming interfaces and developing automated tools that can translate high-level policy objectives into low-level data plane configurations reliably. As the field progresses, the convergence of SDN, programmable data

planes, and edge intelligence will likely define the next generation of IoT networks, enabling real-time processing capabilities that were once confined to centralized data centers. The transition from “compute-after-communicate” to “compute-when-communicate” represents a fundamental shift in network design, promising to unlock new possibilities for scalable, efficient, and responsive IoT ecosystems.

7 Domain-Specific Applications, Emerging Frontiers, and Future Research Directions

7.1 Domain-Specific Deployments in Healthcare, Finance, and Critical Infrastructure

7.1 Domain-Specific Deployments in Healthcare, Finance, and Critical Infrastructure

While theoretical frameworks for Privacy-Preserving Machine Learning (PPML) continue to evolve, the true measure of their efficacy lies in their transition from abstract concepts to tangible deployments within high-stakes sectors. In healthcare, finance, and critical infrastructure, the convergence of Federated Learning (FL) and Differential Privacy (DP) addresses the dual imperative of extracting actionable intelligence from distributed data silos while strictly adhering to rigorous regulatory frameworks such as the GDPR and HIPAA. These domains present unique challenges regarding data heterogeneity, latency constraints, and the catastrophic consequences of privacy breaches, necessitating tailored architectural solutions that balance utility with robust privacy guarantees. This section examines these real-world applications, setting the stage for the subsequent discussion on how these privacy-preserving mechanisms are being scaled to support the next generation of large generative models.

In the healthcare sector, the application of FL and DP has been most vigorous in medical imaging and Electronic Health Record (EHR) analysis. Decentralized medical imaging represents a frontier where institutions can collaboratively train robust diagnostic models without exchanging patient data. Recent frameworks have demonstrated the efficacy of differentially private gradient descent in multi-site semantic segmentation tasks for computed tomography, achieving high performance while successfully mitigating gradient-based model inversion attacks that could otherwise reconstruct sensitive image features [401]. The challenge of preserving utility under strict privacy budgets is further addressed through advanced noise injection strategies and adaptive aggregation mechanisms. For instance, in brain tumor segmentation using multi-modal MRI, the extension of similarity-weighted aggregation to include differential privacy layers has proven effective in minimizing communication costs while safeguarding client data against adversarial inference [402]. Furthermore, the integration of DP into domain transfer learning has shown remarkable resilience; models trained with strong privacy guarantees (epsilon around 1) exhibited negligible performance degradation when validated on external institutional data, suggesting that DP-enhanced models can generalize effectively across diverse clinical settings without compromising patient confidentiality [403].

Beyond imaging, the mining of EHRs for predictive analytics and synthetic data generation has gained traction, foreshadowing the broader role of generative AI in privacy-preserving contexts. Generative models, particularly those leveraging federated GANs and diffusion models, enable the creation of high-fidelity synthetic patient records that preserve statistical properties for research while eliminating re-identification risks [404]. Innovations such as EHR-D3PM and EHRDiff utilize discrete diffusion processes to generate realistic tabular medical data, significantly outperforming traditional baselines in fidelity and utility while reducing membership inference vulnerabilities

[405][406]. However, the deployment of these systems is not without hurdles; the sparsity and heterogeneity of EHR data require specialized handling. Studies indicate that while local differential privacy can severely impact model accuracy in highly sensitive settings, personalized federated learning approaches can recover lost utility, making them viable for tasks such as extracting vaccine adverse event mentions from distributed clinical notes [407]. Additionally, the use of steerable equivariant convolutional networks has been proposed to narrow the privacy-utility gap in medical image analysis by improving feature quality and parameter efficiency under DP constraints [408].

In the financial sector and broader critical infrastructure, the focus shifts toward fraud detection, anomaly identification, and secure data marketplaces. The outsourcing of anomaly detection to third-party analysts poses a unique privacy paradox: detecting anomalies requires identifying deviations, yet privacy mechanisms often obscure such differences. Novel frameworks like DPOAD address this by iteratively disentangling benign from anomalous entries, allowing for accurate detection under differential privacy guarantees without prior knowledge of data labels [409]. This is crucial for cybersecurity applications within critical infrastructures, where FL is employed for data-driven analysis. However, researchers have identified that the noise introduced by DP can be exploited by adversaries to launch adaptive model poisoning attacks, necessitating dynamic defense mechanisms such as reinforcement learning-based privacy level selection to maintain model integrity [223].

Furthermore, the management of data provenance and secure sharing in these sectors increasingly relies on hybrid architectures combining FL with blockchain technology. In healthcare supply chains and data marketplaces, federated ledger frameworks facilitate hierarchical decentralized data exchange, ensuring immutability and auditability while preserving privacy through smart contracts and permissioned domains [171]. Similarly, blockchain-based solutions have been proposed for securing EHRs and managing drug supply chains, offering patient-centric access control and real-time monitoring while addressing scalability through parallel processing [410]. These systems are often complemented by Trusted Execution Environments (TEEs) to protect data during processing at the edge, as seen in secure MQTT brokers for biomedical signal processing, which leverage hardware isolation to prevent data leakage during transmission [411].

Regulatory compliance remains a driving force behind these deployments. The necessity for Data Protection Impact Assessments (DPIAs) has highlighted gaps in current anonymization practices, emphasizing that privacy must be treated as a continuous process involving legal, organizational, and technical measures rather than a one-time static guarantee [412]. In the context of digital health passports and cross-border data flows, decentralized architectures utilizing proof-of-stake consensus have been explored to validate health status without centralizing Protected Health Information, thereby aligning with global privacy mandates while enabling interoperability [413]. Despite these advancements, the tension between stringent privacy requirements and the need for high-accuracy models persists, particularly in safety-critical applications like air traffic management or real-time fraud detection, where even minor utility losses can have severe consequences. As these domain-specific solutions mature, they provide the foundational trust and architectural blueprints necessary for the more complex integration of FL with Large Generative Models, a convergence that promises to further democratize AI while introducing new challenges in hallucination mitigation and scalability.

7.2 The Convergence of Federated Learning and Large Generative Models

7.2 The Convergence of Federated Learning and Large Generative Models

Building upon the domain-specific architectural blueprints established in healthcare and finance, the integration of Federated Learning (FL) with Large Language Models (LLMs) and Generative AI represents a transformative paradigm shift, often termed Federated Foundation Models (FFM). This convergence addresses the critical tension between the data-hungry nature of foundation models and the stringent privacy requirements inherent in the high-stakes sectors previously discussed. While foundation models like LLaMA, BERT, and GPT have demonstrated remarkable success by leveraging vast centralized datasets, optimizing them for specific domains—such as clinical decision support or financial fraud detection—often requires access to sensitive data that cannot be aggregated due to regulatory constraints. The FFM paradigm proposes a collaborative framework where FL enables privacy-preserving learning across multiple end-users, allowing these massive models to benefit from distributed data silos without exposing raw information [414]. This synergy not only democratizes the development of large models but also facilitates computation sharing, thereby reducing the burden on individual participants while fostering inclusivity and innovation [415].

A primary challenge in scaling this convergence is the computational and communication overhead associated with fine-tuning massive models on resource-constrained edge devices, a hurdle particularly acute in the IoT-heavy infrastructures noted in Section 7.1. Traditional full-model training is often infeasible in federated settings due to memory limitations. To mitigate this, recent research has focused on parameter-efficient fine-tuning (PEFT) strategies, such as Low-Rank Adaptation (LoRA), which significantly reduce the number of trainable parameters. For instance, the DP-LoRA algorithm combines differential privacy with low-rank adaptation to ensure strict privacy constraints while minimizing communication overhead during distributed training [416]. Furthermore, industrial-grade frameworks like FATE-LLM have been developed to facilitate federated learning for large language models, promoting efficient training through PEFT methods and protecting intellectual property while preserving data privacy [417]. Beyond standard gradient-based methods, novel approaches like FedMeZO integrate memory-efficient zeroth-order optimization within a federated setting, theoretically proven to converge faster than traditional first-order methods while drastically reducing GPU memory usage to levels comparable to inference [418].

Despite these architectural advancements, the reliability of generative outputs remains a paramount concern, especially given the catastrophic consequences of errors in the critical domains outlined earlier. LLMs are prone to “hallucinations,” generating plausible but factually incorrect content, which poses severe risks in high-stakes environments like healthcare and finance [419]. In a federated context, where models are updated based on heterogeneous local data, the risk of hallucination may be exacerbated by non-IID data distributions and the potential injection of biased local updates. Addressing this requires robust mechanisms to detect and mitigate hallucinations without compromising the privacy guarantees of the federated protocol. Techniques such as Retrieval-Augmented Generation (RAG) have been explored to counter hallucinations by integrating external knowledge; however, RAG itself introduces privacy vulnerabilities, as recent studies indicate that original text can be partially reconstructed from text embeddings, potentially jeopardizing user privacy [420]. Consequently, developing privacy-preserving retrieval mechanisms is essential for safe deployment.

To combat hallucinations specifically within federated ecosystems, researchers are exploring syn-

thetic data generation and advanced decoding strategies. Leveraging foundation models to generate synthetic data can enrich data diversity and reduce overfitting in federated settings, though this must be balanced against the risk of propagating synthetic biases [415]. Innovative approaches like SynTra demonstrate that optimizing system messages on synthetic tasks can effectively reduce hallucinations in real-world downstream tasks, a strategy that could be adapted for federated instruction tuning where direct access to diverse human-written data is limited [421]. Additionally, the concept of Federated Instruction Tuning (FedIT) has emerged as a pioneering approach to leverage heterogeneous user instructions stored on local devices, improving model generalization while preserving privacy [422].

The intersection of FL and generative AI also necessitates a re-evaluation of privacy threats unique to large models. While FL prevents raw data exposure, adversaries may still infer sensitive information from model updates or outputs. Analysis of privacy leakage in federated LLMs has revealed substantial vulnerabilities in popular models, even when protected by state-of-the-art differential privacy mechanisms, highlighting the need for more rigorous privacy accounting tailored to the finite-sample nature of fine-tuning [257]. Frameworks like \ewtune address this by providing finite-sample privacy guarantees for fine-tuning, reducing induced noise and improving performance across natural language understanding tasks [423]. Moreover, the dual-personalizing adapter architecture (FedDPA) has been proposed to handle test-time distribution shifts and personalization simultaneously, ensuring that models remain robust and accurate across diverse client environments without sacrificing global coherence [424].

Looking forward, the path toward trustworthy federated generative AI involves overcoming challenges in communication efficiency, scalability, and security, while simultaneously addressing the lifecycle management of data. The complexity of training foundation models via FL demands novel solutions for contribution evaluation and incentive mechanisms, as outlined in comprehensive surveys of the field [425]. Furthermore, as the community moves toward deploying these systems in critical infrastructure, the ability to detect and mitigate hallucinations in real-time becomes non-negotiable. Methods such as citation-enhanced generation and self-reflection frameworks offer promising avenues for enhancing factuality, yet their integration into decentralized, privacy-preserving workflows remains an open research frontier [426][427]. Ultimately, the successful convergence of FL and large generative models hinges on developing holistic frameworks that simultaneously optimize for utility, privacy, and reliability. However, the deployment of such powerful, data-intensive models introduces a new layer of complexity regarding data retention: if a model learns from distributed data to become a foundation for critical services, how can we ensure that specific data contributions can be removed upon request? This question bridges the gap to the subsequent discussion on Federated Unlearning, Data Sovereignty, and the Right to be Forgotten.

7.3 Federated Unlearning, Data Sovereignty, and the Right to be Forgotten

7.3 Federated Unlearning, Data Sovereignty, and the Right to be Forgotten

While the convergence of Federated Learning (FL) and Large Generative Models discussed in the previous section offers a promising path for privacy-preserving collaboration, it introduces complex lifecycle management challenges, particularly regarding data retention and removal. The emergence of stringent data protection regulations, such as the European Union’s General Data Protection Regulation (GDPR) and the California Privacy Rights Act (CPRA), has fundamentally altered the landscape of machine learning deployment by enshrining the “right to be forgotten.” This legal mandate empowers individuals to request the erasure of their personal data from trained models.

In the context of FL, where data remains decentralized across numerous clients, complying with these mandates presents a unique and formidable challenge. Unlike centralized settings where a server possesses full access to the training corpus, FL operates under constraints of interactivity, stochasticity, and limited data accessibility, rendering traditional unlearning techniques inapplicable [428]. Consequently, the field of Federated Unlearning (FU) has emerged as a critical research frontier, aiming to efficiently remove specific client contributions or data samples from global models without the prohibitive cost of retraining from scratch [429].

The primary technical objective of federated unlearning is to produce a model that is statistically indistinguishable from one trained exclusively on the remaining dataset, thereby ensuring that the influence of the departed client is completely eradicated. The most straightforward approach, often considered the gold standard, involves retraining the global model from scratch using only the data of participating clients. However, this method is computationally expensive and communication-intensive, making it impractical for large-scale, resource-constrained FL ecosystems, a constraint that becomes even more acute when dealing with the massive parameter counts of foundation models [430]. To address this inefficiency, researchers have developed a variety of approximate and exact unlearning strategies. For instance, FedEraser proposes a methodology that leverages historical parameter updates retained by the central server to reconstruct the unlearned model, achieving a significant speed-up of approximately $4\times$ compared to full retraining while maintaining model efficacy [431]. Similarly, other approaches utilize knowledge distillation to restore model performance after subtracting accumulated historical updates, a method that avoids restrictions on neural network types and does not require active client participation during the unlearning phase [432].

Data sovereignty in FL extends beyond simple deletion; it necessitates verifiable proof that the data has been forgotten, a requirement that bridges the gap between technical implementation and regulatory accountability. Recent advancements emphasize the need for “verifiable federated unlearning,” where the leaving participant is granted the right to verify the unlearning effect. Frameworks like VeriFi introduce a systematic approach where clients inject carefully designed markers to fingerprint their contribution and subsequently check the global model’s performance on these markers to confirm erasure [433]. This shifts the paradigm from blind trust in the server to an auditable process, addressing the opacity that often plagues distributed systems. Furthermore, blockchain technology has been integrated into FU frameworks to provide immutable audit trails and prove data removal through cryptographic mechanisms like Chameleon hash functions, ensuring that data contributions are eliminated even from other clients’ models [434]. These verification mechanisms are crucial precursors to the formal compliance and auditing frameworks discussed in the following section.

The complexity of FU is further compounded by the heterogeneity of data distributions and the variety of application domains. In scenarios involving Human Activity Recognition (HAR) on IoT devices, lightweight unlearning methods have been proposed that utilize third-party datasets and KL divergence to align probability distributions, achieving speedups ranging from hundreds to thousands of times compared to retraining [435]. For recommendation systems, where user interaction data is highly dynamic, frameworks like FRU employ log-based rollback mechanisms inspired by database transactions to efficiently erase user contributions while preserving the integrity of the recommender [436]. Moreover, the challenge of unlearning in non-IID settings has led to the development of stability-aware algorithms, such as FATS, which modify the classical FedAvg algorithm to ensure total variation stability, thereby enabling exact unlearning with theoretical guarantees [437].

Despite these advancements, significant challenges remain regarding the trade-offs between privacy, utility, and efficiency. Approximate unlearning methods, while efficient, may leave residual traces of the forgotten data, potentially exposing the system to membership inference attacks. Conversely, exact unlearning methods often incur high computational or storage overheads. The interplay between unlearning and fairness is also critical; research indicates that data heterogeneity can significantly impact the stability and fairness of the unlearned model, necessitating optimization frameworks that manage these trade-offs explicitly [438]. Additionally, the strategic behavior of users regarding data revocation introduces game-theoretic complexities, where users’ decisions to withdraw data are coupled with model performance and unlearning costs [439].

Looking forward, the integration of federated unlearning with emerging technologies such as Large Language Models (LLMs) and generative AI presents new frontiers that directly impact the viability of foundation models in regulated environments. The sheer scale of foundation models makes retraining infeasible, prompting the exploration of in-context unlearning and parameter-efficient fine-tuning strategies that can operate without full model access [440]. Furthermore, the concept of “deletion-as-control” offers a novel formalism that allows data to be used freely prior to deletion while imposing strict post-deletion constraints, potentially reconciling the tension between data utility and privacy rights [441]. As the regulatory landscape continues to evolve towards the rigorous standards outlined in the subsequent section, the development of standardized, certifiable, and efficient federated unlearning mechanisms will be paramount for establishing trustworthy and compliant AI ecosystems that respect data sovereignty.

7.4 Regulatory Compliance, Legal Frameworks, and Standardization Efforts

7.4 Regulatory Compliance, Legal Frameworks, and Standardization Efforts

While Section 7.3 detailed the technical mechanisms for enforcing the “right to be forgotten” and ensuring data sovereignty through Federated Unlearning, the broader deployment of Privacy-Preserving Machine Learning (PPML) is equally constrained by a rapidly evolving global regulatory landscape. As these technologies transition from theoretical frameworks to critical infrastructure in healthcare, finance, and public sectors, the alignment between technical capabilities—such as Differential Privacy (DP) noise injection and federated aggregation—and legal mandates has become a paramount concern. The European Union has emerged as a primary architect of this environment, spearheading initiatives that demand rigorous compliance, transparency, and accountability. Central to this shift is the EU AI Act, which establishes a risk-based approach to AI regulation, categorizing systems based on their potential impact on fundamental rights and safety. However, a critical analysis reveals significant gaps between the current state of PPML research and the specific requirements imposed by emerging legislation and standardization bodies, setting the stage for the quantifiable auditing metrics discussed in Section 7.5.

The EU AI Act represents a pioneering legislative effort to create a harmonized legal framework for AI, yet its application to decentralized and privacy-preserving architectures presents unique challenges that directly intersect with the unlearning strategies previously described. The Act mandates strict conformity assessments and post-market monitoring for high-risk AI systems, effectively proposing a Europe-wide ecosystem for AI auditing [442]. For PPML systems, this creates a complex compliance landscape where the very mechanisms designed to protect privacy can obscure the transparency required for regulatory verification. The Act’s provisions on data quality and human oversight have been critiqued as potentially problematic for foundation models and complex distributed systems, suggesting that a purely risk-based ex-ante approach may need to be

supplemented by robust liability frameworks [443]. Furthermore, the interpretation of “high-risk” within the Act requires precise methodology to map legal requirements to measurable technical characteristics, a task that remains difficult for non-IID and heterogeneous federated environments where data provenance is often obscured [444].

Parallel to legislative efforts, the standardization community is racing to provide actionable guidelines for AI management to bridge the gap between abstract legal principles and technical implementation. The introduction of ISO 42001:2023 marks a significant milestone as the first international standard specifically designed for Artificial Intelligence Management Systems (AIMS). Comparative analyses indicate that while frameworks like COBIT 2019 align closely with the EU AI Act’s risk management principles, ISO 42001 offers the most comprehensive facilitation for integrating Large Language Models (LLMs) and addressing multifaceted risks [445]. For PPML, these standards are crucial because they provide the structural backbone for documenting the governance of decentralized training processes. However, current evaluations suggest that even these advanced frameworks possess inadequacies in overseeing LLM-specific risks, highlighting a critical need for continuous evolution to address the nuances of privacy-preserving generative AI [445].

A profound gap exists in the realm of liability attribution, a challenge that is exacerbated by the technical complexities of unlearning and data removal discussed in the preceding section. Current legal liability systems often struggle to assign responsibility when harms arise from unforeseeable behaviors in AI systems, a problem compounded by the opacity of federated models and the stochastic nature of differential privacy. The unpredictability of judicial outcomes regarding AI liability creates an environment of uncertainty that stifles innovation and investment [446]. Recent proposals for European AI liability directives have been described as a “half-hearted approach,” primarily relying on disclosure mechanisms and narrow presumptions of fault rather than establishing a comprehensive framework for end-to-end accountability [447]. This is particularly detrimental to PPML, where the separation of data and model updates across multiple jurisdictions complicates the determination of causality and defectiveness, especially when verifying whether a specific client’s data has been successfully erased. To bridge this gap, scholars advocate for distinguishing between endogenous and exogenous sources of harm and carefully allocating liability between developers and deployers, potentially through industry-funded guarantee schemes [443][446].

Moreover, the intersection of privacy regulations like the GDPR with sustainability goals and fairness mandates adds another layer of complexity to the compliance landscape. There is an emerging discourse on “Sustainable AI Regulation,” which argues that current frameworks must be adjusted to account for the environmental footprint of training large models, even suggesting that certain individual rights, such as the right to erasure, might need reinterpretation if they conflict with broader sustainability objectives [448]. In the context of FL, this raises questions about the energy efficiency of privacy-preserving protocols and whether compliance certification should include carbon footprint metrics, creating a tension between the computational cost of unlearning and ecological goals. Additionally, the challenge of ensuring fairness while maintaining privacy is acute; competing notions of algorithmic fairness are often mutually incompatible, and regulatory frameworks must provide guidance on how to navigate these trade-offs without violating non-discrimination laws [449].

The path forward requires a move towards standardized compliance certification that transcends national borders, converting the abstract legal mandates discussed here into the concrete, quantifiable benchmarks explored in the subsequent section. Proposals for a contextual, coherent, and commensurable (3C) framework suggest bifurcating the AI lifecycle and adopting international industry standards to convert abstract principles into quantifiable metrics [450]. Such standardization is

essential for creating a trustworthy AI ecosystem where PPML techniques can be audited not just for their privacy guarantees, but for their adherence to a holistic set of legal and ethical norms. Without unified standards and clear liability attribution, the promise of privacy-preserving AI may remain unrealized, hindered by regulatory fragmentation and legal ambiguity. Consequently, future research must focus on developing automated auditing tools and dual governance models that can satisfy the rigorous demands of the EU AI Act, ISO 42001, and emerging global statutes while preserving the core utility of federated and differentially private systems, thereby enabling the metric-driven accountability frameworks detailed in Section 7.5 [451].

7.5 Metrics for Trustworthiness, Accountability, and Holistic Auditing

7.5 Metrics for Trustworthiness, Accountability, and Holistic Auditing

While Section 7.4 established the complex regulatory and legal landscape governing Privacy-Preserving Machine Learning (PPML), a critical gap remains in translating these abstract mandates into quantifiable, technical realities. The deployment of Federated Learning (FL) and Differential Privacy (DP) demands a shift from isolated technical guarantees toward a unified framework for evaluating trustworthiness that satisfies both legal requirements and ethical imperatives. As the industry matures, the fragmentation of evaluation strategies—where privacy, fairness, and robustness are assessed in silos—often leads to suboptimal deployments where mitigating one risk exacerbates another. To address this, recent literature advocates for a comprehensive taxonomy of trustworthiness that integrates dimensions such as explainability, data governance, safety, accountability, and reproducibility, forming the bedrock of a meta-decision process for strategic AI adoption [452]. This multidimensional perspective is essential for converting the regulatory principles discussed previously into the quantifiable benchmarks necessary for reliable PPML systems.

A central challenge in operationalizing these metrics lies in the inherent tensions between different pillars of trustworthiness, a theme that bridges the legal constraints of the previous section and the ethical dilemmas of the next. For instance, while Differential Privacy provides rigorous mathematical bounds on privacy leakage, the introduction of noise can disproportionately impact underrepresented groups, thereby degrading fairness—a conflict that regulators struggle to adjudicate without precise metrics. Similarly, efforts to enhance model robustness against adversarial attacks may conflict with privacy preservation goals. Acknowledging these trade-offs, researchers have proposed information-theoretic frameworks that analytically optimize the balance between privacy, interpretability, and transferability, offering a rigorous method to study these interactions rather than treating them in isolation [453]. Furthermore, the concept of “conformity” has been introduced as a sixth pillar alongside explainability, robustness, transparency, fairness, and privacy, providing probabilistic assurances that a system will behave as expected within defined boundaries [454]. Integrating such diverse metrics requires a structured approach that connects governance requirements to the entire AI system lifecycle, ensuring that trust is maintained from data acquisition through to deployment and monitoring [455].

To bridge the gap between theoretical frameworks and the compliance needs outlined in Section 7.4, the development of automated auditing tools and standardized metrics is paramount. Current auditing practices often suffer from a lack of agreed-upon procedures, making it difficult to verify claims of compliance or safety in decentralized environments. In response, the community is moving toward “criterion audits” modeled after financial auditing practices, which provide assurance to stakeholders regarding an organization’s ability to govern its algorithms effectively [456]. However,

the ecosystem of available tools remains fragmented, with many open-source solutions failing to support the full scope of accountability goals, ranging from harms discovery to advocacy [457]. To bridge this gap, unified frameworks like ComplAI have been developed to enable the observation, analysis, and quantification of multiple responsible AI attributes—including fairness, robustness, and drift susceptibility—culminating in a single “Trust Factor” that facilitates actionable recourse for model improvement [458]. Additionally, the creation of a comprehensive metrics catalogue that delineates process, resource, and product metrics is essential for operationalizing accountability, particularly in the context of large-scale generative models [459].

The scope of holistic auditing must also expand to include environmental sustainability, a dimension often overlooked in traditional PPML evaluations yet increasingly mandated by emerging regulations. The substantial carbon footprint and water consumption associated with training large-scale models pose significant ethical challenges that intersect with privacy and security concerns. Regulatory frameworks are beginning to recognize this, suggesting that sustainability metrics, such as greenhouse gas emissions disclosure, should be integrated into the broader accountability toolkit alongside privacy and safety standards [448]. Ignoring the environmental cost of privacy-preserving techniques, such as the computational overhead of homomorphic encryption or secure multi-party computation, risks creating systems that are private but unsustainable. Therefore, a truly trustworthy AI ecosystem must adopt a tripartite evaluation framework that balances procedural integrity, resource efficiency, and product quality [459].

Finally, achieving robust accountability requires the adoption of dual governance models that synergize the centralized regulation discussed in Section 7.4 with decentralized, crowdsourced safety mechanisms, setting the stage for the human-centric collaboration explored in Section 7.6. While centralized regulations like the EU AI Act provide a necessary legal baseline, they often lack the agility to address rapid technological advancements or specific contextual nuances. Conversely, decentralized approaches offer flexibility but may struggle with enforcement and standardization. A proposed “Dual Governance” framework suggests a cooperative synergy where government regulations set the overarching safety standards, while community-driven tools and participatory auditing mechanisms provide continuous, real-time monitoring and verification [460]. This approach is complemented by the concept of “herd audits,” which leverage collective intelligence to create credible accountability threats for developers, incentivizing the upholding of privacy and ethical claims even in the presence of epistemic disparities among auditors [461]. By integrating unified metrics, automated tooling, environmental considerations, and hybrid governance structures, the field can move toward a future where PPML systems are not only mathematically secure but also socially accountable and holistically trustworthy, providing the necessary foundation for the ethical frontiers of bias mitigation and explainability.

7.6 Ethical Frontiers: Bias Mitigation, Explainability, and Human-AI Collaboration

7.6 Ethical Frontiers: Bias Mitigation, Explainability, and Human-AI Collaboration

Building upon the dual governance models and holistic auditing frameworks introduced in Section 7.5, the focus now shifts to the specific ethical imperatives that these mechanisms must uphold: bias mitigation, explainability, and meaningful human-AI collaboration. While Section 7.5 established the metrics and governance structures necessary for accountability, this section delves into the substantive ethical challenges that arise when privacy-preserving generative systems are deployed within federated learning architectures. The intersection of differential privacy, federated aggregation, and generative AI creates unique vectors for bias propagation, where noise injection and

data heterogeneity can skew model outcomes against underrepresented groups. Consequently, ensuring that privacy mechanisms do not inadvertently compromise fairness or transparency becomes paramount as these systems permeate critical domains such as healthcare, finance, and autonomous decision-making. Addressing these frontiers requires a holistic approach that integrates bias mitigation strategies directly into the privacy-preserving pipeline, enhances explainability without violating data sovereignty, and establishes robust human-in-the-loop workflows.

Bias mitigation in federated environments is particularly complex due to the non-IID (non-independent and identically distributed) nature of client data and the obscuring effects of differential privacy noise. Standard federated learning protocols may inadvertently amplify existing disparities if local datasets reflect historical prejudices or demographic imbalances. Research indicates that fairness and privacy are often in tension, necessitating frameworks that optimize for both simultaneously. For instance, specific approaches have been developed to incorporate fairness metrics directly into federated learning models alongside differential privacy, revealing an empirical interplay where accuracy, fairness, and privacy must be balanced carefully [462]. Furthermore, the use of generative content to mitigate data heterogeneity has shown promise; by diversifying private data through generative models, systems can prevent clients from overfitting to biased local distributions, thereby alleviating group unfairness while maintaining privacy [463]. However, the introduction of synthetic data itself carries risks, as generative models can hallucinate or reinforce stereotypes if not rigorously audited. To counter this, comprehensive auditing frameworks are essential to evaluate synthetic datasets for bias, fidelity, and privacy preservation across diverse modalities, ensuring that the “trustworthiness” of the generated data is quantifiable before deployment [464]. In recommendation systems, which are highly susceptible to demographic bias, novel graph neural network architectures have been proposed to mitigate group unfairness significantly without compromising the privacy of sensitive user attributes [465].

Parallel to the challenge of bias, explainability remains a formidable barrier in privacy-preserving federated systems, where the “black-box” nature of deep learning is compounded by the inability to inspect raw training data or centralized model weights. The requirement to protect user privacy often conflicts with the need for transparency, making it difficult to diagnose why a model makes certain decisions or to attribute errors to specific data sources. This opacity is particularly concerning in high-stakes applications where accountability is legally mandated, echoing the compliance needs discussed in previous sections. To address this, researchers are exploring methods to provide transparent source attribution and in-situ explanations within multimodal intelligence frameworks, leveraging visual analytics to balance automation with human understanding [466]. Moreover, the development of ethical guidelines specifically for generative AI in scientific research emphasizes the need for transparency and reproducibility, arguing that users must understand model limitations regarding truthfulness and bias to maintain research integrity [467]. In the context of large language models (LLMs), which are increasingly integrated into federated ecosystems, the risk of generating harmful misinformation or biased content necessitates advanced filtering and safety mechanisms that can operate without exposing private prompts [468]. The challenge lies in designing explanation interfaces that reveal the logic of federated updates without leaking information about individual contributions, a gap that current regulatory frameworks like the EU AI Act are beginning to address but have not yet fully resolved.

To navigate these trade-offs between privacy, fairness, and transparency, human-AI collaboration emerges as the cornerstone of ethical deployment, shifting the paradigm from fully autonomous systems to those governed by “experts-in-the-loop” or community-driven oversight. The complexity of these ethical dilemmas often exceeds the capacity of automated algorithms to resolve op-

timally, requiring human judgment to adjudicate conflicting objectives. For example, dynamic workflows have been proposed to transform complex privacy policies into accessible question-and-answer formats through conversational AI, facilitated by continuous collaboration between legal experts and designers [469]. This collaborative approach ensures that privacy safeguards are not only technically sound but also legally compliant and user-centric. Furthermore, the concept of “dual governance” suggests a synergistic model where centralized regulations are complemented by crowdsourced safety mechanisms, allowing communities to actively participate in identifying and mitigating harms in generative AI systems [460]. In scenarios involving moral dilemmas, such as autonomous vehicle decision-making, privacy-preserving crowd-guided approaches enable the aggregation of ethical preferences without exposing individual voters’ sensitive moral stances, thus fostering a collective ethical framework [470]. Additionally, the integration of human feedback loops is critical for refining safety in open-domain chatbots, where human-and-model-in-the-loop frameworks have demonstrated superior performance in mitigating toxic behaviors compared to purely automated filters [471].

Looking forward, the ethical frontier of privacy-preserving machine learning demands a reconfiguration of how we view the relationship between technology and society, setting the stage for the scalable and sustainable ecosystems discussed in Section 7.7. It is insufficient to treat bias, explainability, and human oversight as add-on features; they must be foundational design principles. Future research must focus on developing standardized metrics for trustworthiness that encompass privacy, fairness, and robustness simultaneously, moving beyond siloed evaluations [472]. As generative models become more powerful, the risk of exacerbating health disparities and social inequalities grows, necessitating proactive ethical frameworks like “GREAT PLEA” that encompass governance, reliability, equity, and empathy specifically for healthcare applications [473]. Ultimately, the goal is to create ecosystems where privacy preservation empowers rather than obscures, enabling human-AI collaboration that is both ethically grounded and technically robust, thereby providing the necessary social license for the scalable, edge-centric deployments explored in the following section.

7.7 Future Research Directions Toward Scalable and Sustainable Privacy-Preserving Ecosystems

7.7 Future Research Directions Toward Scalable and Sustainable Privacy-Preserving Ecosystems

Building upon the ethical imperatives of bias mitigation, explainability, and human oversight discussed in the preceding section, the trajectory for future research must now pivot toward constructing ecosystems that are not only ethically grounded but also technically scalable, sustainable, and rigorously privacy-preserving. As artificial intelligence shifts from centralized cloud repositories to distributed edge environments, the convergence of Federated Learning (FL) and Differential Privacy (DP) provides a robust theoretical foundation; however, practical deployment across heterogeneous networks faces significant hurdles. A primary avenue for investigation lies in the development of ultra-lightweight protocols specifically designed for resource-constrained Edge-IoT devices. Current deep learning models often demand computational resources exceeding the capabilities of typical sensors and actuators, creating a barrier to ubiquitous intelligence. Research must therefore focus on algorithm-hardware co-design strategies that enable efficient on-device training and inference without compromising privacy guarantees. As noted in recent surveys, the success of Edge-AI hinges on innovative circuits and hardware designs capable of supporting limited learning in autonomous systems, emphasizing the need for mixed-signal circuits and neuro-inspired computing systems to

realize energy-efficient ASICs [474]. Furthermore, the integration of quantization, pruning-aware training, and sparsification techniques is essential to expand the functional range of these devices, allowing them to manage complex machine learning tasks using only local resources while maintaining the fidelity required for fair outcomes [475].

To address the explainability challenges highlighted previously, another critical frontier is the integration of neuro-symbolic AI within privacy-preserving frameworks. While deep learning excels at pattern recognition, it often lacks the interpretability and logical reasoning capabilities required for high-stakes decision-making in decentralized settings. Future ecosystems should explore hybrid architectures that combine the data-driven power of neural networks with the rule-based transparency of symbolic AI. This approach aligns with the broader vision of the “Cognitive Internet,” where connected objects independently acquire knowledge and understanding while preserving decision-making autonomy [476]. By embedding symbolic reasoning into federated nodes, systems can achieve more robust privacy auditing and explainability, ensuring that DP mechanisms do not obscure the logical basis of model updates—a crucial step for applications requiring strict regulatory compliance where the “black-box” nature of current models remains a significant adoption barrier.

Complementing the need for transparent reasoning, decentralized identity management represents a third pillar for future research, essential for establishing trust in multi-stakeholder environments and facilitating the human-AI collaboration models described earlier. As IoT ecosystems evolve into complex socio-technical systems, the traditional reliance on centralized authorities for identity verification becomes a single point of failure and a privacy bottleneck. Future architectures must leverage blockchain and distributed ledger technologies to create self-sovereign identity frameworks that allow devices and users to authenticate and interact without revealing sensitive attributes. This aligns with the proposal of hierarchical integrated federated ledgers aimed at guaranteeing decentralized security and privacy-preserving properties in multi-domain resource orchestration [477]. Moreover, the concept of “socialized learning,” which amplifies collaborative capacity through social principles, suggests that identity management should also account for behavioral trust and reputation systems, optimizing networking processes across diverse platforms to support community-driven oversight [478].

Achieving global interoperability and equity in AI infrastructures requires addressing the profound heterogeneity of devices and network conditions, ensuring that privacy benefits are not restricted to well-resourced domains. The path forward involves standardizing communication protocols and aggregation mechanisms that can seamlessly operate across terrestrial and non-terrestrial networks, including satellites and UAVs. Research into Terrestrial and Non-Terrestrial Integrated Networks (TaNTIN) highlights the necessity of collaborative technologies to ensure quality-of-service guarantees for futuristic services, suggesting that privacy-preserving protocols must be adaptable to varying latency and bandwidth constraints [479]. Additionally, the vision of an “open metaverse” and decentralized digital society emphasizes the need for open-source tooling and federated environments that foster inclusivity and prevent the concentration of power in proprietary platforms, thereby supporting the equitable access called for in ethical frameworks [480].

Finally, sustainability remains a defining challenge for these future ecosystems, as the environmental cost of distributed intelligence cannot be overlooked. The energy consumption of distributed training and the carbon footprint of massive IoT deployments must be mitigated through green AI initiatives. Future research should investigate energy-centric federated management strategies and immersion cooling technologies to ensure that the push for intelligence does not come at an unacceptable environmental cost [481]. The dichotomy between the resource-intensive nature of deep learning and the limited battery life of edge devices necessitates a holistic design approach that

balances privacy, utility, and energy efficiency [482]. Ultimately, the realization of scalable and sustainable privacy-preserving ecosystems depends on a unified framework that integrates lightweight protocols, neuro-symbolic reasoning, decentralized identity, and equitable interoperability. By addressing these open challenges, the research community can pave the way for a trustworthy AI future where intelligence is ubiquitous, secure, and beneficial to all stakeholders, transcending the limitations of current siloed approaches [483].

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